Methods for studying the brain

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COMPUTATIONAL APPLICATION TO COGNITIVE NEUROSCIENCE

Overview

We are going to investigate non-invasive brain imaging methods:

- Magnetic Resonance Imaging
- Functional Near-Infrared Spectroscopy
- ElectroEncephaloGraphy
- MagnetoEncephaloGraphy
- Positron Emission Tomography
- Intracranial recordings

Overview

We are going to investigate non-invasive brain imaging methods:

- Based on blood-oxygen-level-dependent (BOLD):
  - Magnetic Resonance Imaging
  - Functional Near-Infrared Spectroscopy
- Based on electromagnetism:
  - ElectroEncephaloGraphy
  - MagnetoEncephaloGraphy
- Invasive (based on multiple principles):
  - Positron Emission Tomography
  - Intracranial recordings

"Old" methods

Broca's area

- Determine the functions of cerebral areas
- Still essential in many surgical operations

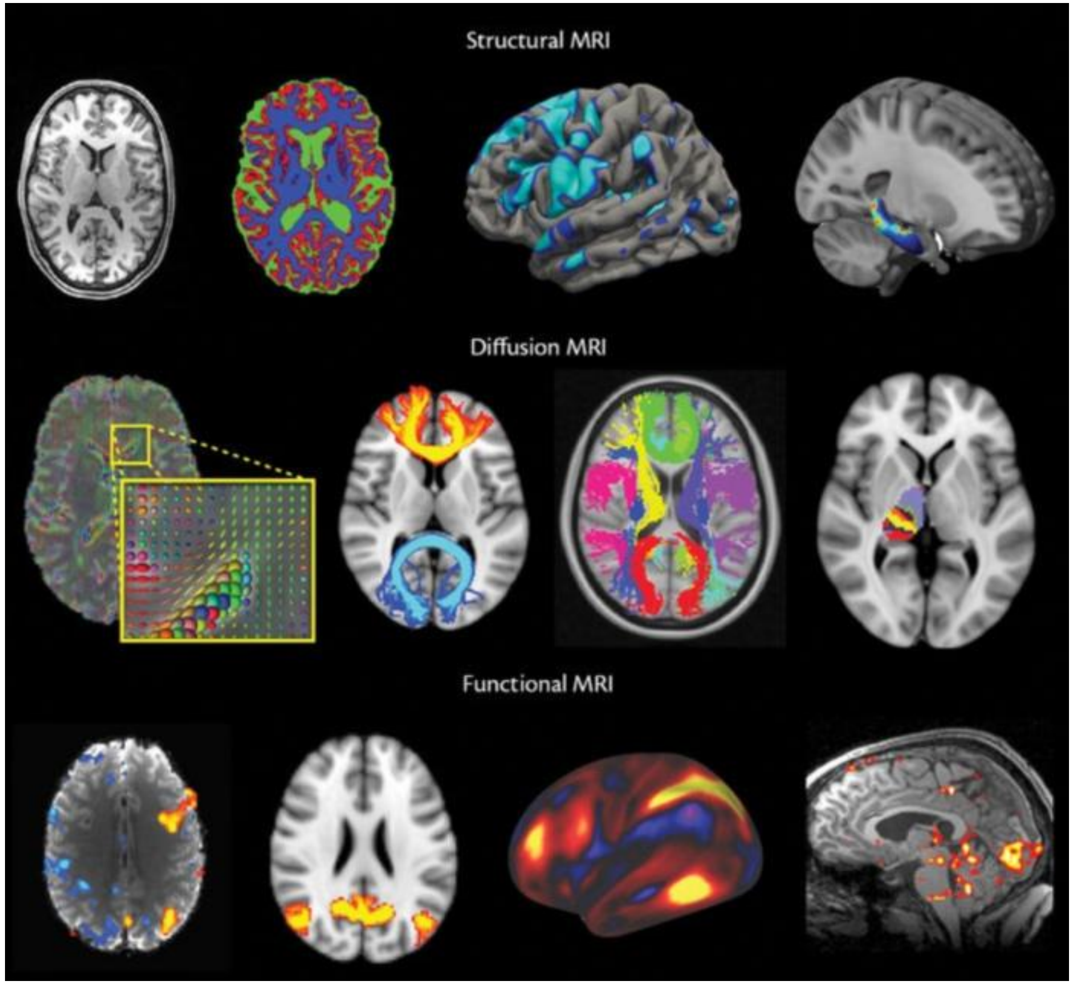

theguardian

Magnetic Resonance Imaging (MRI)

- ❑ it is possible to investigate many modalities → structural, functional, molecules concentrations, ...

- ❑ It is not invasive

- ❑ MRI is not always accessible and is very expensive



First medical images: X-rays

Discovered in 1895

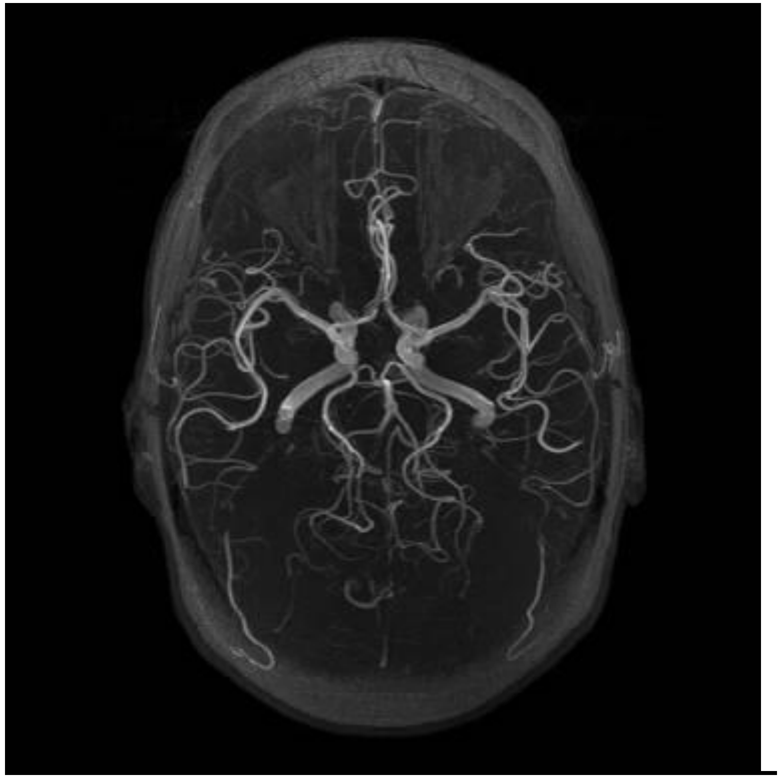
Images of bones

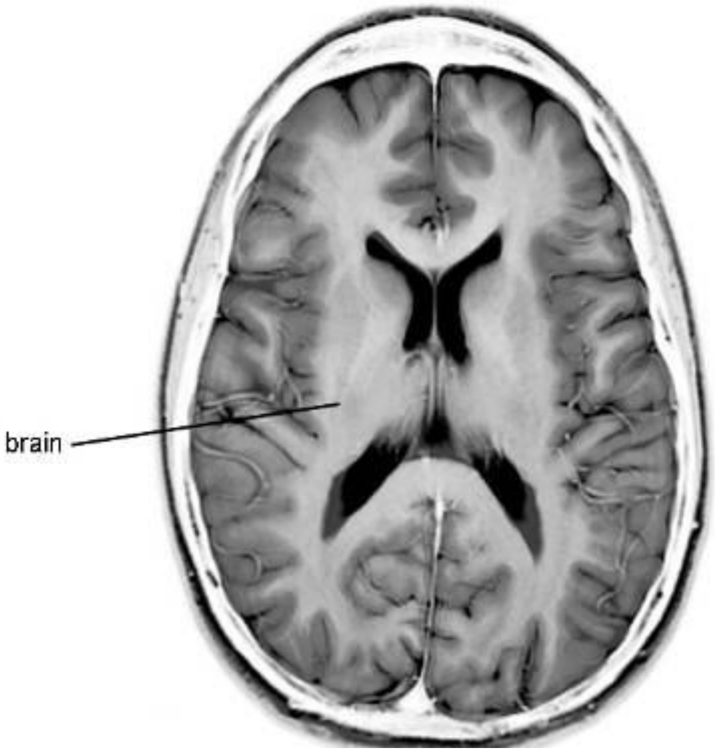
What part of the body is this?

Magnetic Resonance Imaging (MRI)

MRI add more information:

- soft tissues (muscles, blood vessels, ligaments and internal organs).
- blood flow
- iron content
- ...

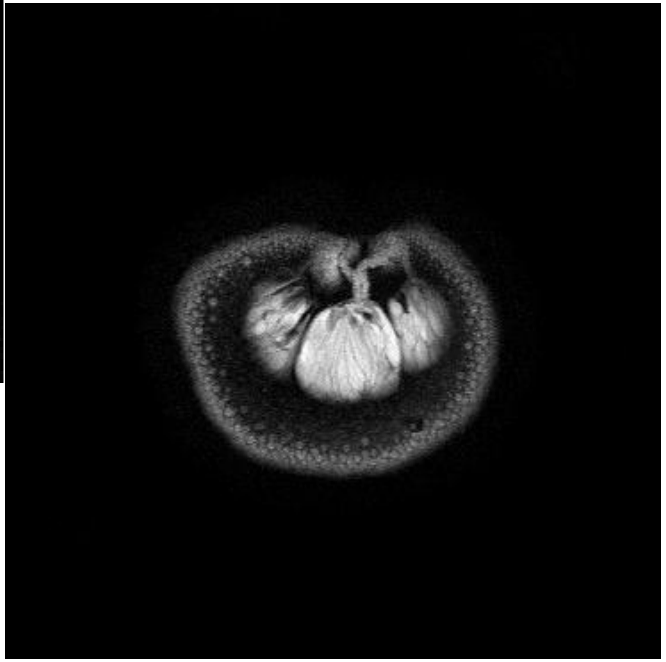




Magnetic Resonance Imaging

MRI scanners capture sections through the body.

Sections are used to make three-dimensional images.

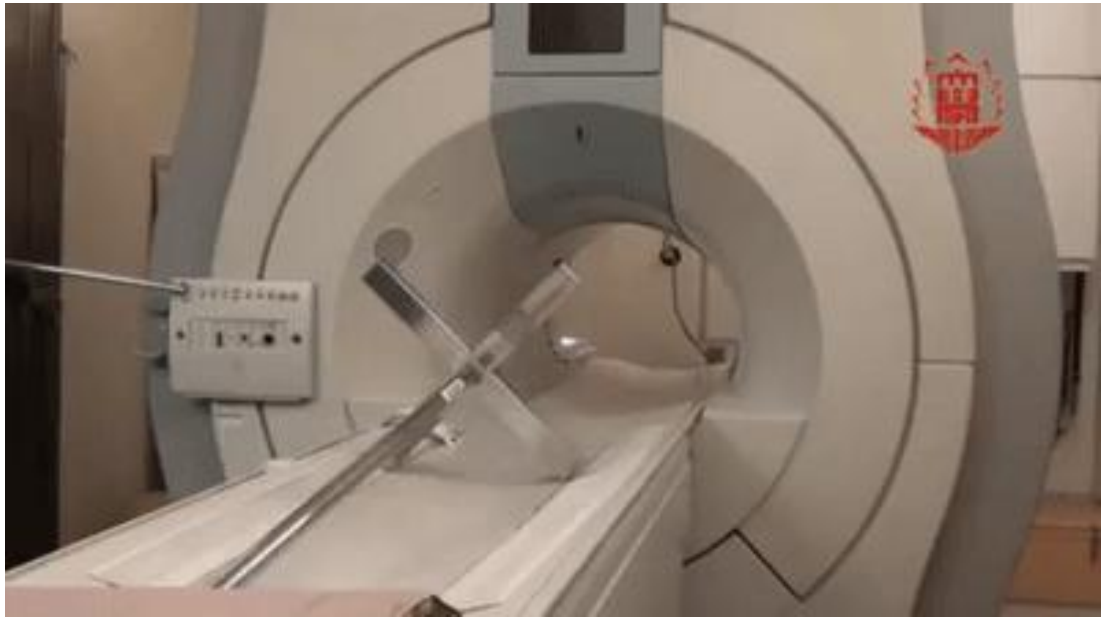


MRI physics

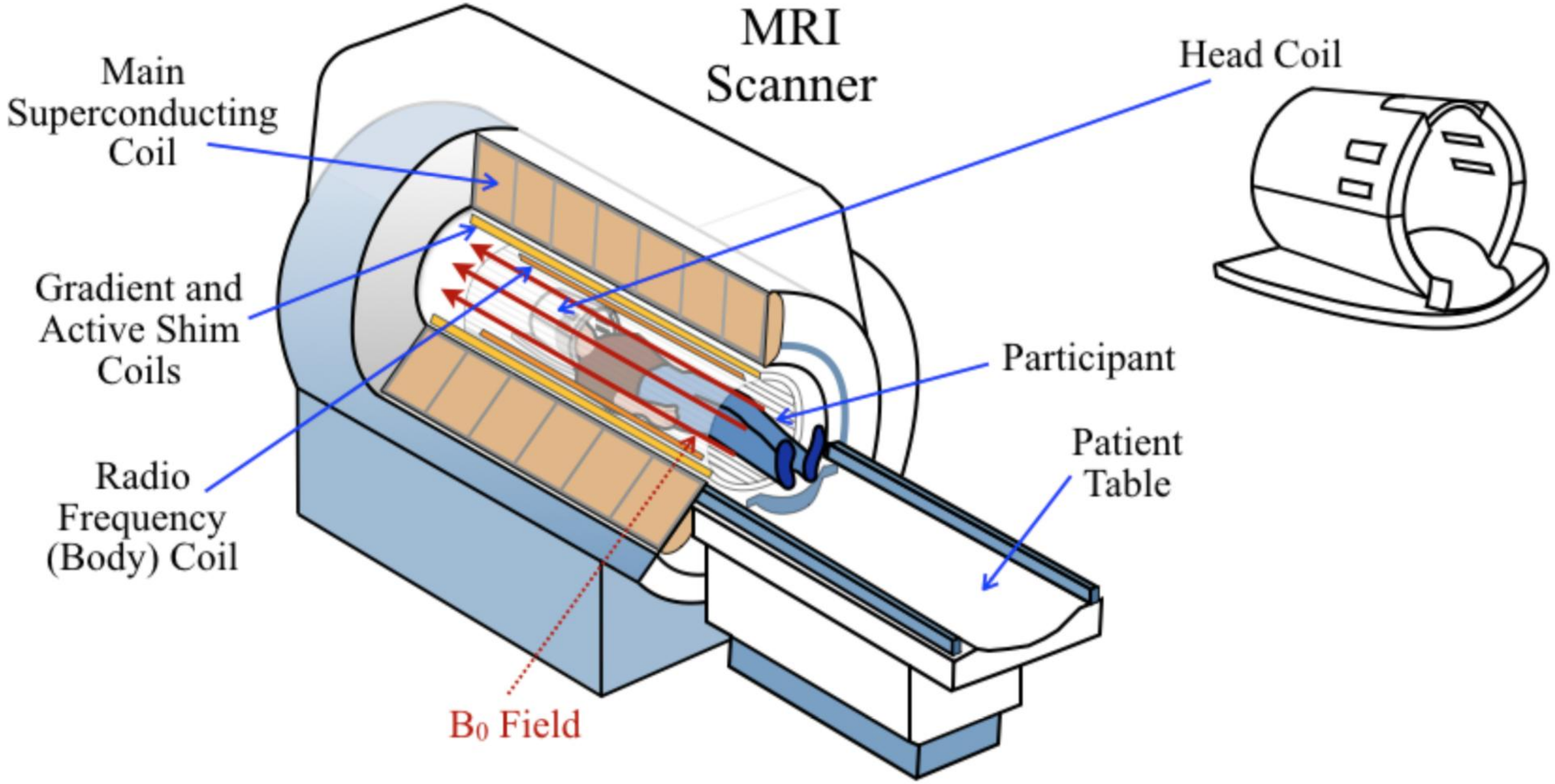
It's all about strong magnets.

MRI scanner strength = 3 Tesla

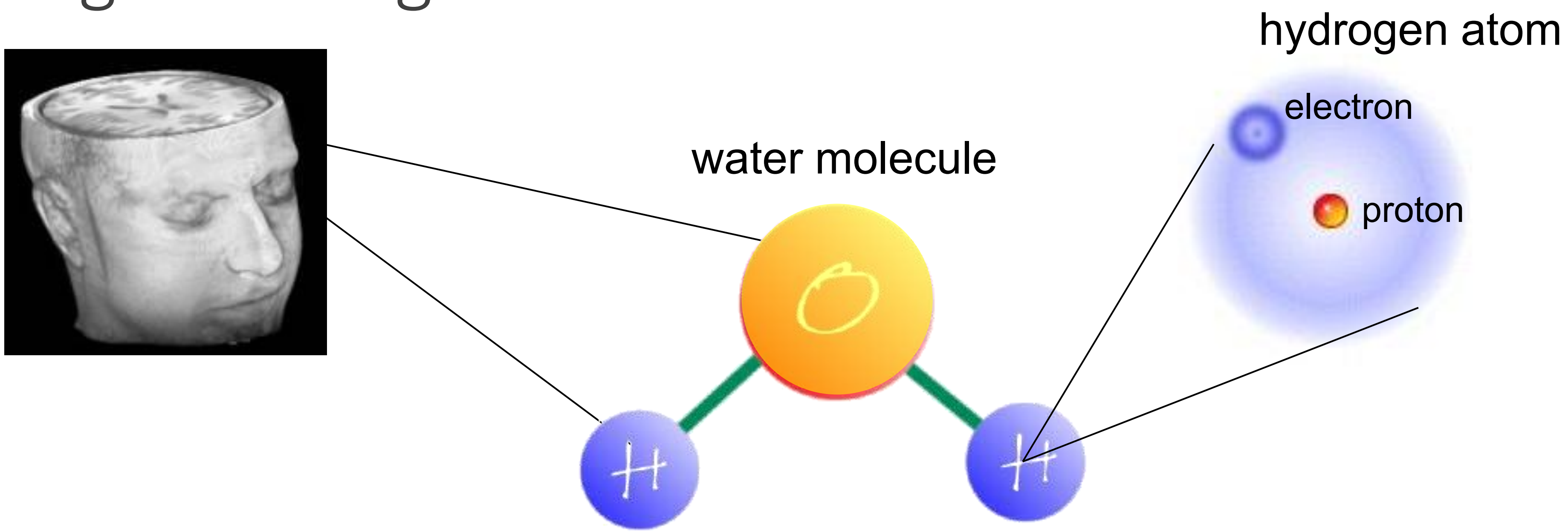
3 Tesla does not sound much but it's very strong.



How does MRI work?



Seeing with magnetism

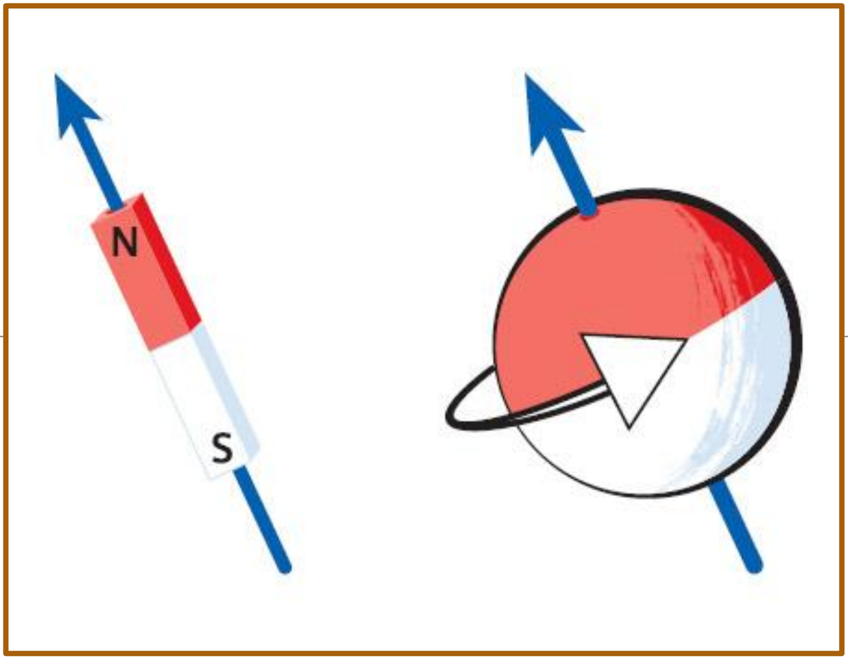


An MRI scanner uses magnetism to ‘see’ the position of hydrogen atoms in water molecules inside the body.

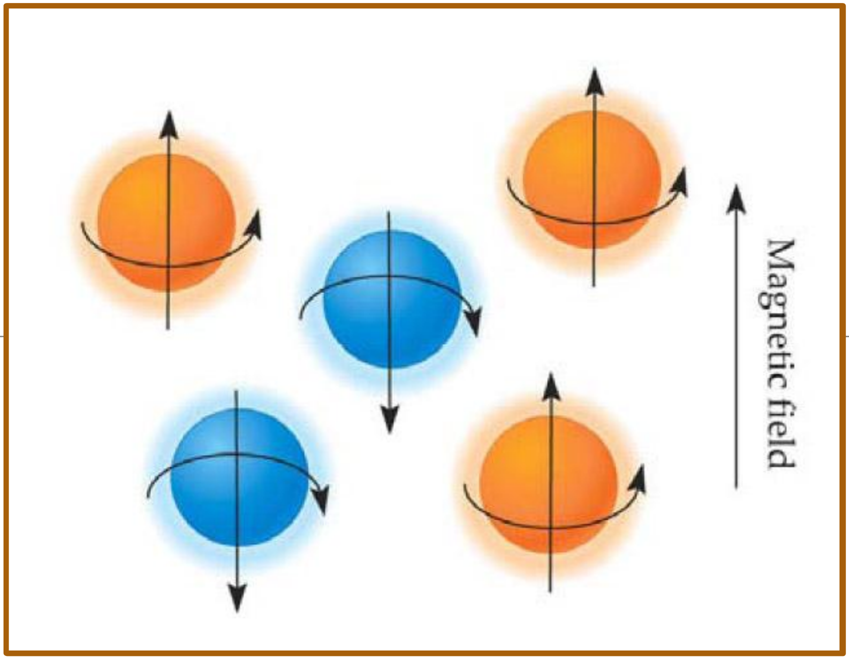
This allows it to build up an image of the internal structures.

Where are our “magnets”?

Protons have a spin and therefore a magnetic dipole moment (MDM)

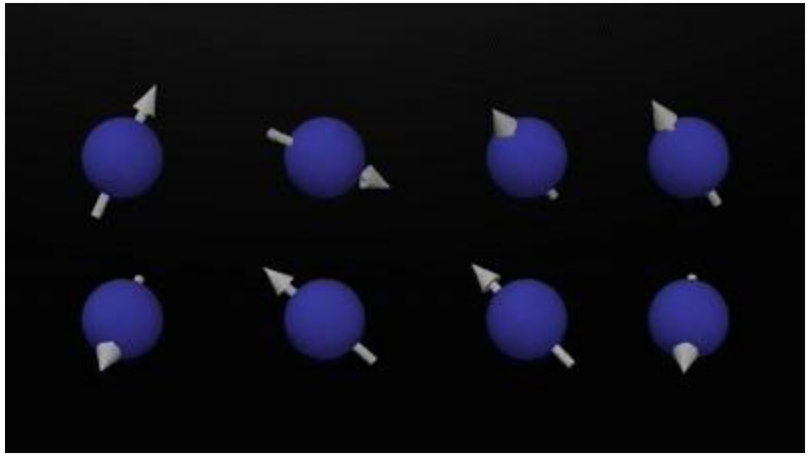


They can align to external magnetic fields in two ways: parallel or anti-parallel

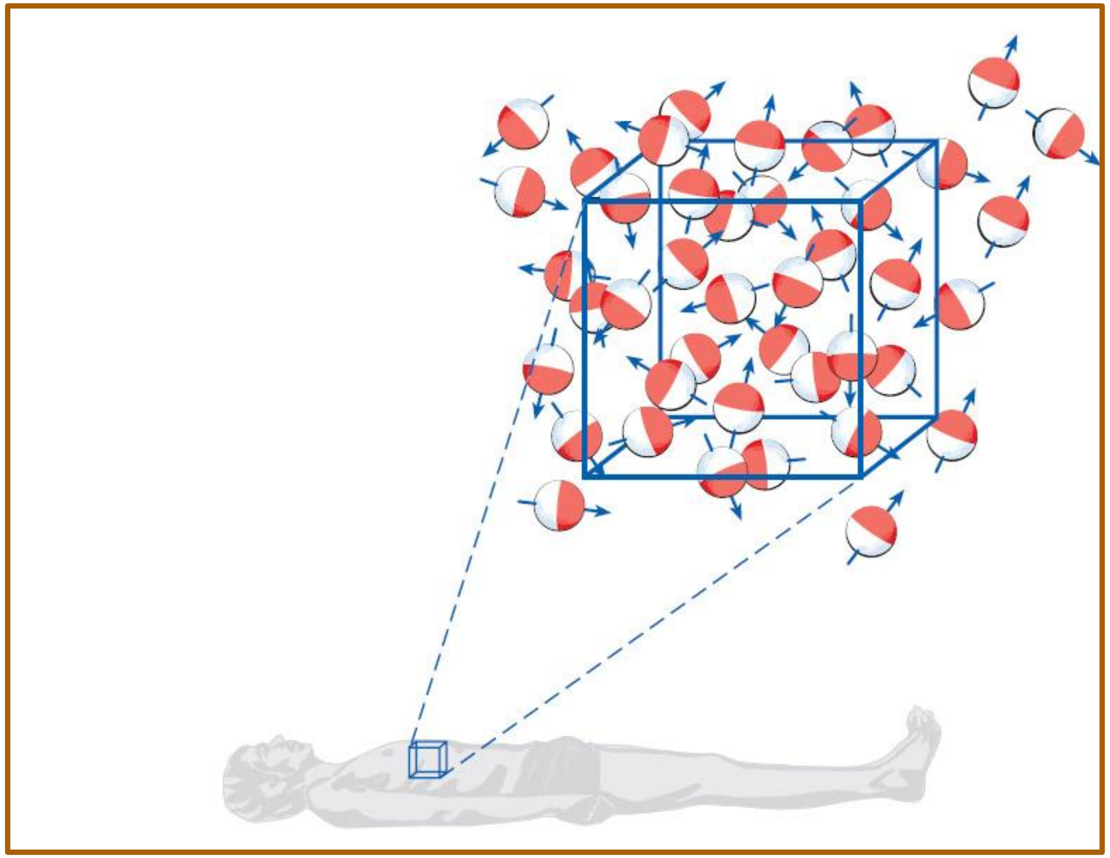


What happens in the scanner?

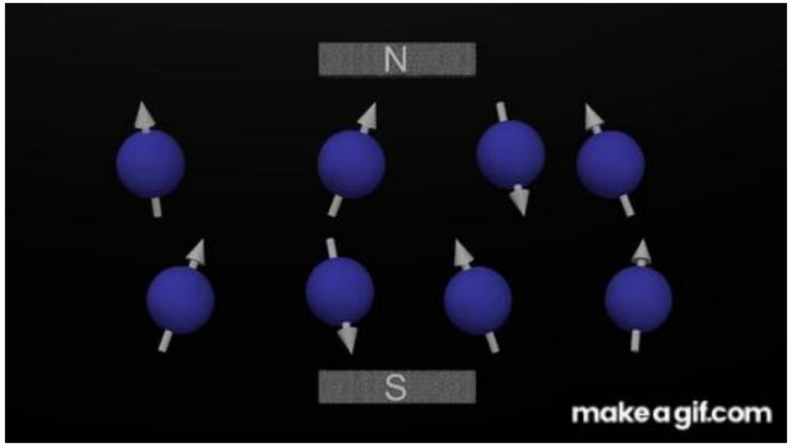
- All the atoms are pointing in different directions
- Net Magnetization Vector = 0



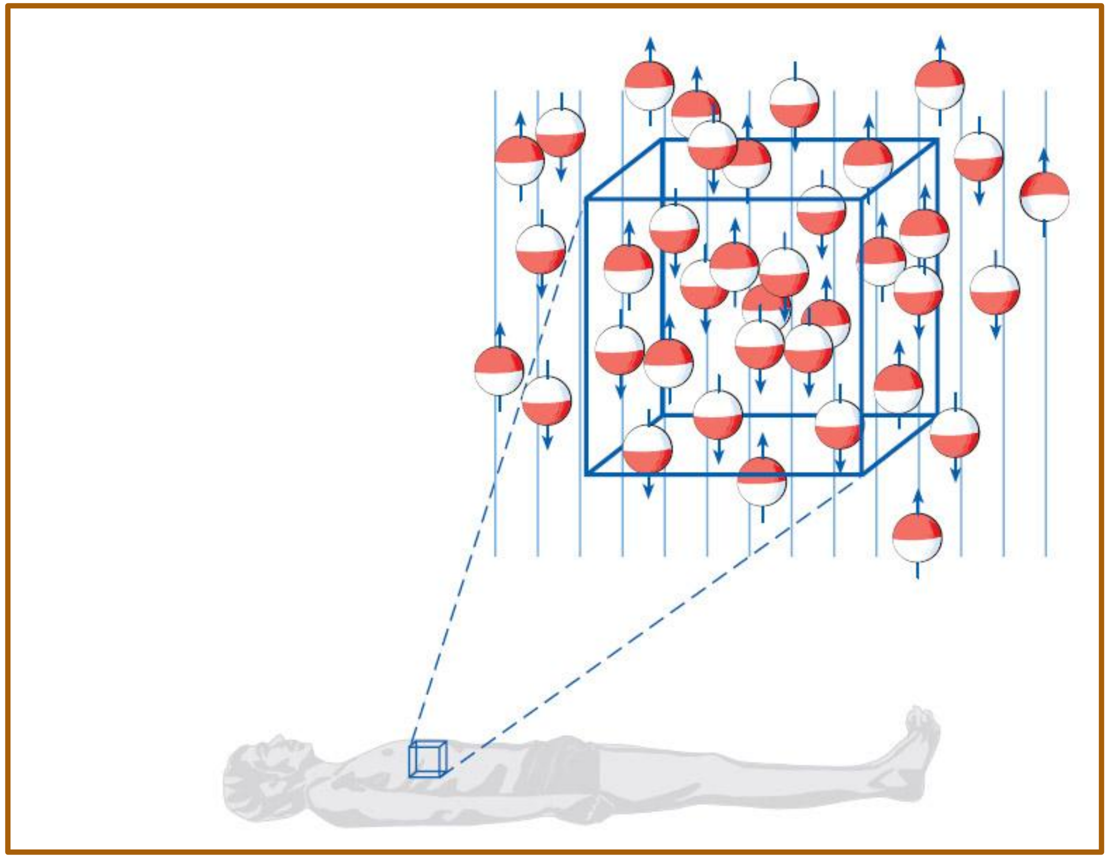
Outside scanner



What happens in the scanner?

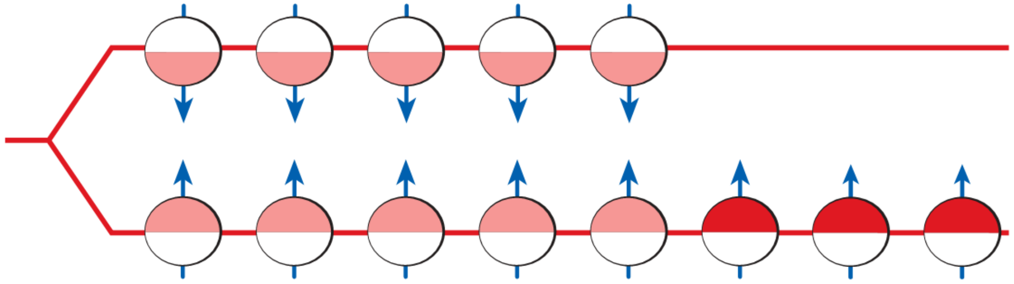


Inside scanner (B0)

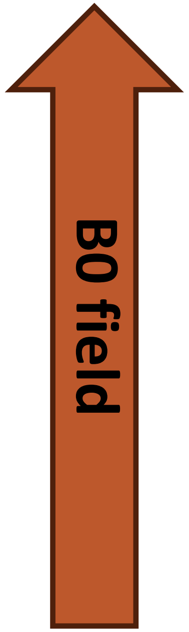


The protons of the H_2O molecules in our body align along B0

Are there more up-spins than down-spins?

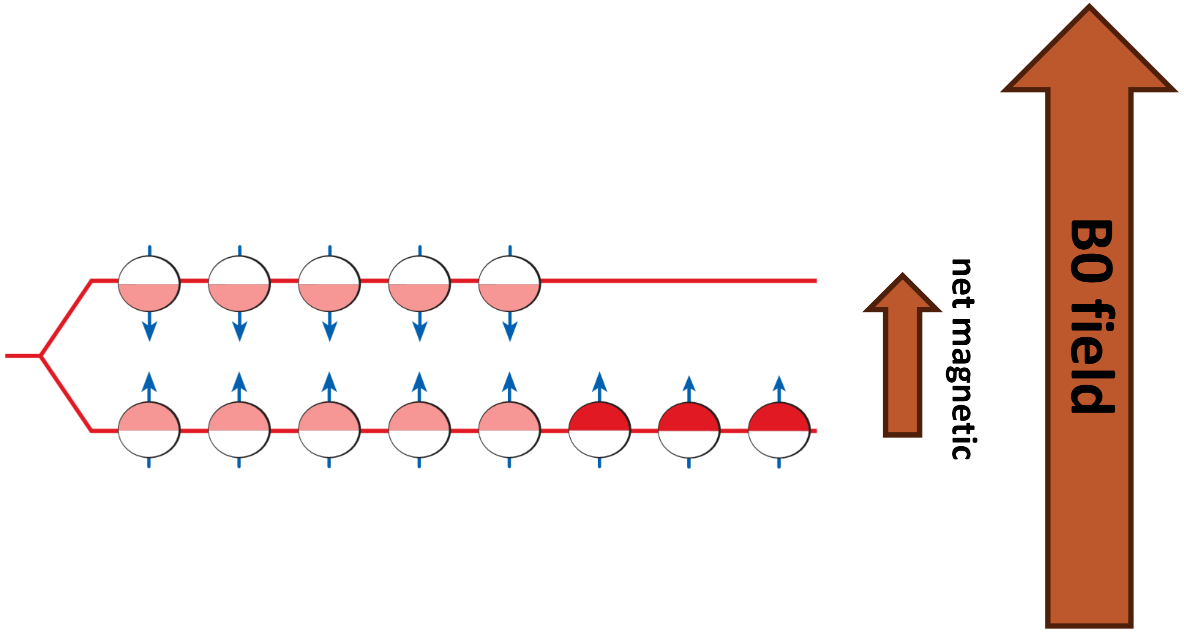


Yes and the excess spins create a new magnetic field M which we can measure



Higher Tesla → Stronger magnetic field → More signal from tissue (more protons alignment).

Mmmm, we have a huge problem



Excitation

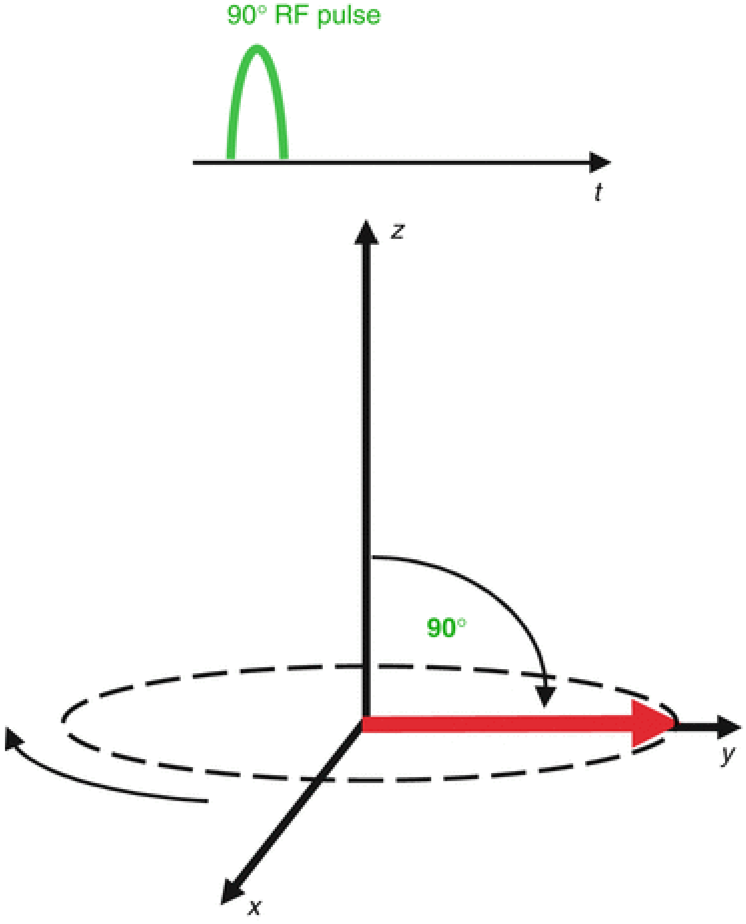
$$\omega_0 = \gamma B_0$$

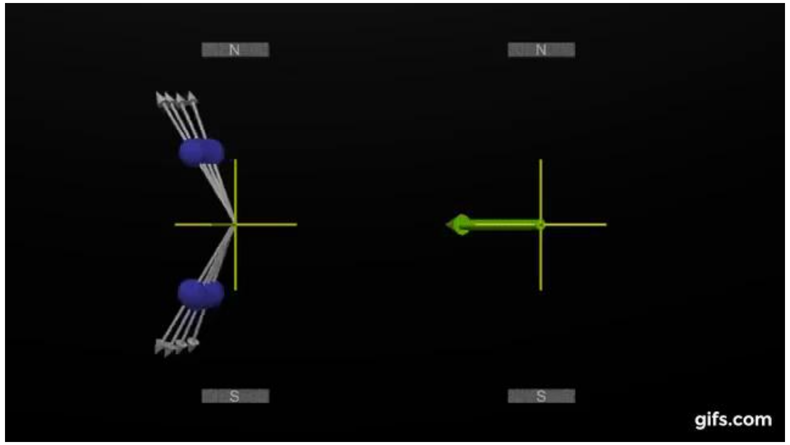
where:

- ω_0 is the Larmor frequency.
- γ is the gyromagnetic ratio of the proton, which is approximately $42.58\ \mathrm{MHz/T}$ (megahertz per tesla).
- B_0 is the strength of the external magnetic field applied in teslas (T).

Away from the B0 field:

- away from the longitudinal (parallel) to B0 (spin synchro)
- new plane is the transvers plane
- the Radio Frequency Pulse (B1 field) is released according to Larmor frequency of the H proton
- For a 3T would be 127.74 MHz

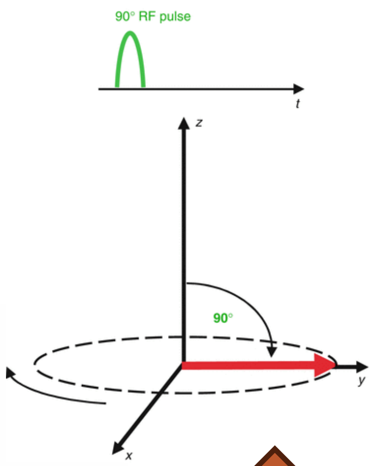


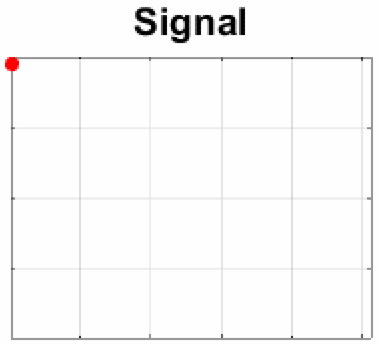


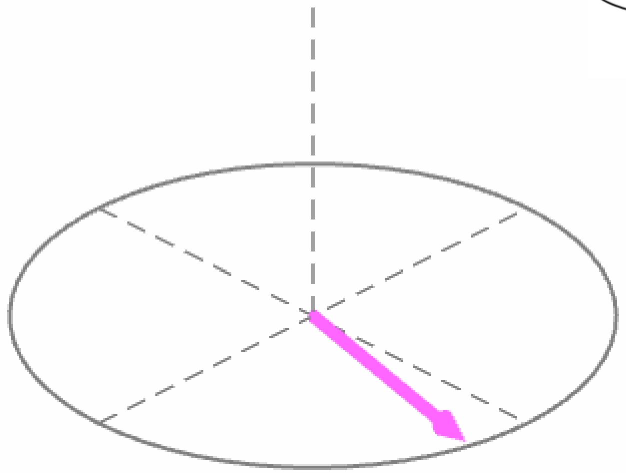
Excitation - Relaxation

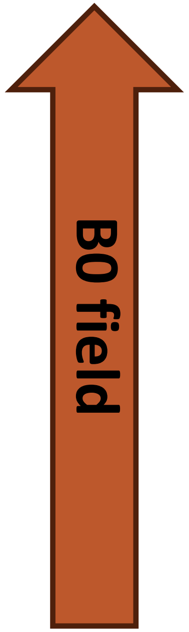
Once we applied the RF, we can receive the signal...

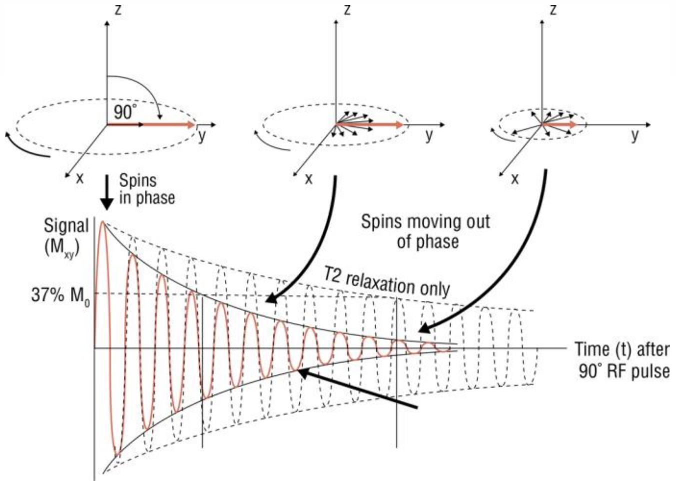
- After the RF pulse, max signal but homogeneous
- **Free induction decay (FID)** is the fading signal emitted by spinning protons after being disturbed by a magnetic field in MRI.
- FID is caused by the desynchronization of spin (T2 decay) → local inhomogeneity









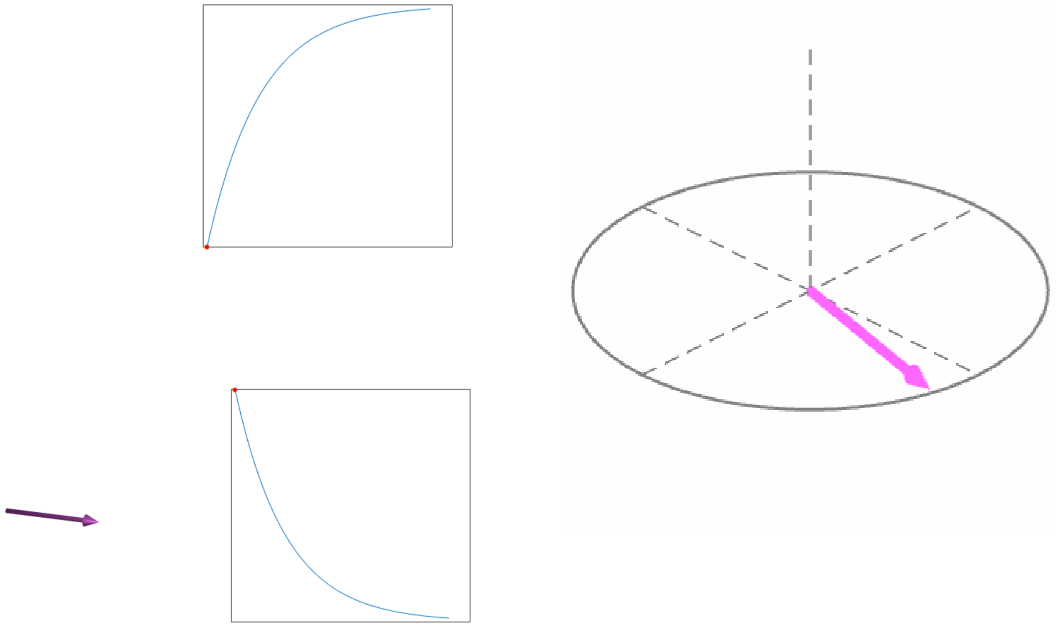


MRI contrasts

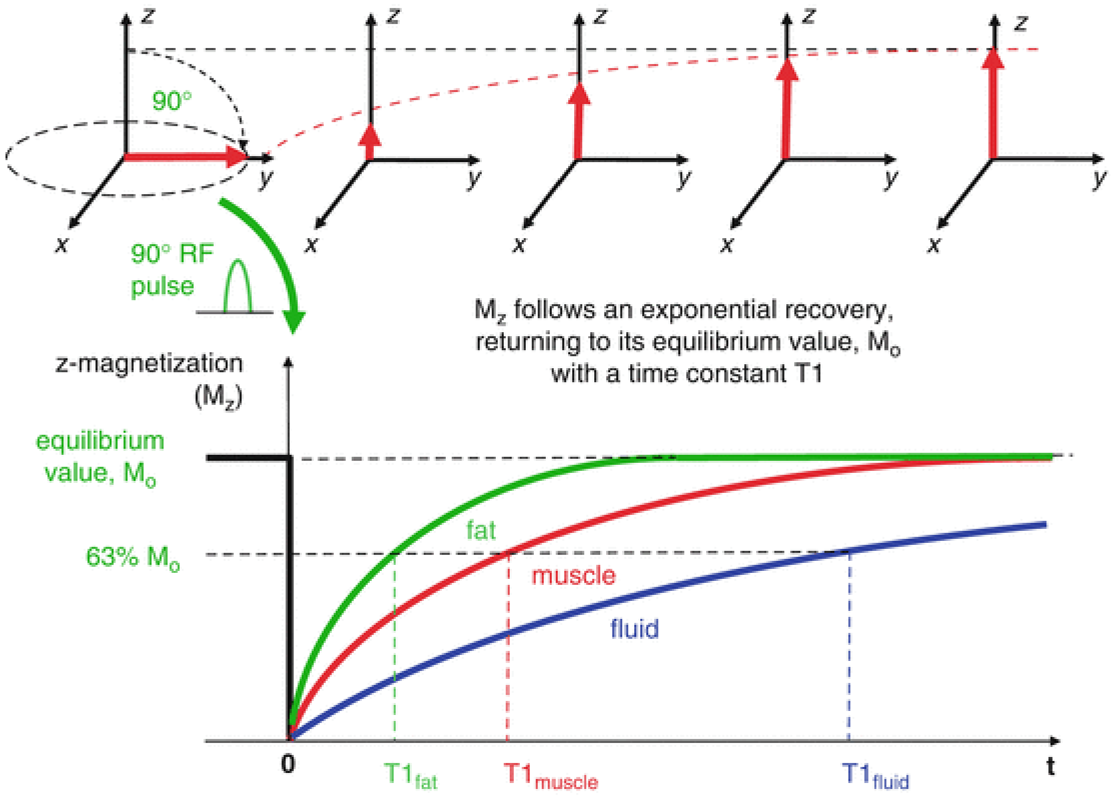
It is the decay and the restoration in the tissue that give us the contrast of the tissues:

T(ime)1 Restore (or longitudinal relaxation): Protons realign with the magnetic field (recovery).

T2 Decay (or transverse relaxation): : Protons lose synchronization with each other (dephasing).

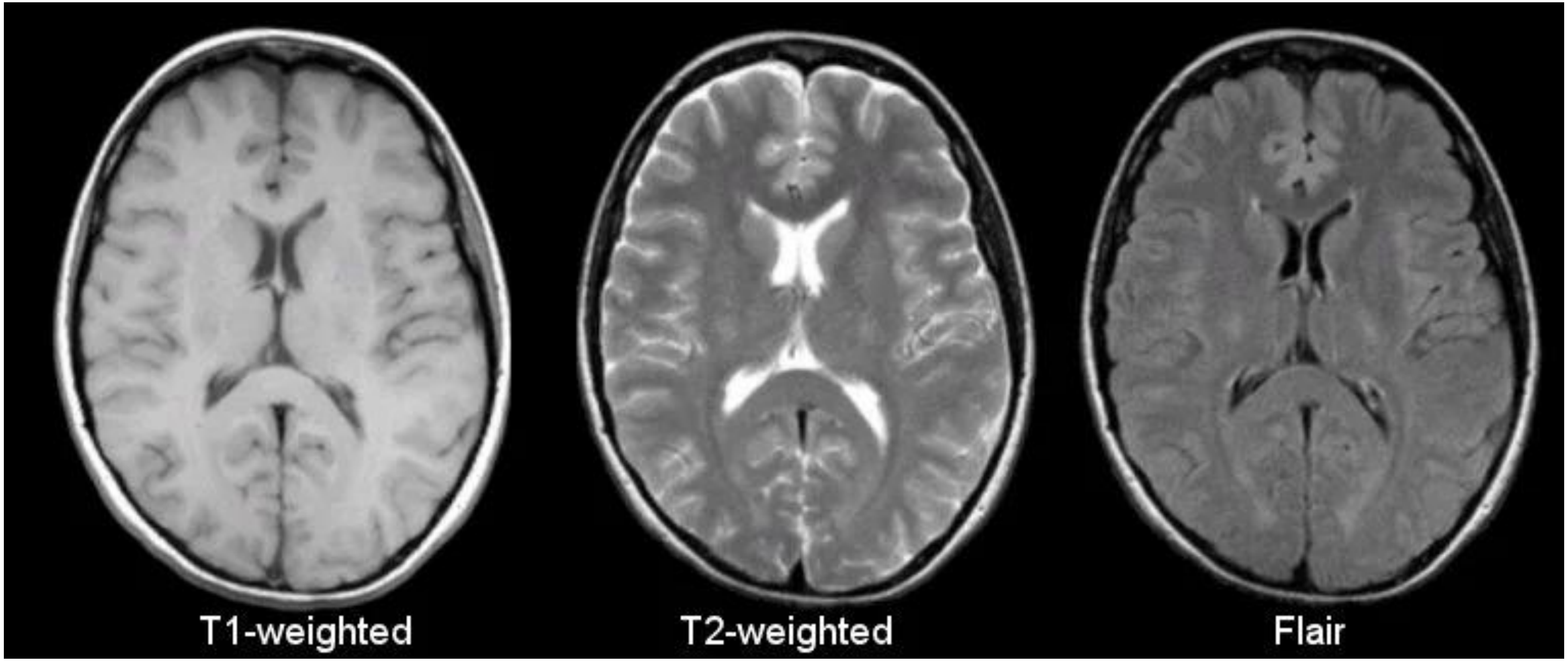


MRI Contrast



Tissue	T1 (msec)	T2 (msec)
Water/CSF	4000	2000
Gray matter	900	90
Muscle	900	50
Liver	500	40
Fat	250	70
Tendon	400	5
Proteins	250	0.1- 1.0
Ice	5000	0.001

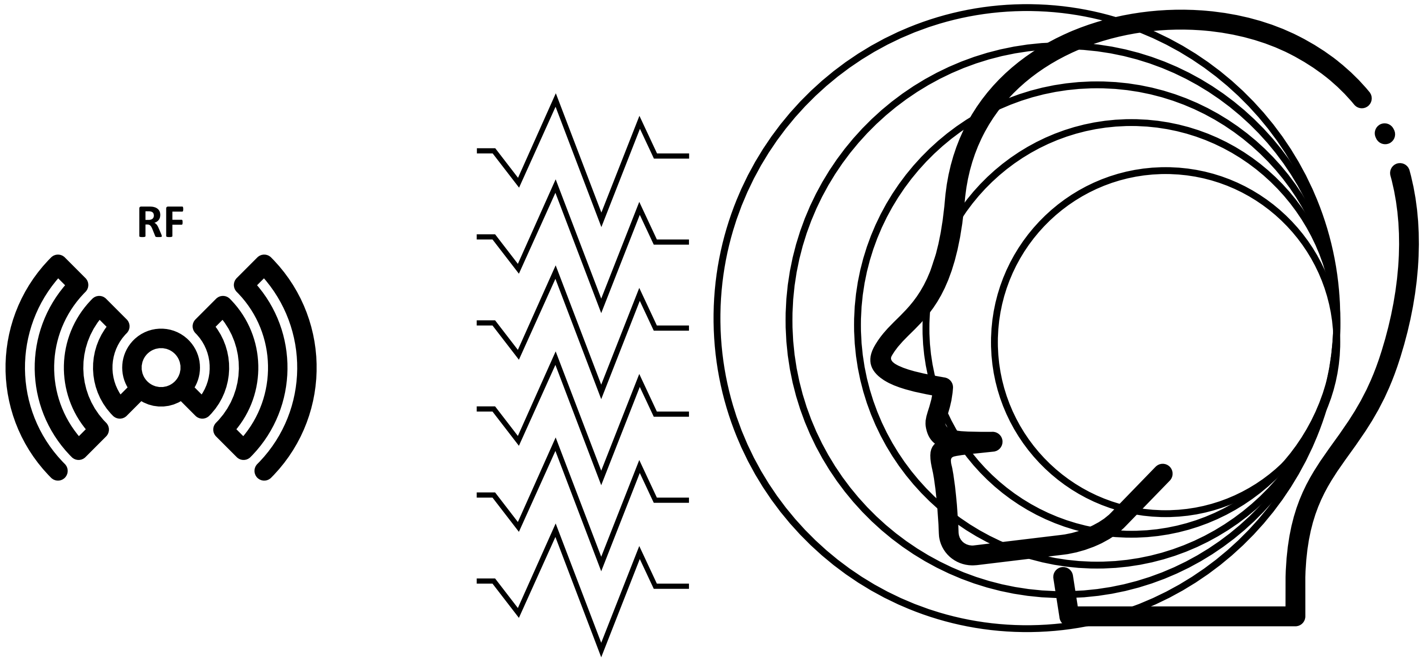
MRI Contrast



Now, a new problem!

The RF pulse excites all the protons:

- How can I spatially locate the echo?



Now, a new problem!

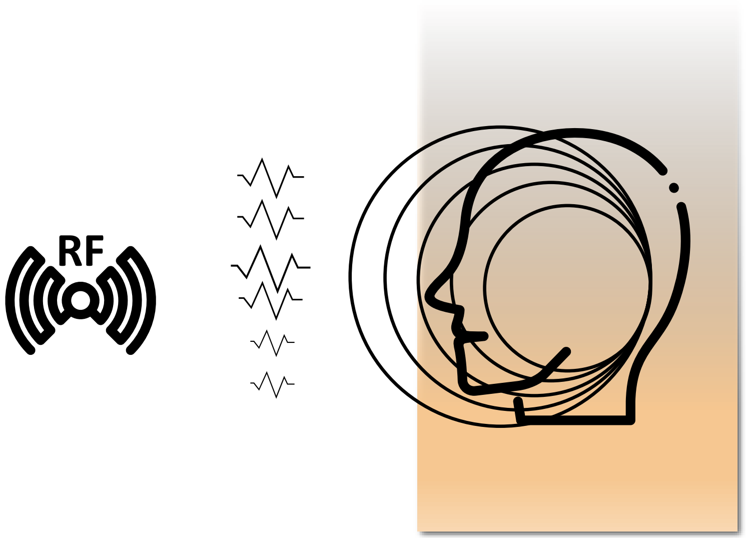
By applying 3 magnetic field gradient along 3 axis:

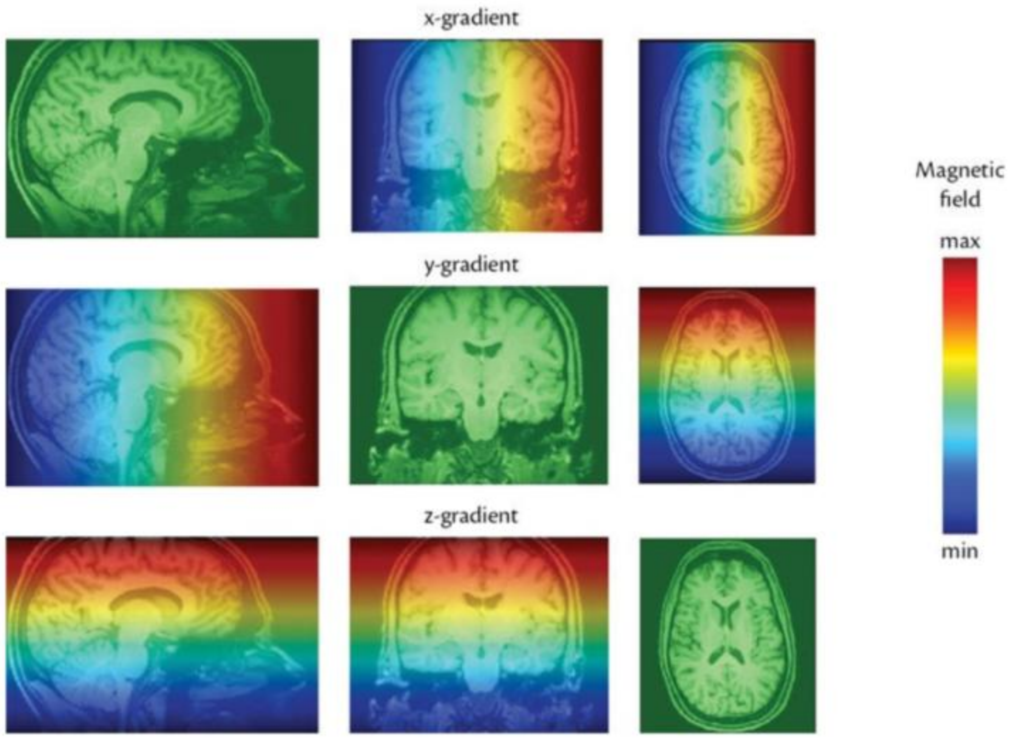
Slice Selection Gradient

Frequency Encoding Gradient

Phase Encoding Gradient

<u>**Now when know where we are!**</u>

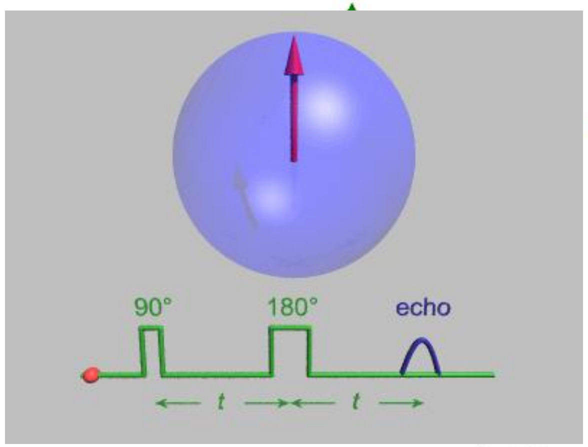


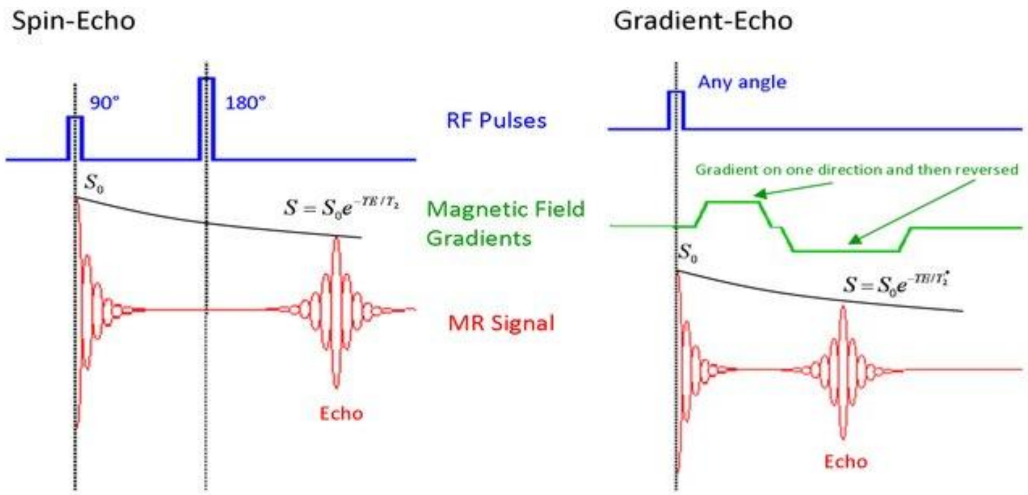


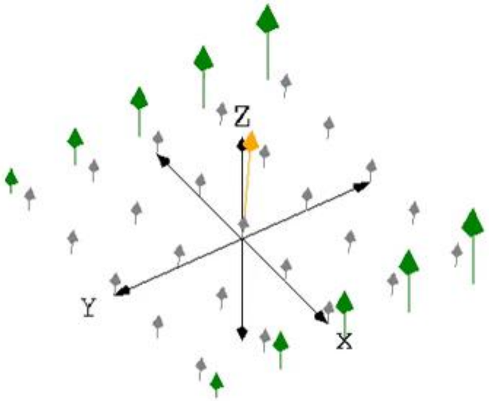
Example of a MR sequence → AKA how to get the echo

- **Gradient echo imaging (GRE)** → Field gradients are not just useful for discriminating between signals spatially, they can also be used to create an echo.
- **Spin Echo (SE)** → produces high-quality images by correcting for signal loss due to magnetic field inhomogeneities.

- TR: time repetition
- TE: time echo



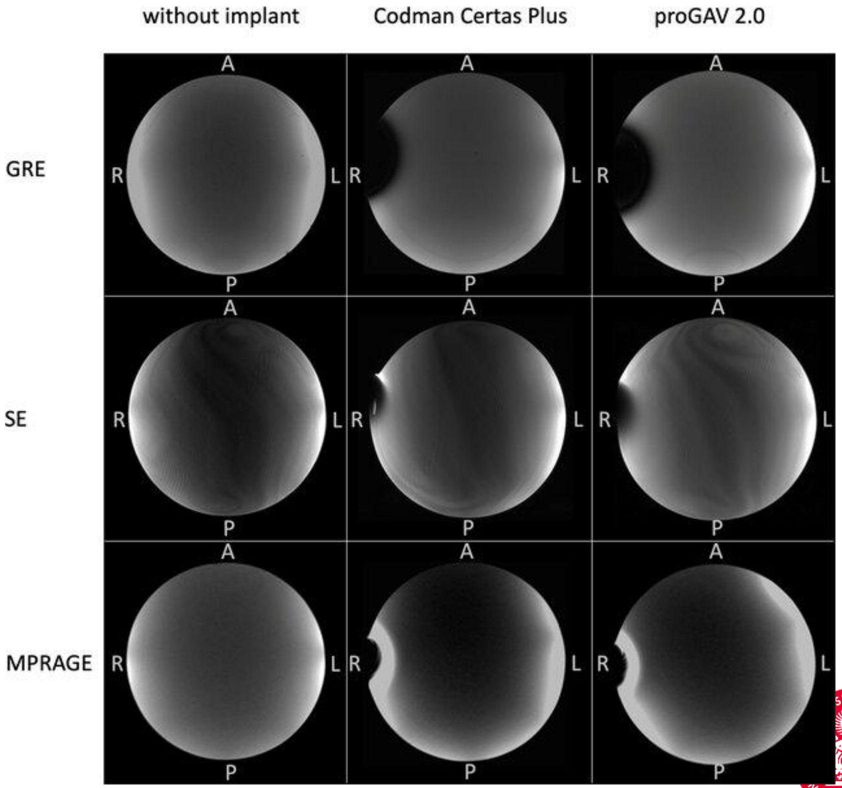




Tyler Moore, 2011

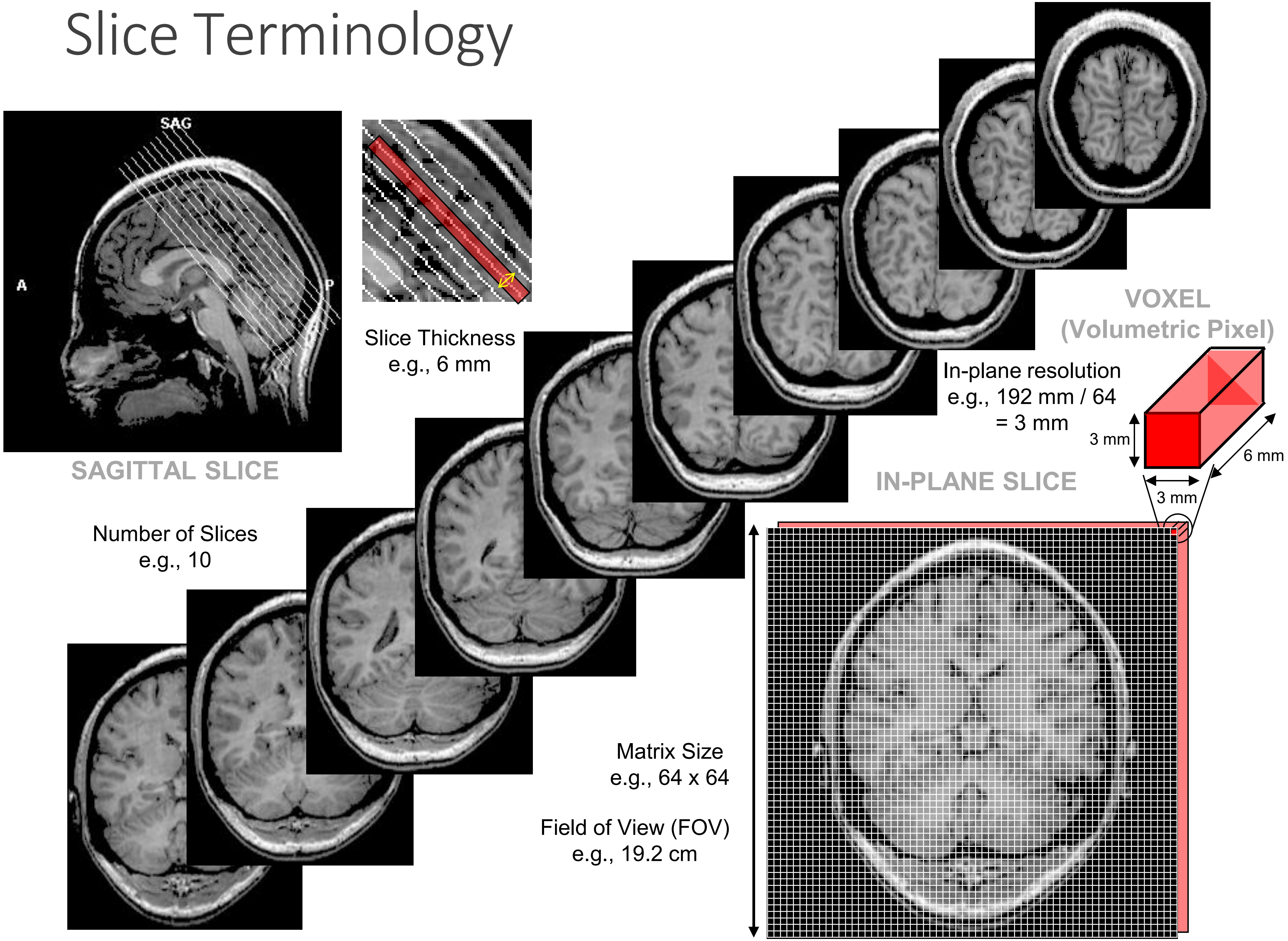
SE vs GRE

Feature	Spin Echo (SE)	Gradient Echo (GRE)
Echo Type	Spin Echo (from 180° RF pulse)	Gradient Echo (from gradient rephasing)
TE	Typically longer	Shorter (allows for faster imaging)
TR	Typically longer	Can be very short
Artifacts	Less affected by magnetic field inhomogeneities	More sensitive to magnetic susceptibility (e.g., metal, air-tissue interfaces)
Image Contrast	True T1, T2, or PD weighting	T1 and T2* weighting
Speed	Slower	Faster imaging
Applications	High-resolution, artifact-free imaging	fMRI, perfusion studies, fast imaging, T2*-sensitive studies



Chen et al., 2023

Slice Terminology



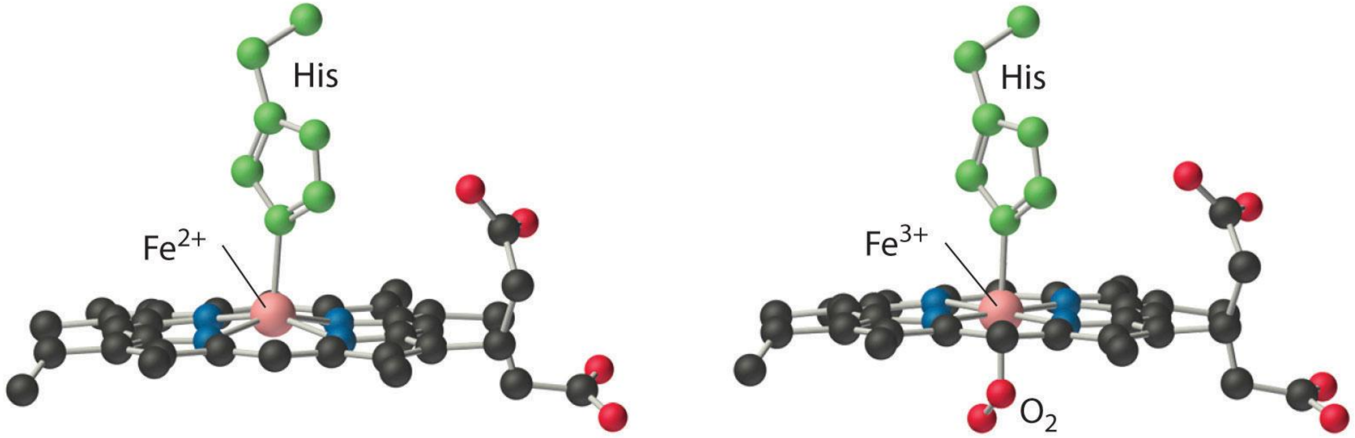
functional MRI

It aims to determine the neurobiological correlation of behavior by identifying the brain regions (or “functioning modules”) that become “active” during the performance of specific tasks in vivo.

It extends traditional anatomical imaging to functional imaging.

- observe both the structures and which structures participate in specific functions
- improving our understanding of a variety of brain pathologies.
- such as the addictive behaviors of gambling or drug abuse, are without structural brain changes.

functional MRI - Basics



(a) Deoxymyoglobin **(b) Oxymyoglobin**

Deox has 4 unpaired electrons → paramagnetic

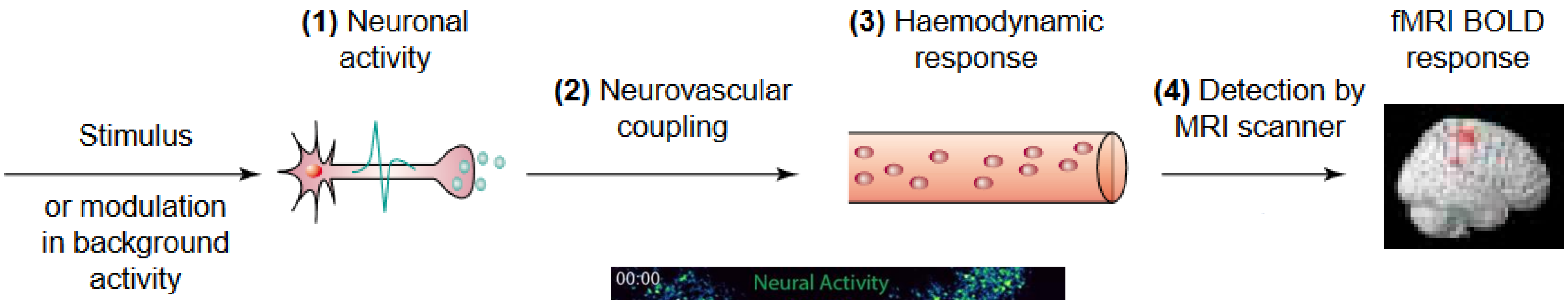
Oxy → **diamagnetic**

OXYHEMOGLOBIN VERSUS DEOXYHEMOGLOBIN

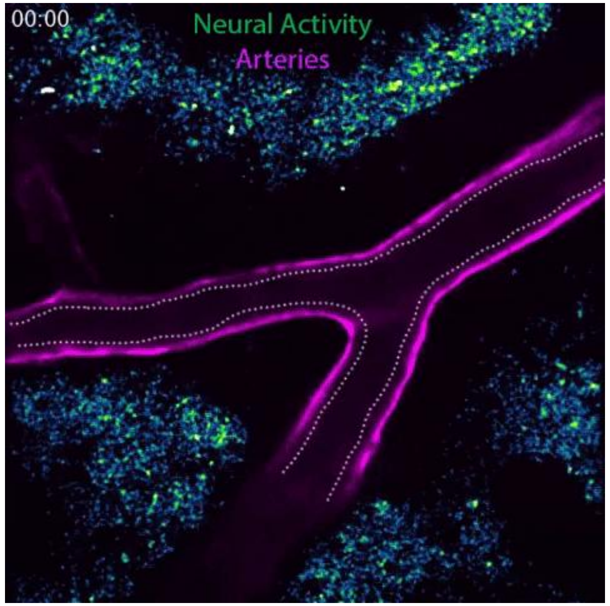
OXYHEMOGLOBIN	DEOXYHEMOGLOBIN
Bright red substance formed by the combination of hemoglobin with oxygen	Hemoglobin not combined with oxygen
Carries four oxygen molecules at its saturated stage	Carries no oxygen molecules
Relaxed state of hemoglobin	Tensed state of hemoglobin
Has a significantly lower absorption, which occurs at 660 nm	Has a higher absorption, which occurs at 940 nm
Diamagnetic, weakly repulsed by the magnetic field	Paramagnetic, weakly attracted by the magnetic field
Occurs in oxygenated blood	Occurs in deoxygenated blood
Bright red	Purplish blue
Transported away from the heart	Transported towards the heart

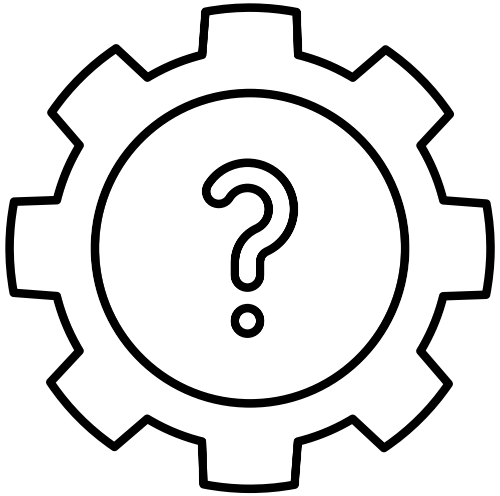
Visit www.PEDIAA.com

functional MRI - in our brain



Fine Motor Skills





Arthurs et al, 2002

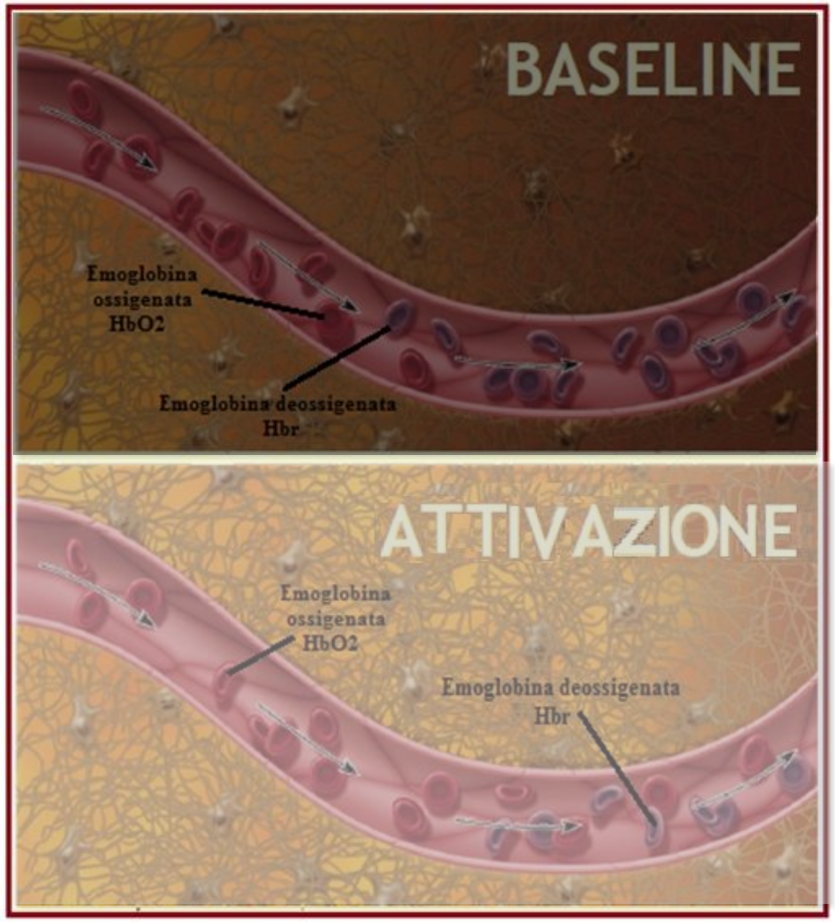
Gu Lab/Harvard Medical School

functional MRI - Basics

fMRI is based on

Blood **O**xygenation **L**evel **D**ependent (**BOLD**) signal

→ Indirect measurement of neuronal activity

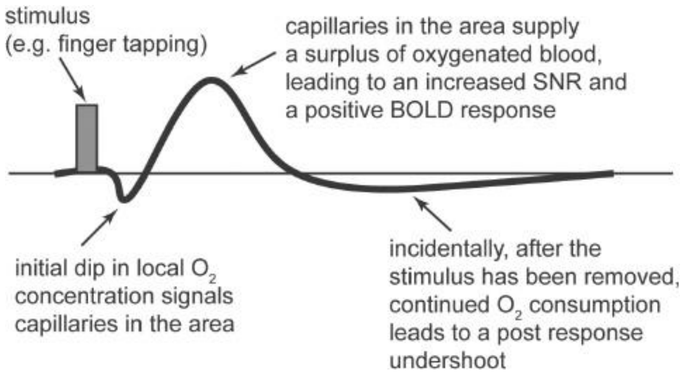


Huettel, S. 2006

↑ Neural activity

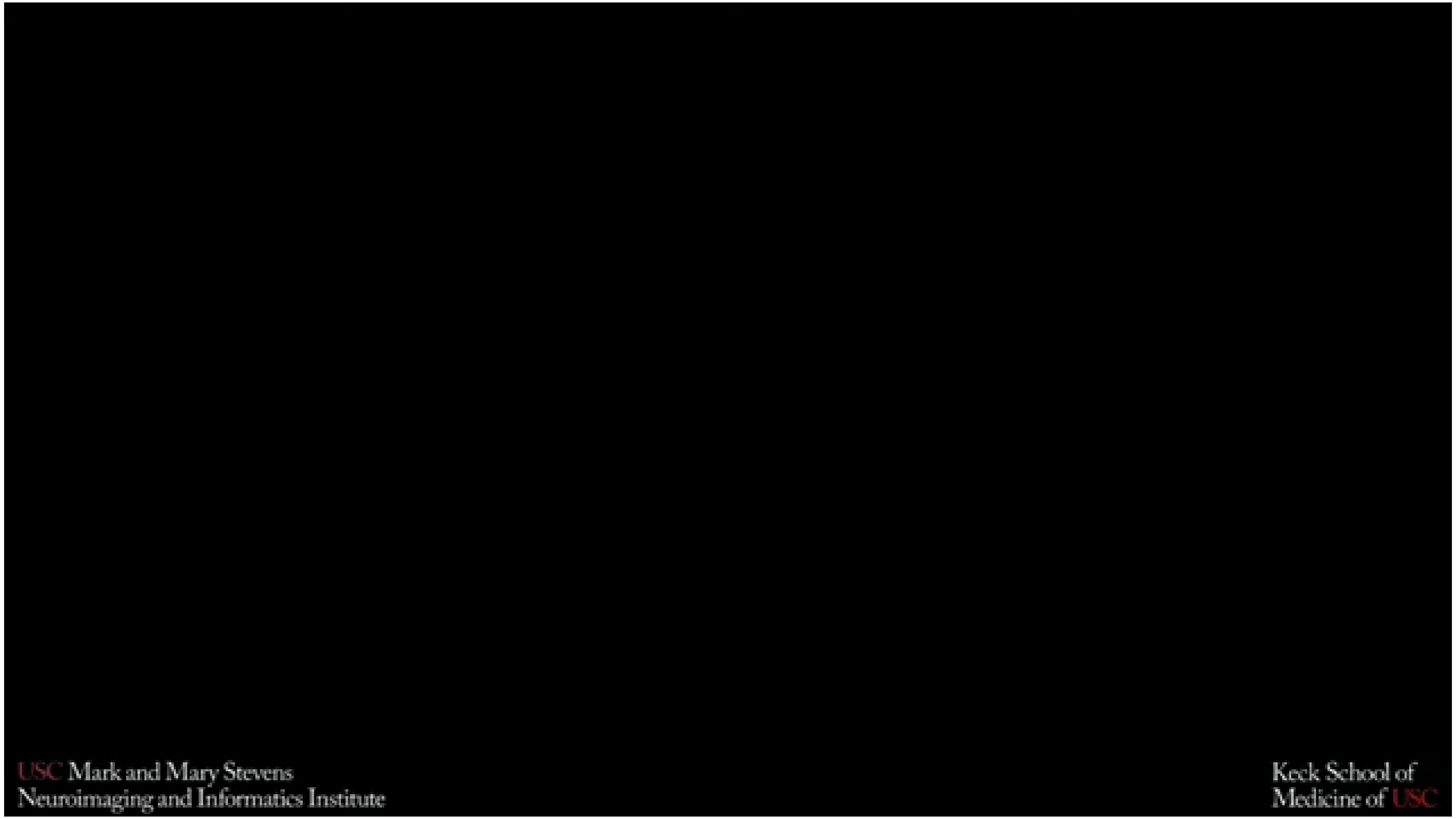
↑ Oxygenated blood

↑ fMRI Signal

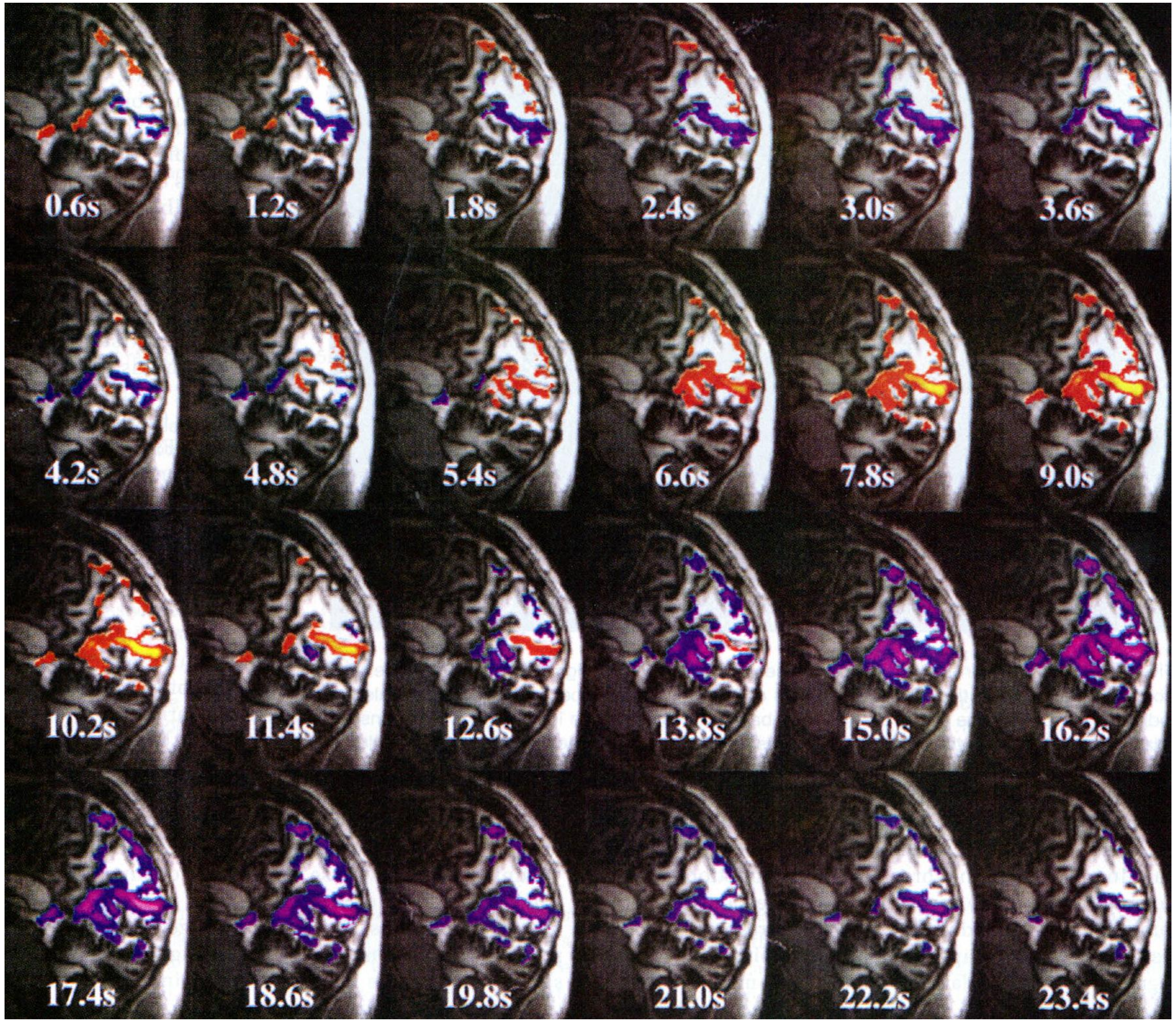


→ Hemodynamic Response Function (HRF)

BOLD


USC Mark and Mary Stevens
Neuroimaging and Informatics Institute
Keck School of
Medicine of USC

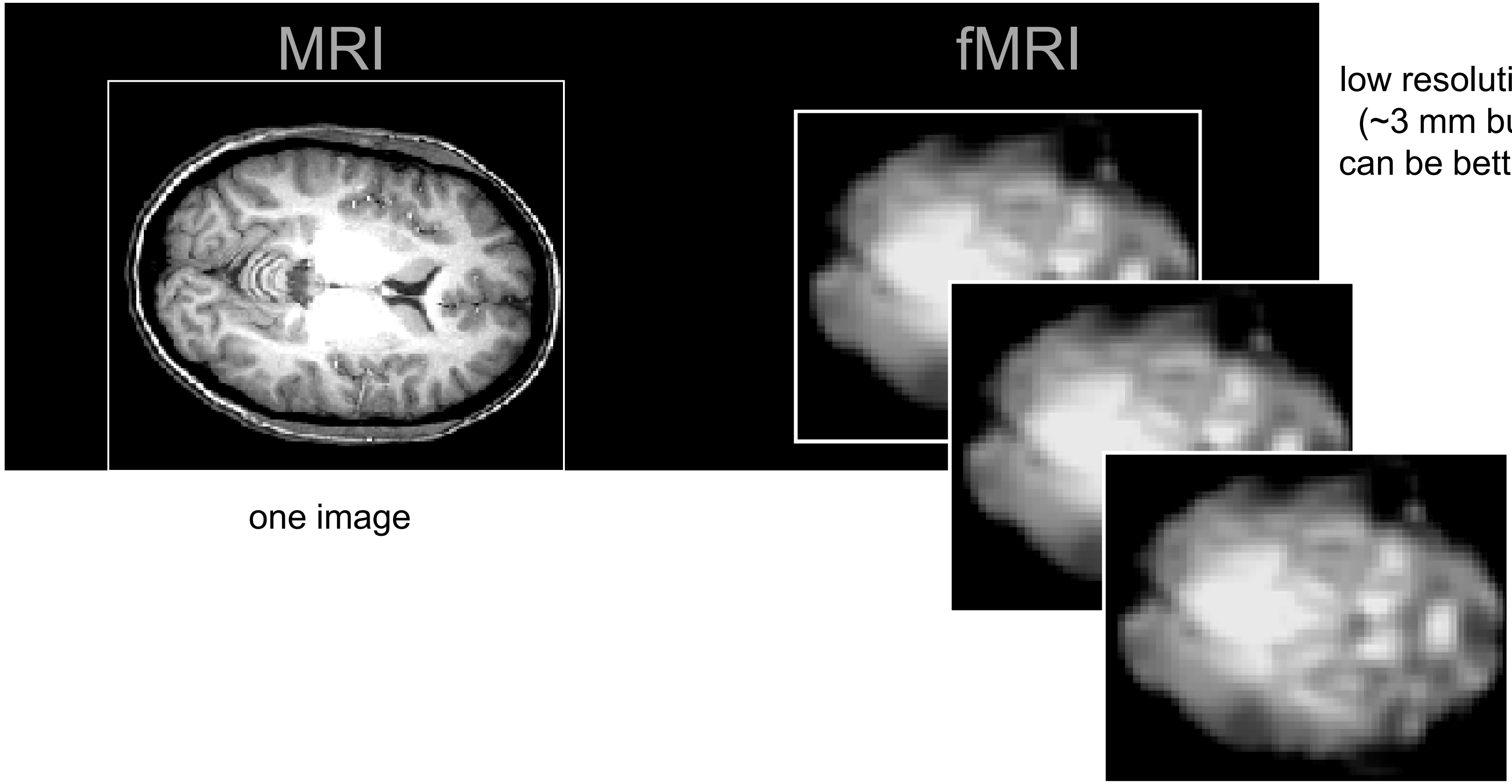
Evolution of BOLD Response



Hu et al., 1997, MRM

MRI vs. fMRI

high resolution (1 mm)



low resolution (~3 mm but can be better)

one image

. . .

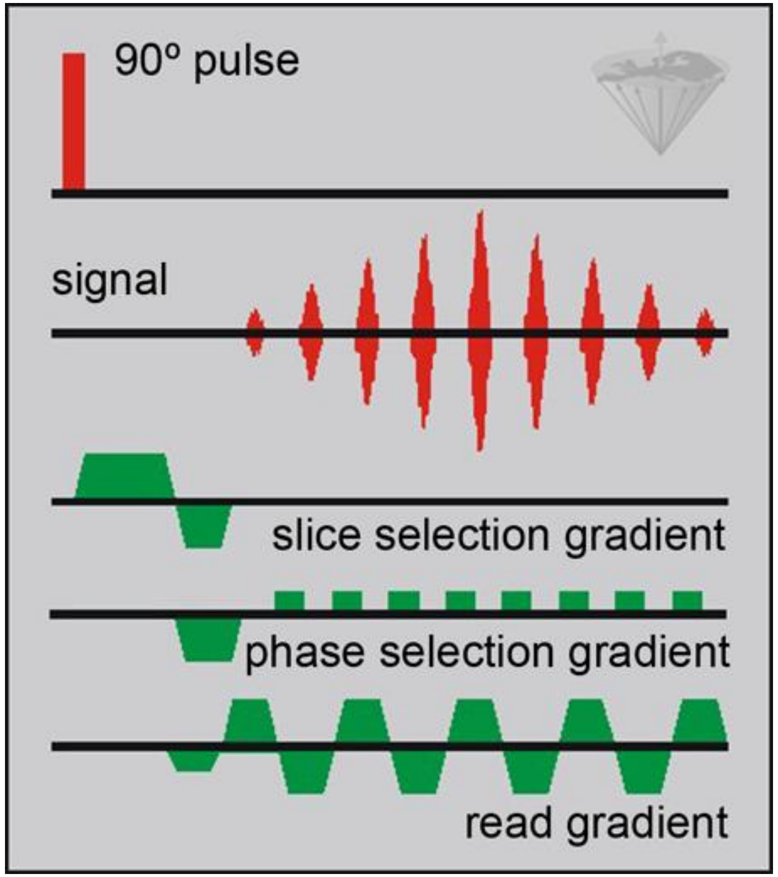
many images

↑ neural activity ➔ ↑ blood oxygen ➔ ↑ fMRI signal

We need "fast" images to record BOLD

The are many methods sensible to BOLD

- most famous is Echo Planar Imaging (EPI)

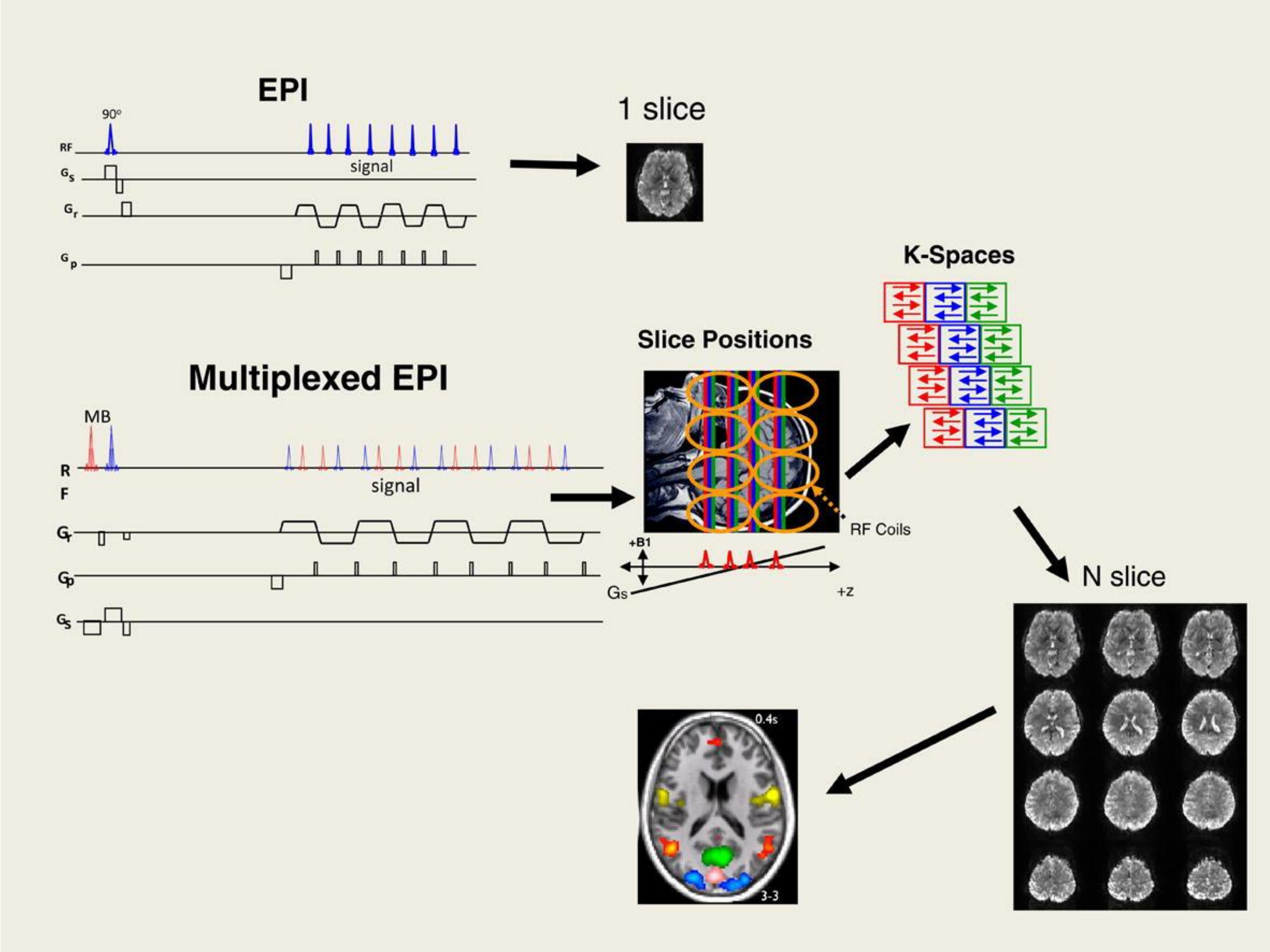


TR classically from 2-3 s

We need "fast" images to record BOLD

The are many methods sensible to BOLD

- most famous is Echo Planar Imaging (EPI) → PIMPED (multiband)
- the entire brain is collected in 0.635 s



Feinberg et al., 2013

Could we read the mind with fMRI?

Hunt ... Gallant et al Nature Neuroscience 2016

Advantages of MRI for Brain Study:

High-resolution imaging

Functional MRI (fMRI)

Non-invasive and radiation-free

Can detect early signs of neurological conditions like stroke, multiple sclerosis, and Alzheimer's

Disadvantages of MRI for Brain Study:

Expensive compared to other imaging methods (e.g., CT scans).

Time-consuming, requiring 30-60 minutes of stillness.

Artifacts: near bone and air-filled areas can lower image quality

Claustrophobia

Noise

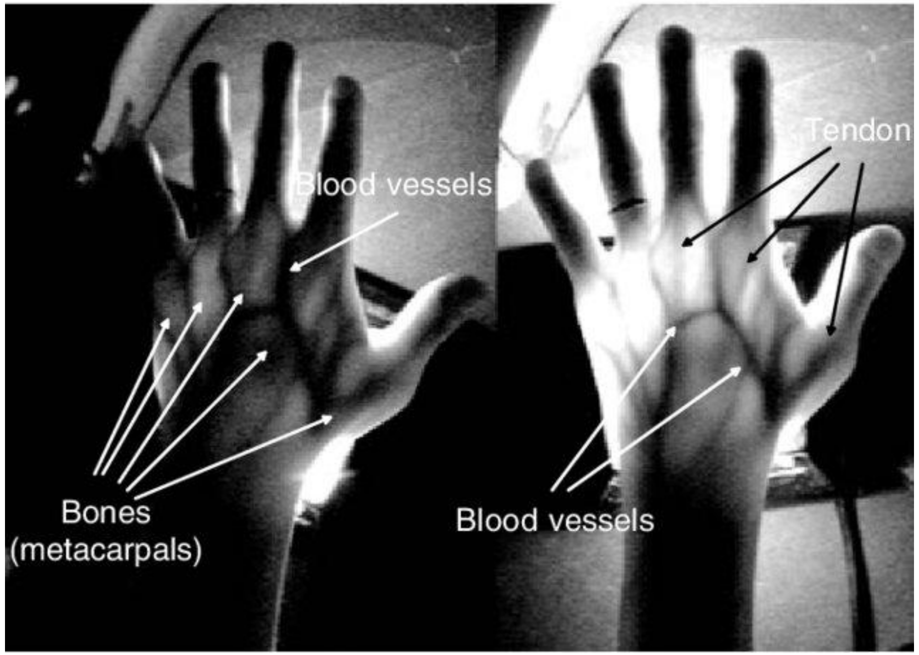
fNIRS (functional Near-Infrared Spectroscopy)

What is fNIRS?

- Non-invasive brain imaging technique

- Measures brain activity by detecting changes in blood oxygenation

- Uses near-infrared light (650–900 nm) to penetrate the scalp and skull

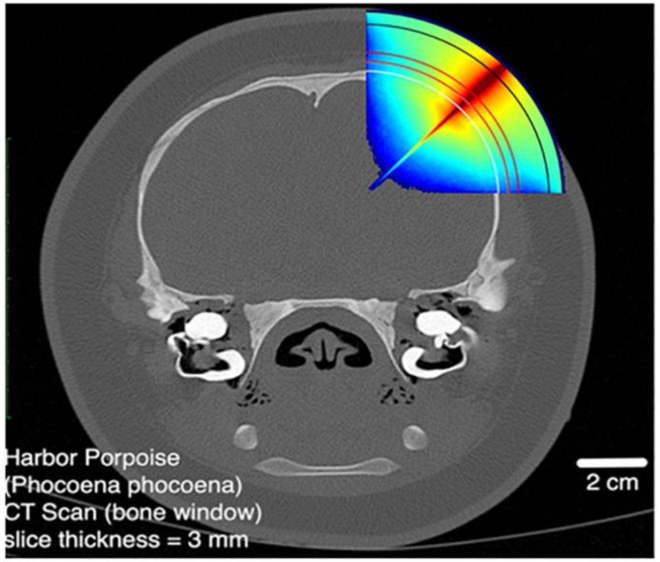


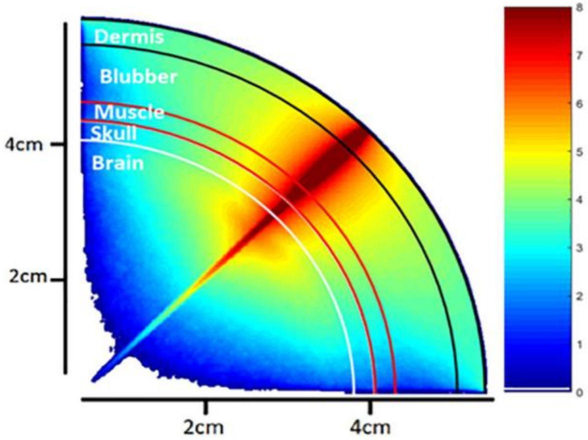
Dohen et al., 2006

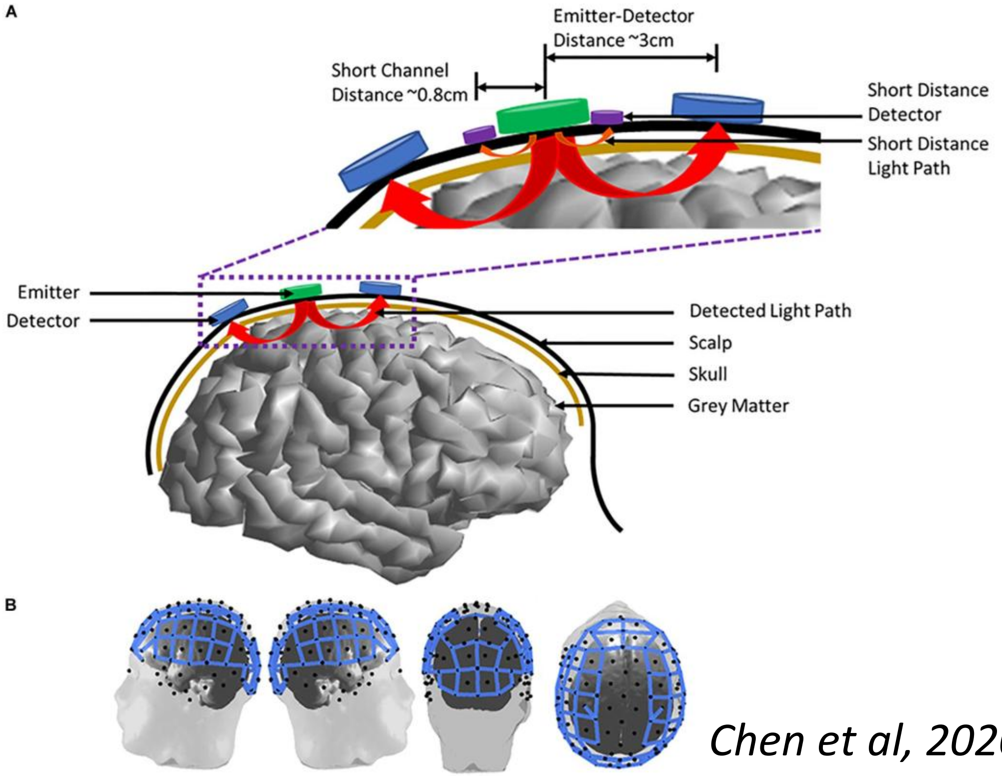
How Does fNIRS Work?

- Near-Infrared Light: Shines light into the scalp

- Light Detectors: Measure how much light returns to the sensor



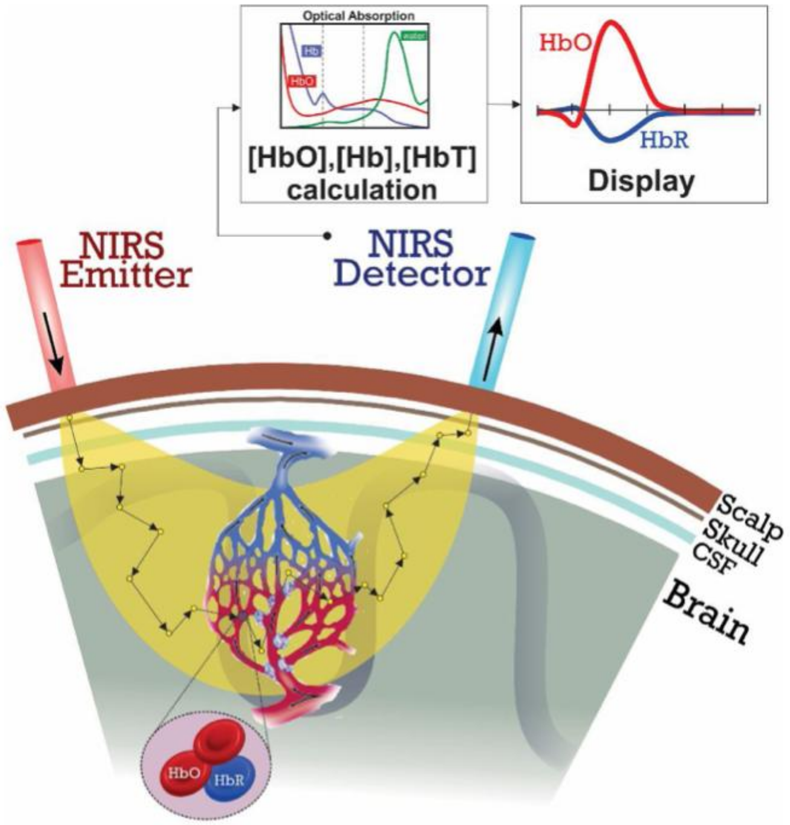


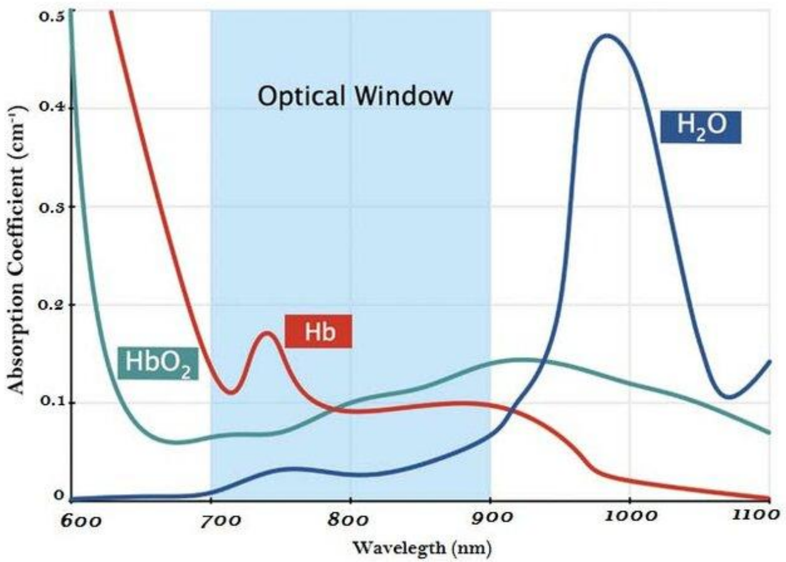


Chen et al, 2020

How Does fNIRS Work?

- Near-Infrared Light: Shines light into the scalp
- Light Detectors: Measure how much light returns to the sensor
- Absorption of Light: Light is absorbed differently by oxygenated vs. deoxygenated haemoglobin
- Brain Activity: Increased neural activity → increased oxygen consumption → changes in light absorption

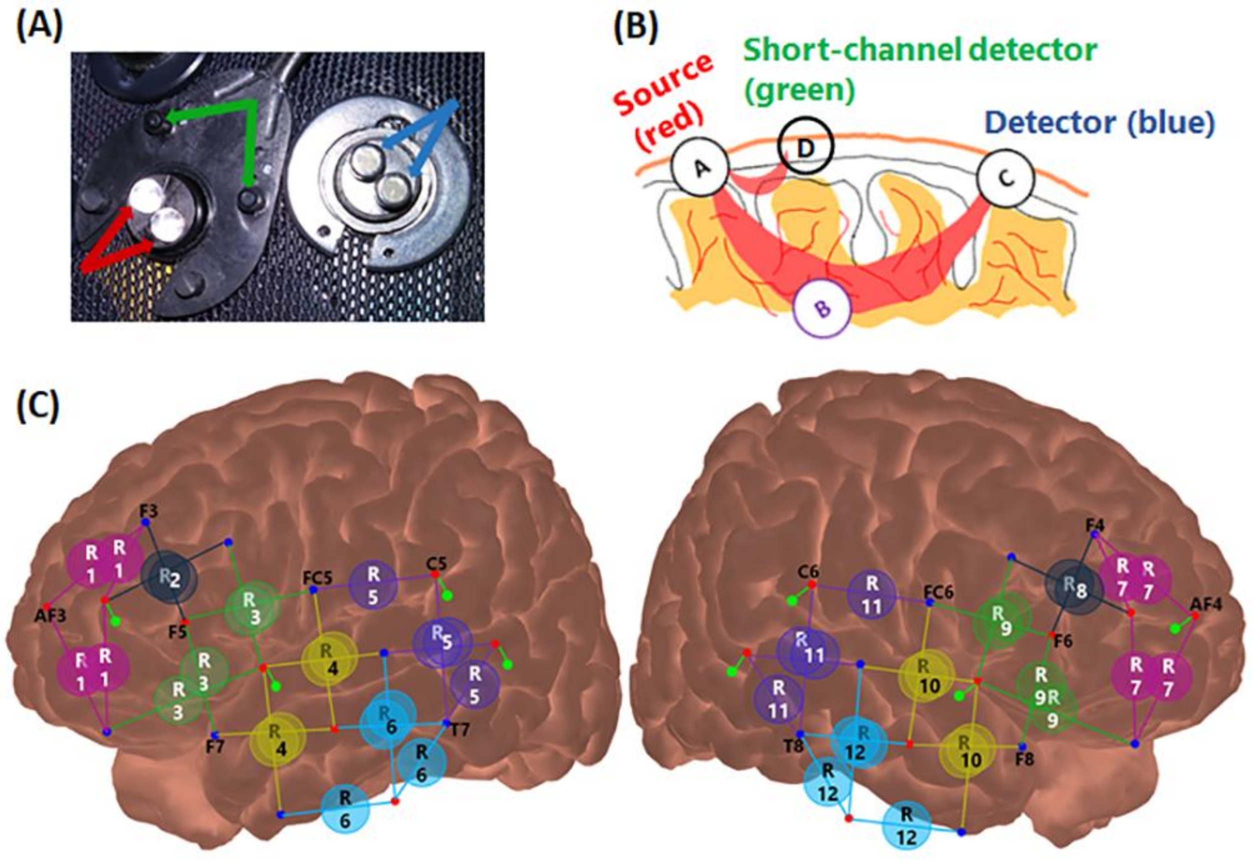




Nejad et al, 2018

Components of an fNIRS System

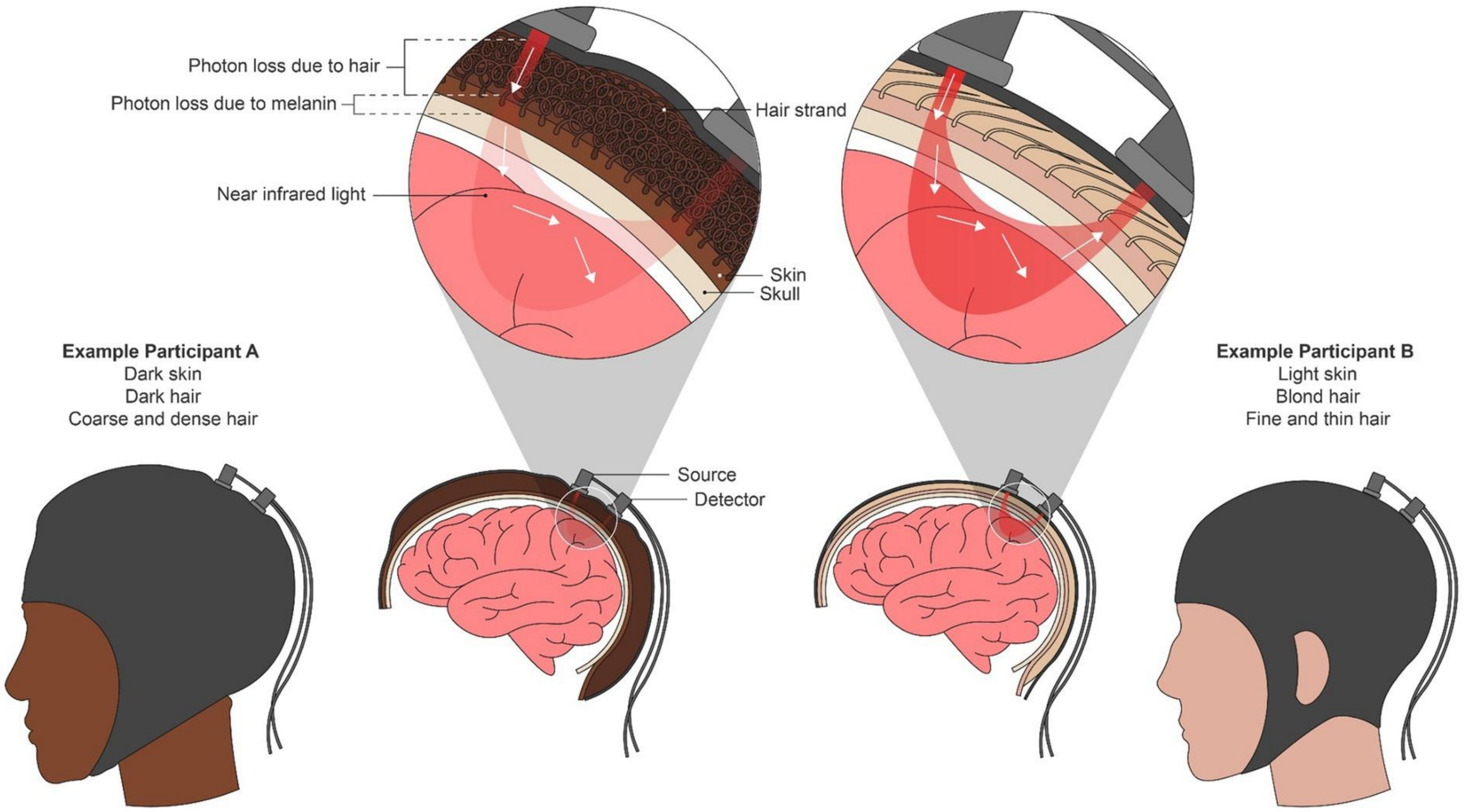
- **Source:** Emits near-infrared light
- **Detectors:** Captures reflected light
- **Optodes:** Devices that combine sources and detectors
- **Data Acquisition System:** Converts light intensity into signals
- **Computer Software (YOU!):** Processes signals to visualize brain activity



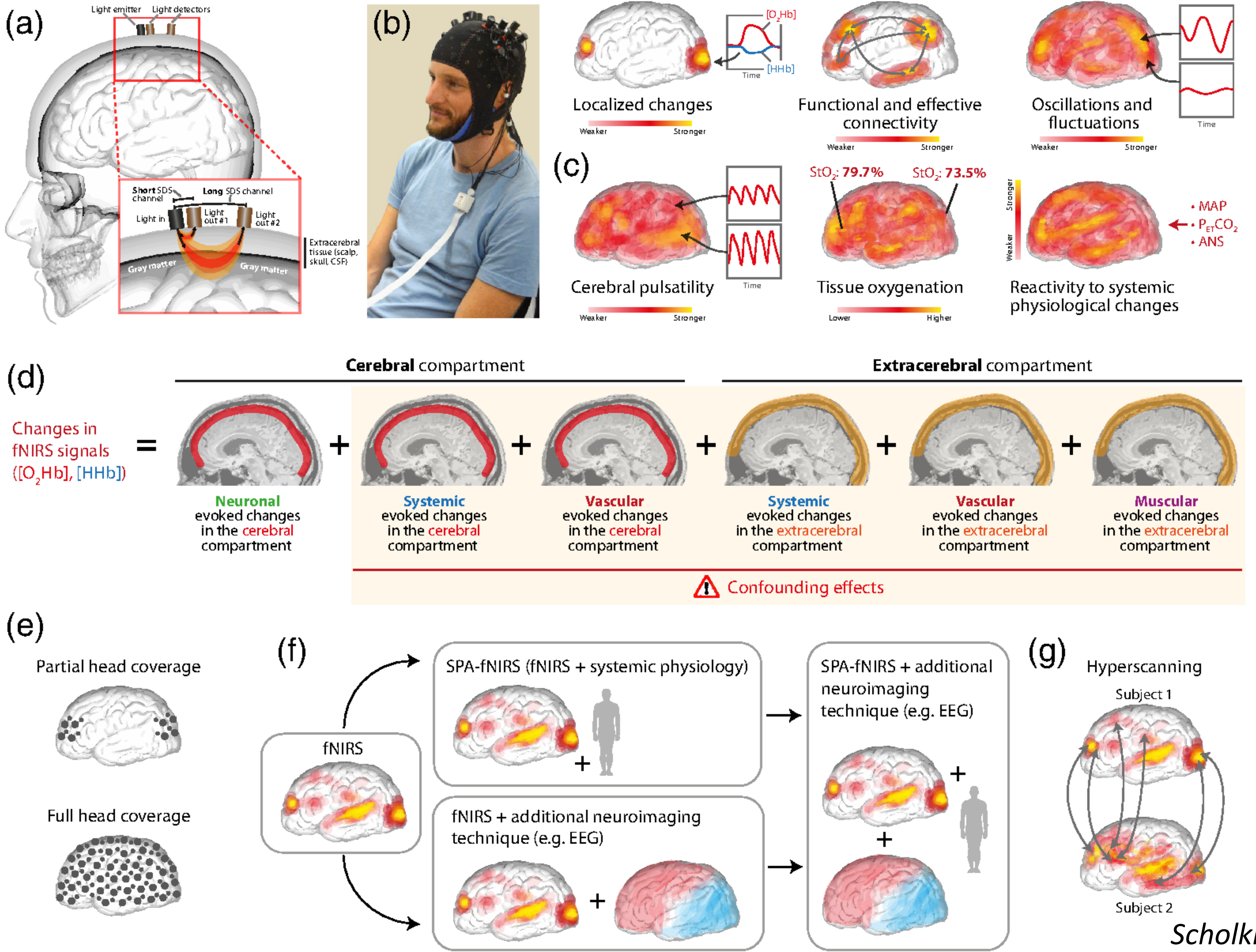
Zhou et al, 2020

Limits of the fNRIS

- affected by many factors not related to neuronal activity
- low spatial resolution
- low temporal resolution: sample at 10 Hz but based on BOLD
- only cortical



Kwasa et al., 2023

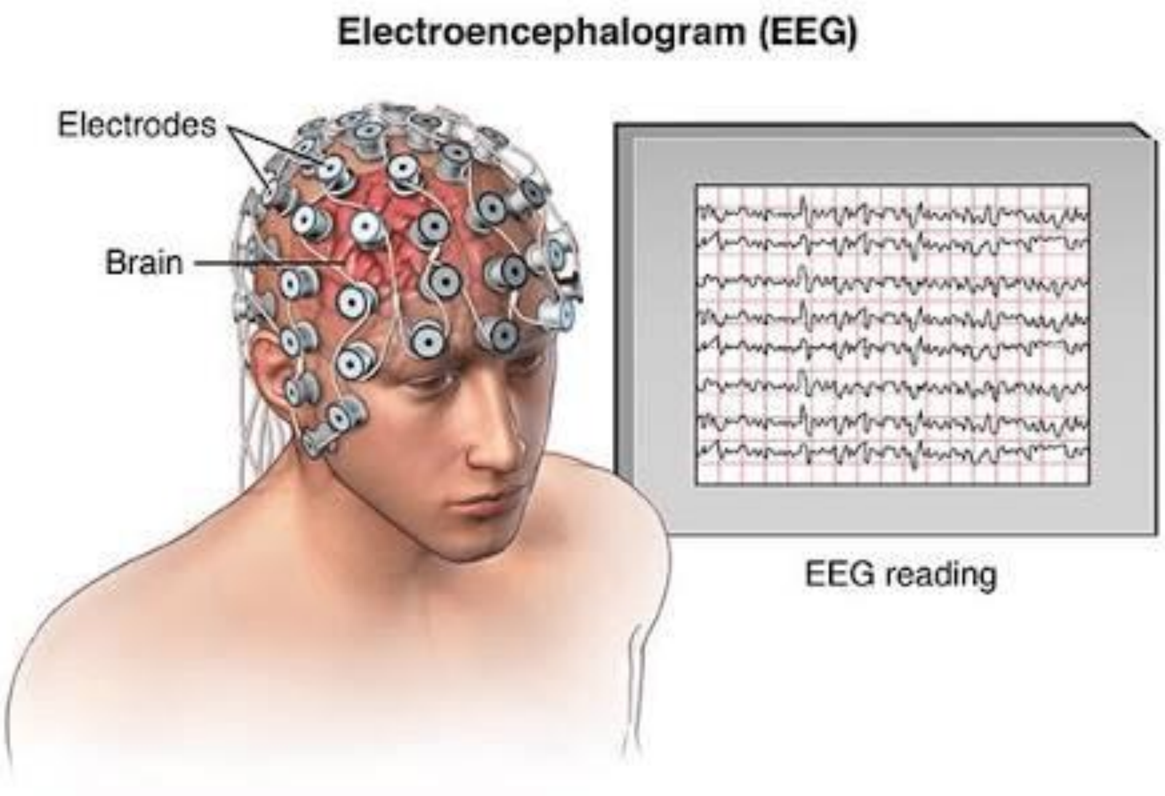


Scholkmann et al., 2020

EEG (Electroencephalography)

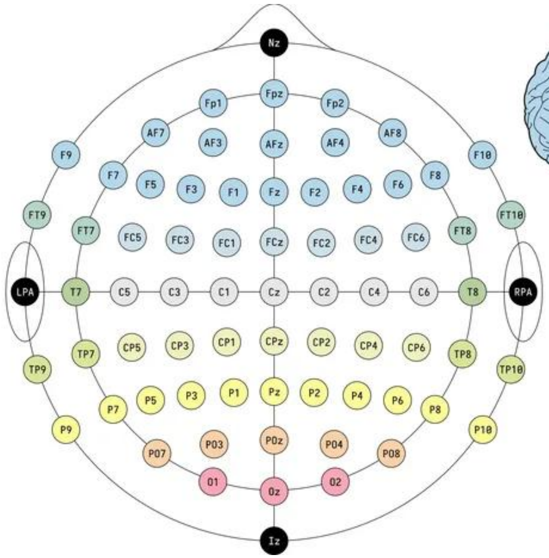
What is EEG?

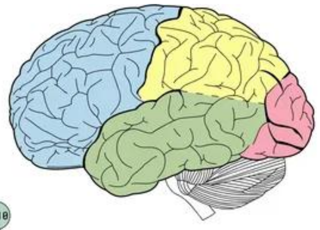
- Measures brain's electrical activity
- Non-invasive
- Used in clinical and research settings

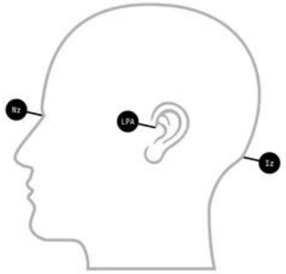


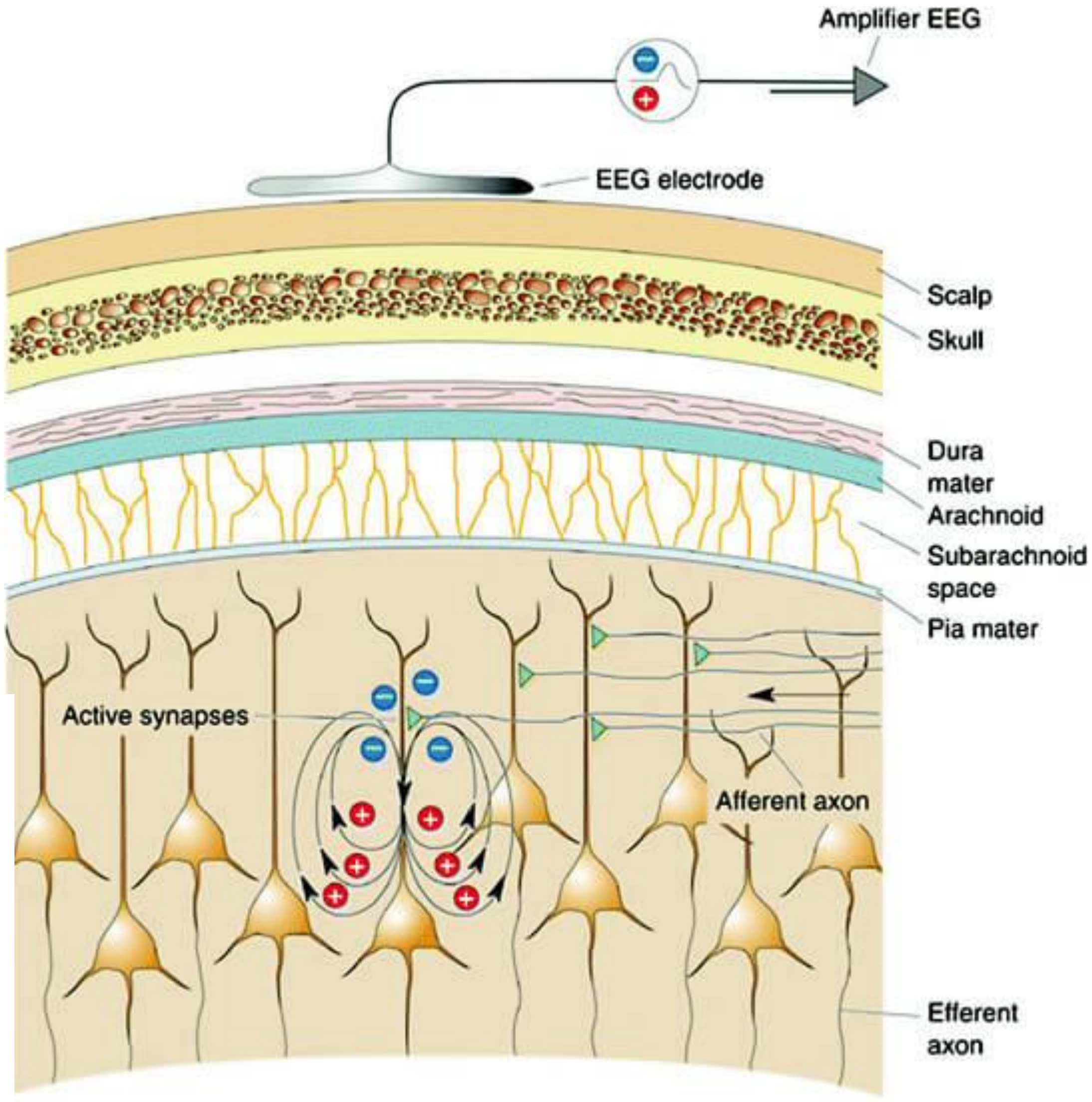
How Does EEG Work?

- Electrodes on the scalp detect brainwaves
- Measures voltage fluctuations from neuron activity (many neurons together)
- Uses the 10-20 electrode placement system



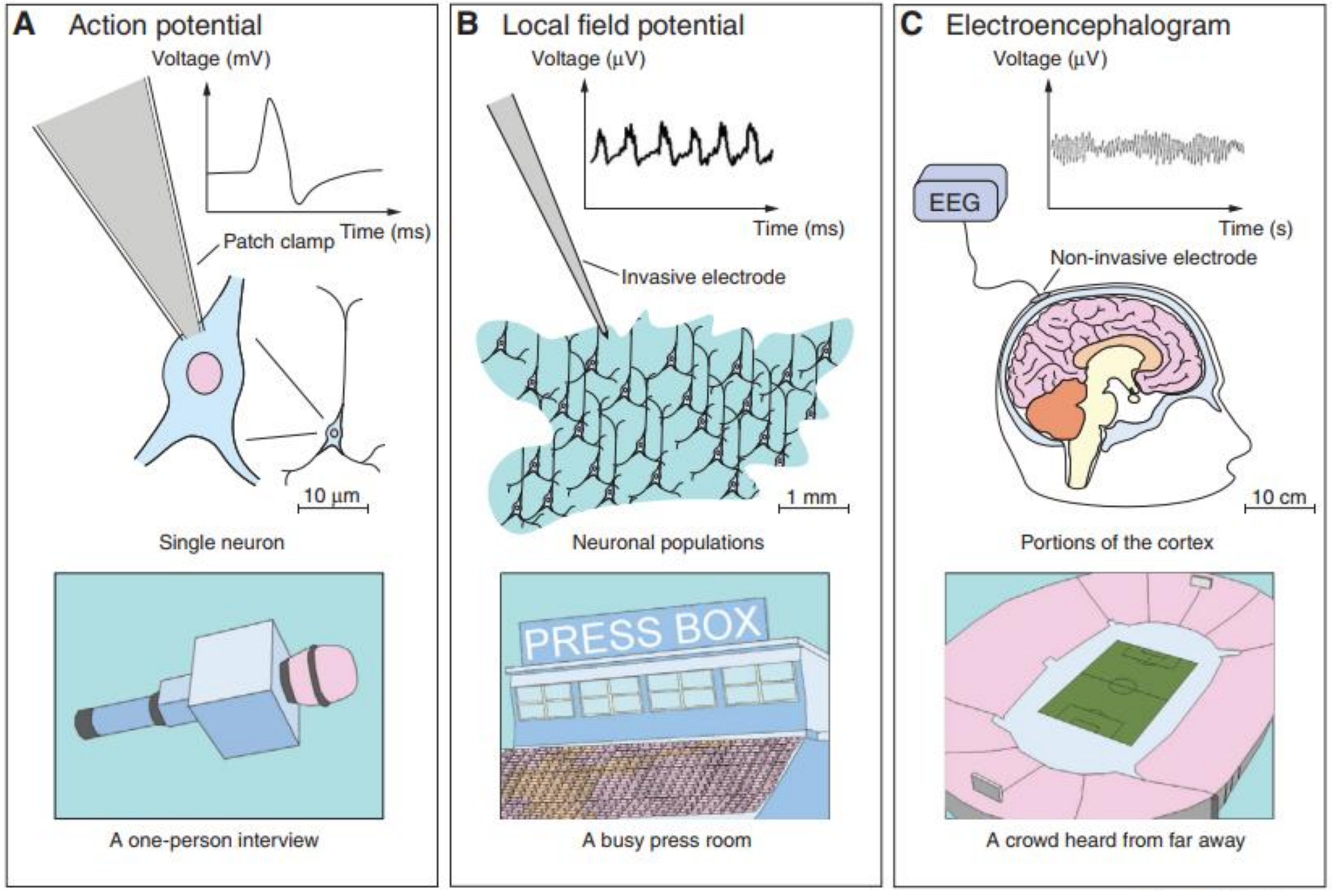






Siuly, et al. 2016

EEG (Electroencephalography)



Current Biology

How Does EEG Work?

- The DIPOLE in EEG

What Does EEG Measure?

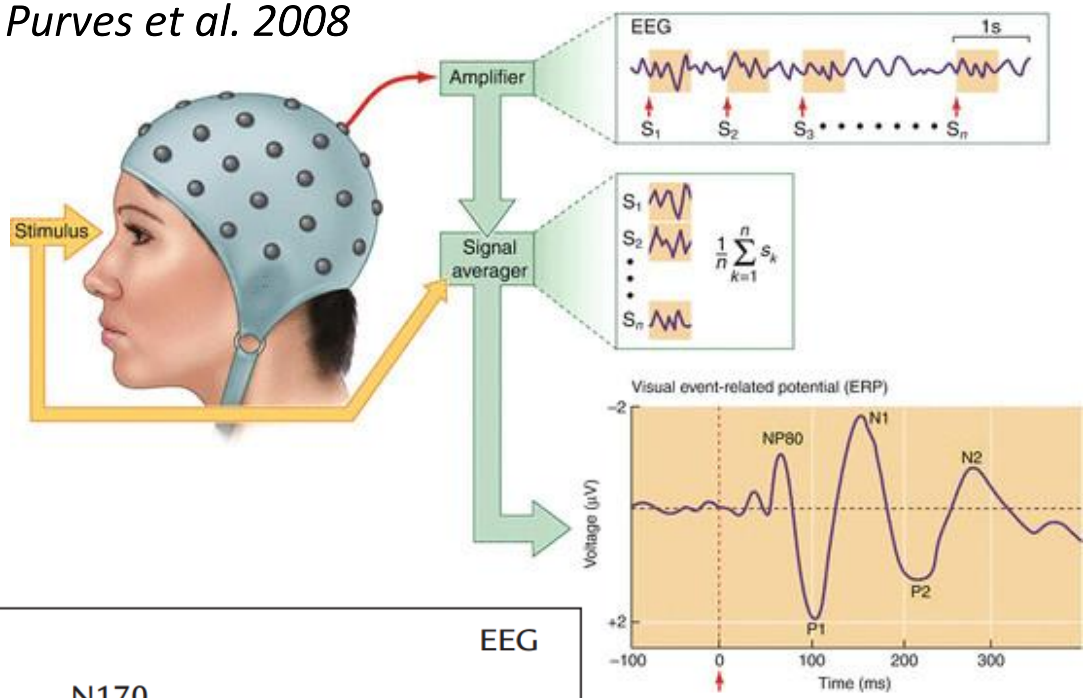
- Brainwave patterns:
  - Delta
  - Theta
  - Alpha
  - Beta
  - Gamma

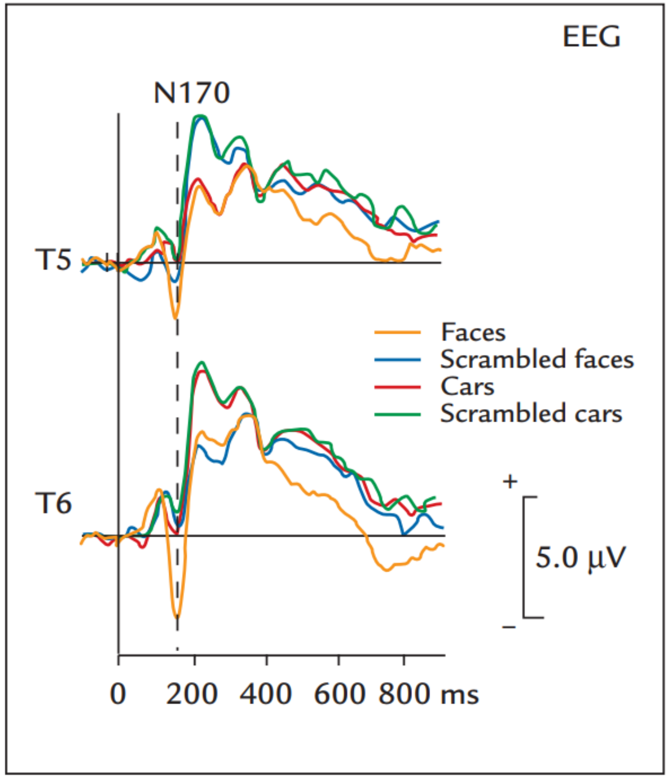
What Does EEG Measure?

- Event Related Potential (ERP)

- Example of visual ERP (VEP):
  - P1: A positive wave that occurs around 100 ms after the stimulus, associated with early visual processing.
  - N1: A negative wave around 150-200 ms, linked to attention and the detection of visual stimuli.
  - P3: A positive wave around 300 ms, often associated with higher-level processing like decision-making or categorization.

Purves et al. 2008

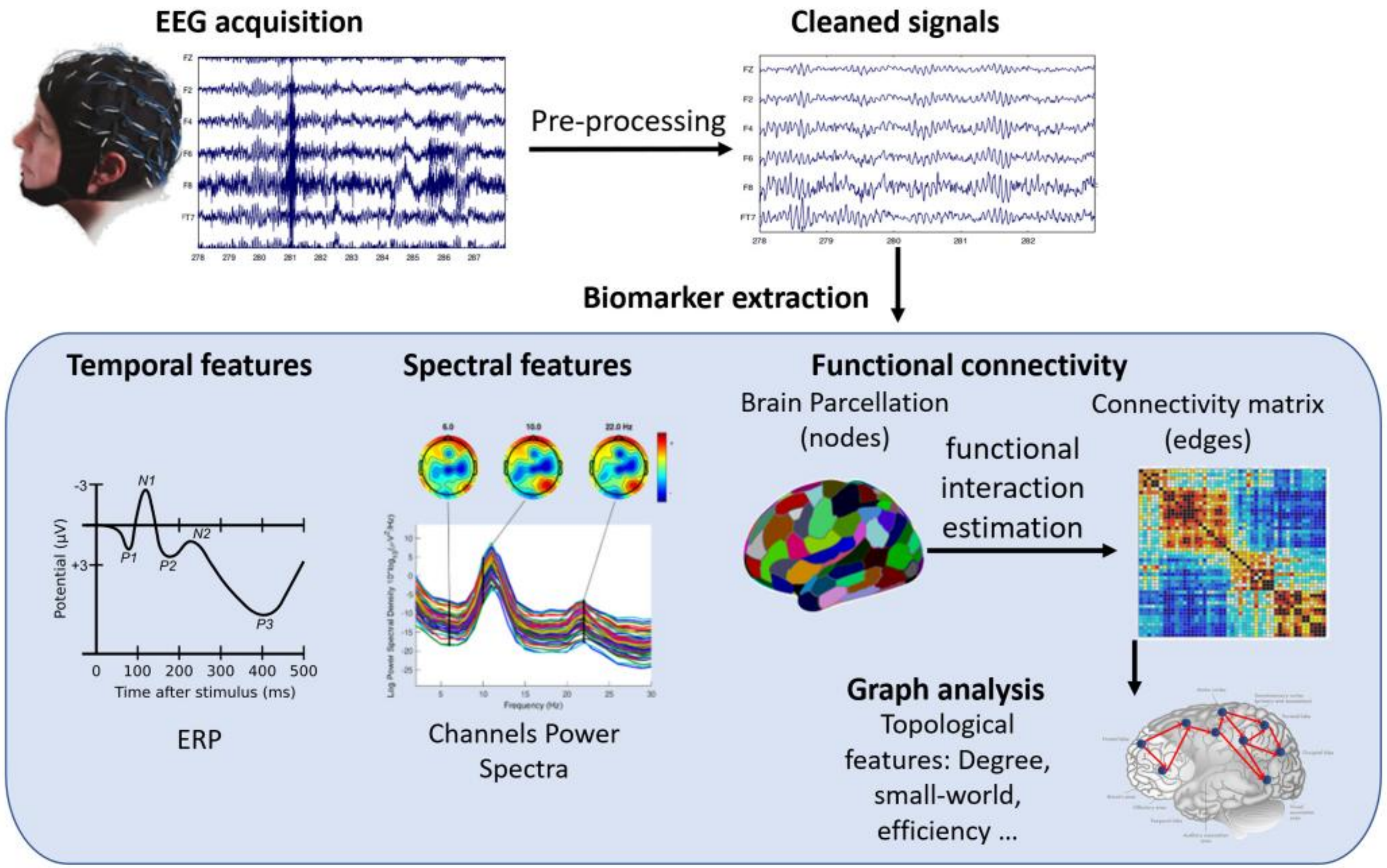




Purves et al. 2008

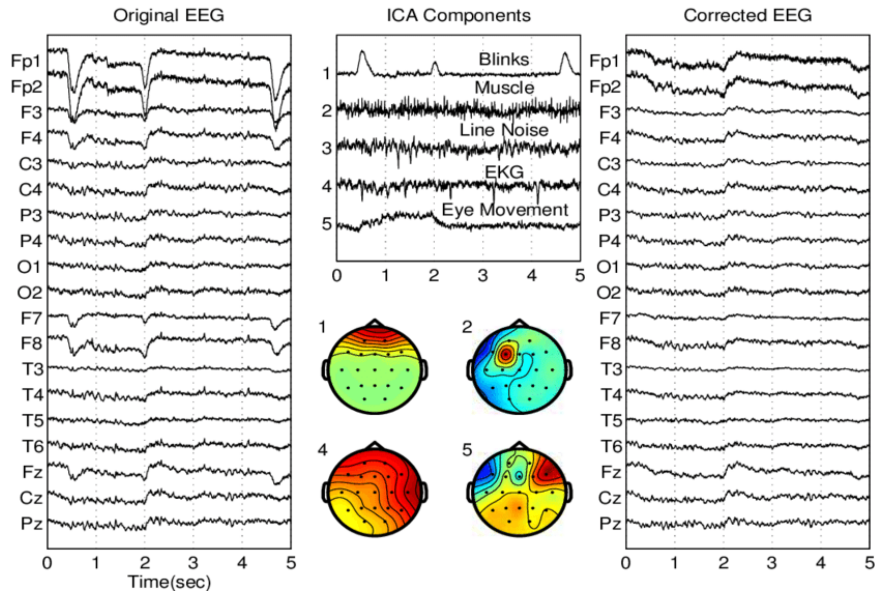
EEG Signal Processing

- Of course, depends on your hypothesis!



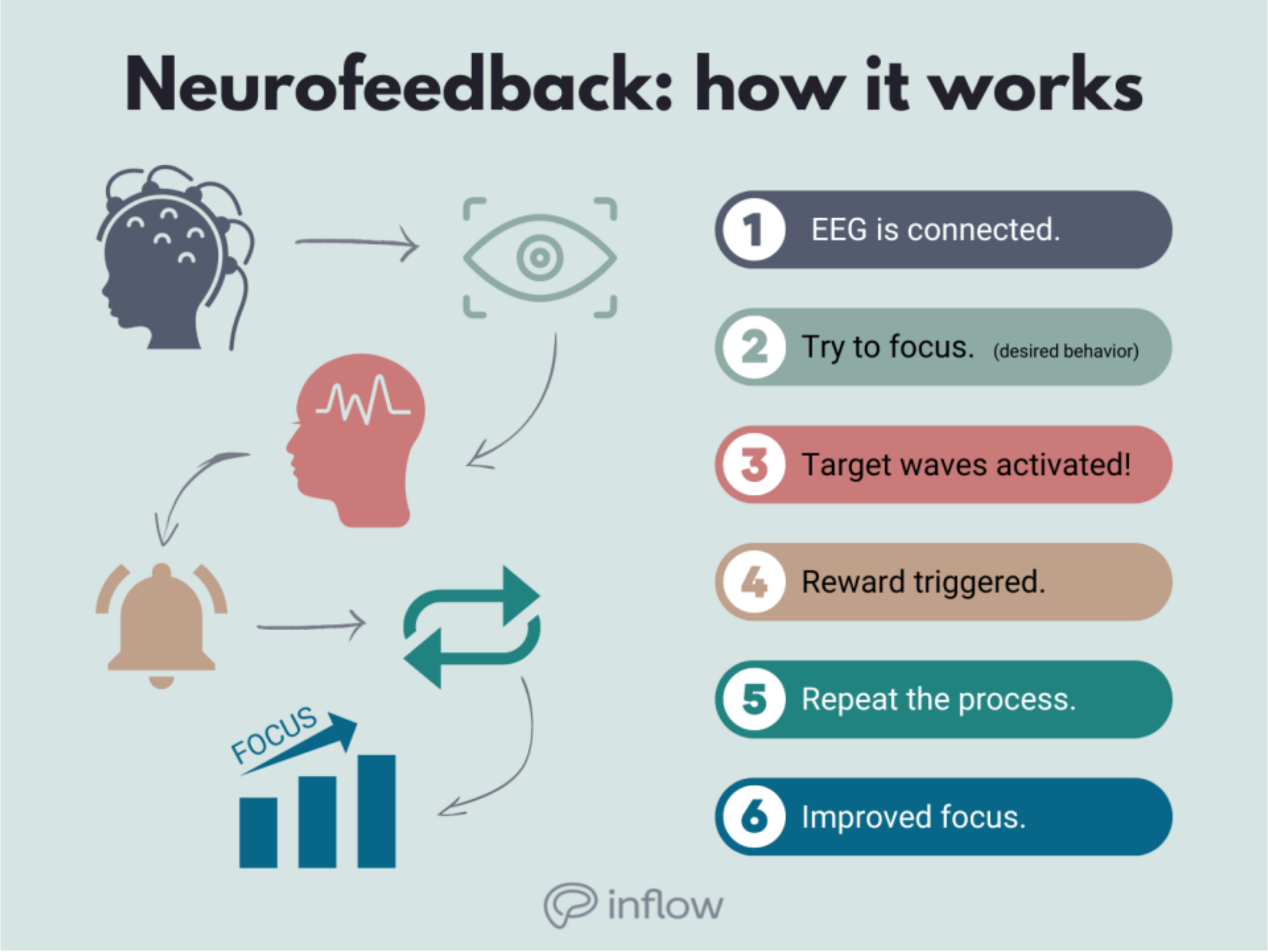
- Preprocessing the data:

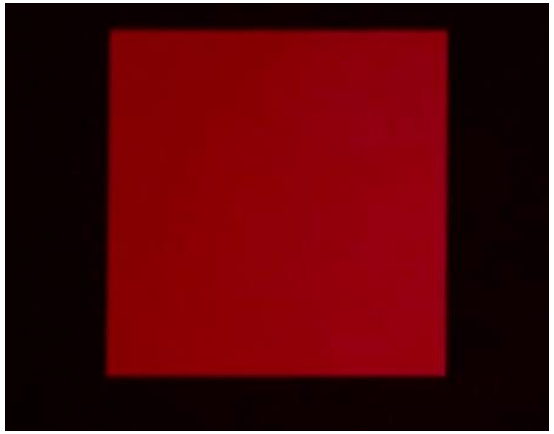
lot of artefacts coming from multiple sources



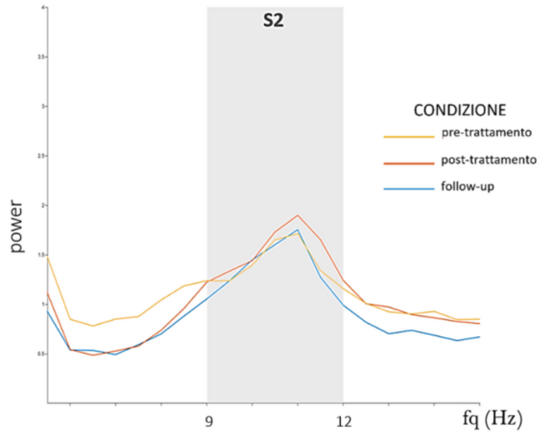
Scano et al., 2023

Possible application: NEUROFEEDBACK





- the subject need to reach and maintain the red square → <u>gamma band</u>



Advantages of EEG:

- **High Temporal Resolution**: Captures brain activity in real-time with millisecond precision.
- **Non-Invasive and Safe**: Easy to use across all age groups and health conditions.
- **Relatively Affordable**: Cheaper and more portable than other imaging techniques like fMRI.
- **Direct Measure of Neural Activity**: Tracks brain's electrical signals directly, providing real-time insight into brain function.

Disadvantages of EEG:

- **Low Spatial Resolution**: Cannot precisely pinpoint the location of brain activity.
- **Prone to Noise**: Sensitive to artifacts like muscle movements or eye blinks.
- **Surface-Level Measurement**: Primarily detects activity from the brain's outer layer (cortex), missing deeper structures.

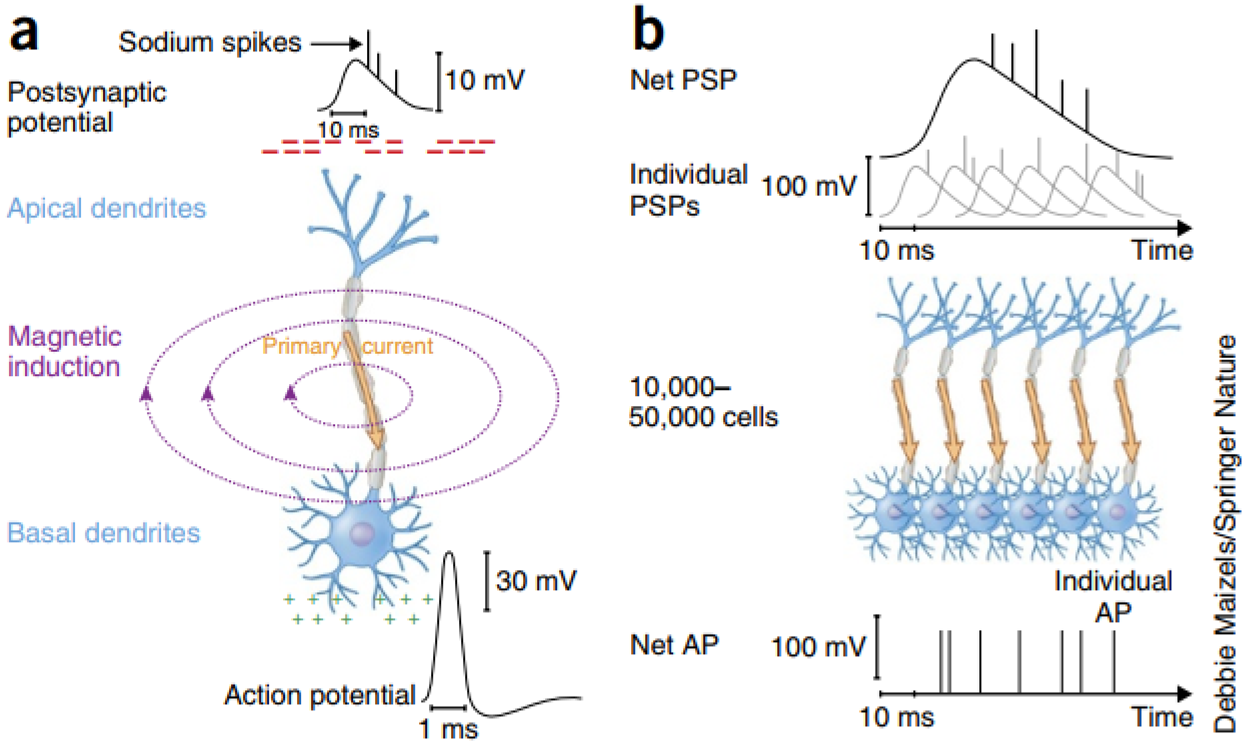
MEG (magnetoencephalography)

- Measures brain's magnetic fields
- Non-invasive
- Real-time recording of brain activity

How Does MEG Work?

- Measures magnetic fields from neuron activity → current of ions induce a magnetic field

- MEG like EEG relies mainly on postsynaptic potential



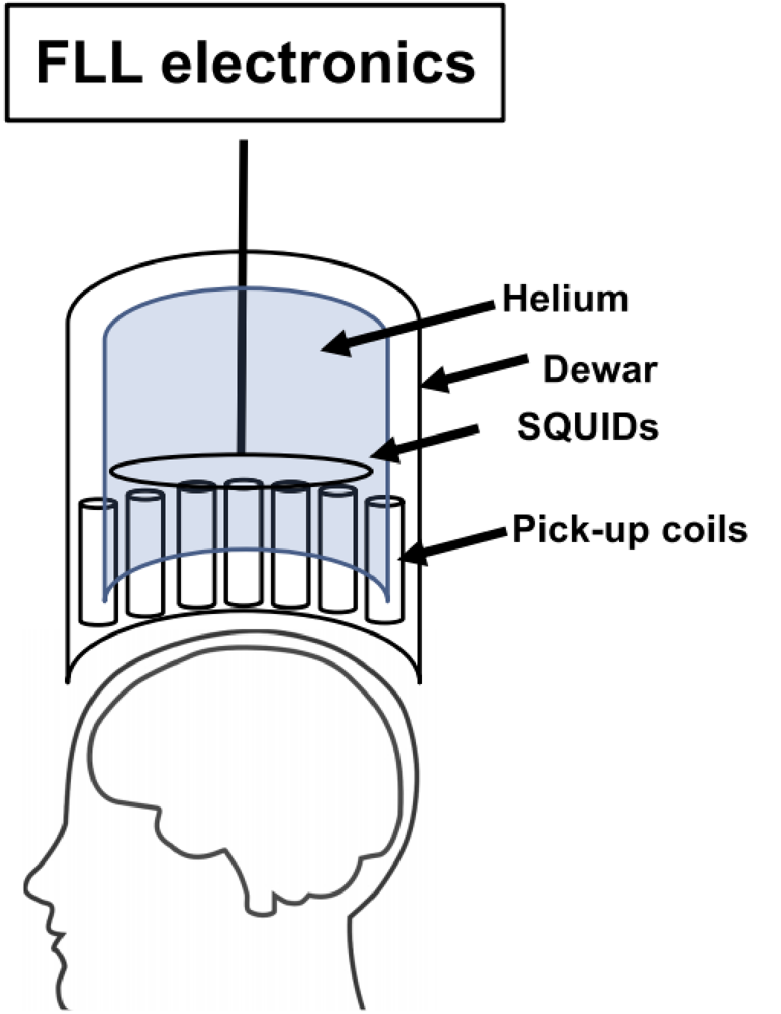
Baillet 2017

How Does MEG Work?

- In order to detect the small magnetic field we need SQUID (superconducting quantum interference device)

- FLL flux-locked loop

- Dewar → container isolated from the world with liquid helium



Kim et al., 2021

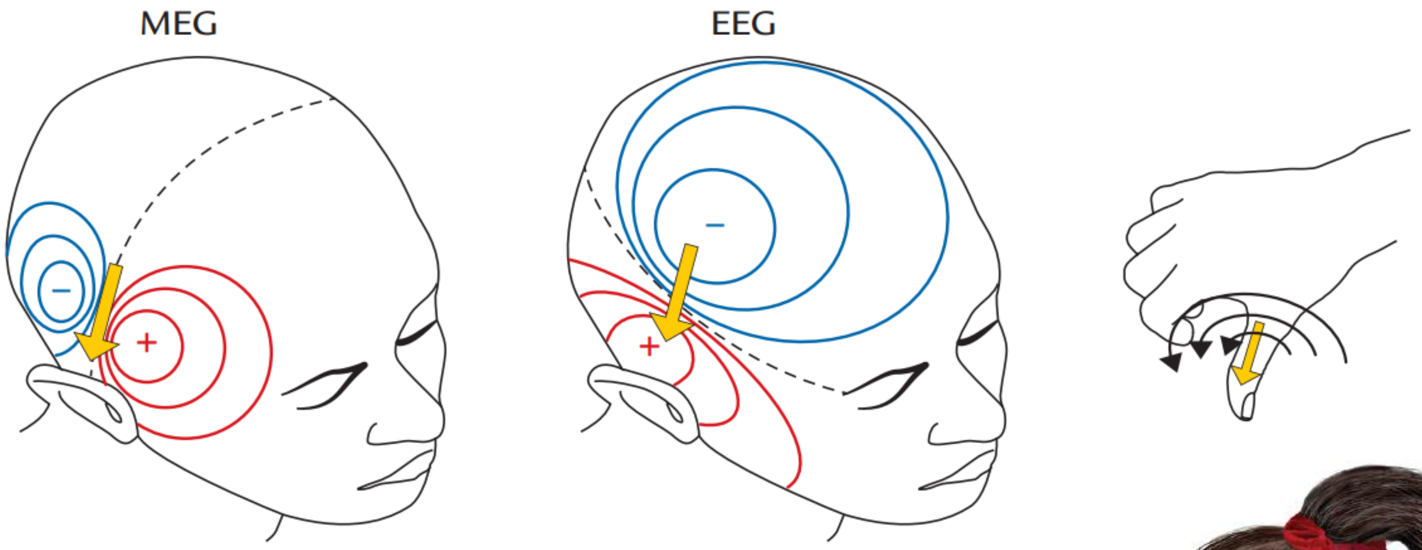
MEG vs EEG

	EEG	MEG
Signal magnitude	Large signal (10 mV), easy to detect	Tiny signal (10 fT), difficult to detect
Cost	Cheap	Expensive
What does signal index?	Measures secondary (volume) currents	Measures fields generated by primary currents
Signal purity	Affected by skull, scalp, etc.	Unaffected by skull, scalp, etc.
Temporal Resolution	~ 1 ms	~ 1 ms
Spatial Localization	~ 1 cm	~ 1 mm
Experimental Flexibility	Allows some movement	Requires complete stillness
Dipole Orientation	Sensitive to tangential and radial dipoles	Sensitive only to tangential dipoles

MEG vs EEG

- Example with auditory stimuli

- MEG and EEG are sensible for different spatial sources and neural configuration



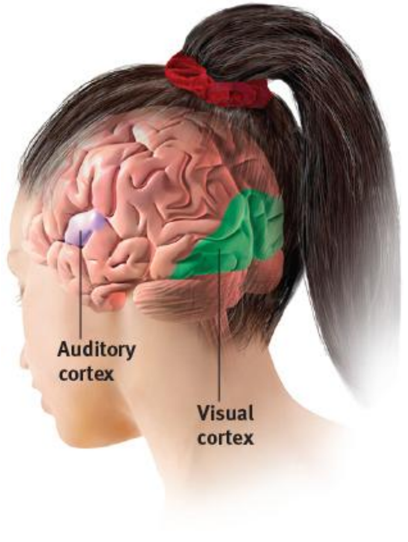
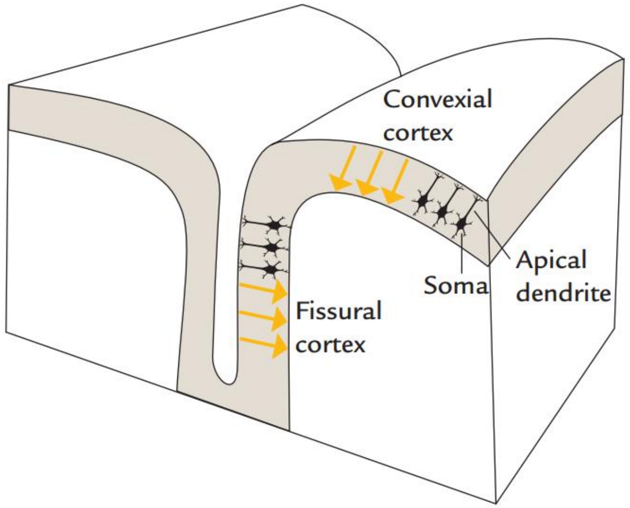
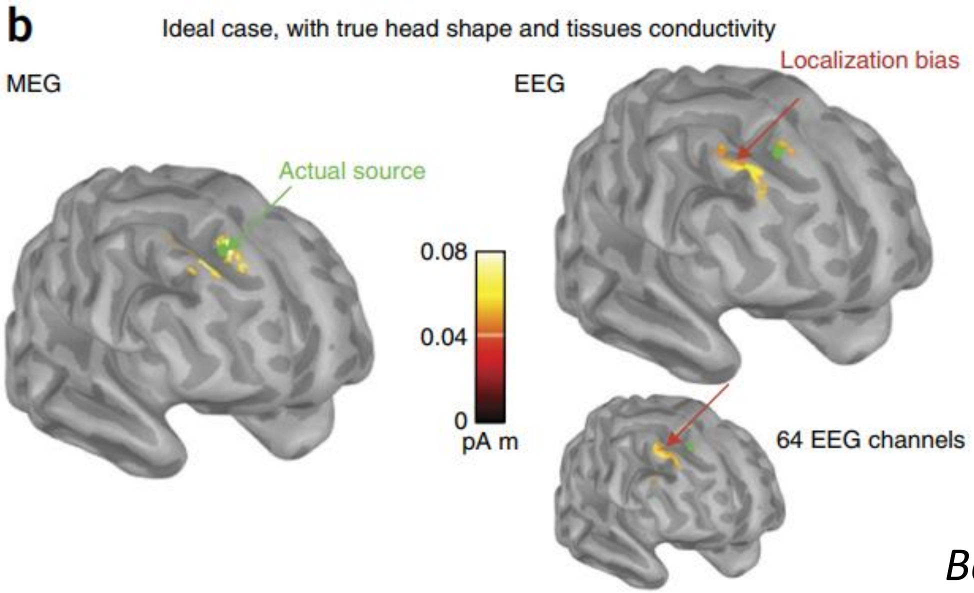


Figure 2.18
Myers/DeWall, *Psychology in Everyday Life*, 4e, © 2017 Worth Publishers





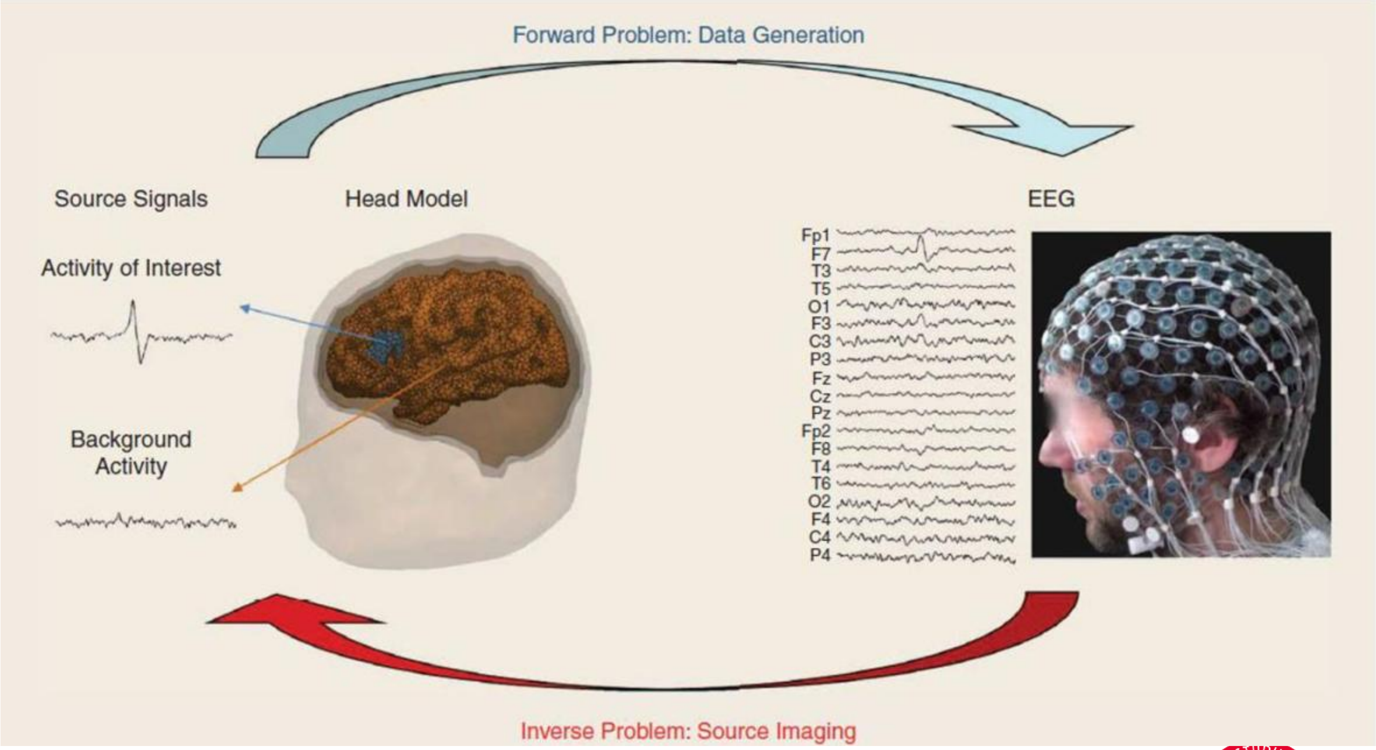
Baillet 2017

The Foward and Inverse Problem in EEG - MEG

The EEG/MEG signals measured on or around the scalp do not directly reflect the activated neurons in the brain. To reconstruct the actual activity in the brain, source reconstruction techniques are used

- ❑ **Forward problem:** Given a known distribution of neural activity inside the brain, compute the expected electrical potentials (for EEG) or magnetic fields (for MEG) that would be measured at the scalp or around the head.

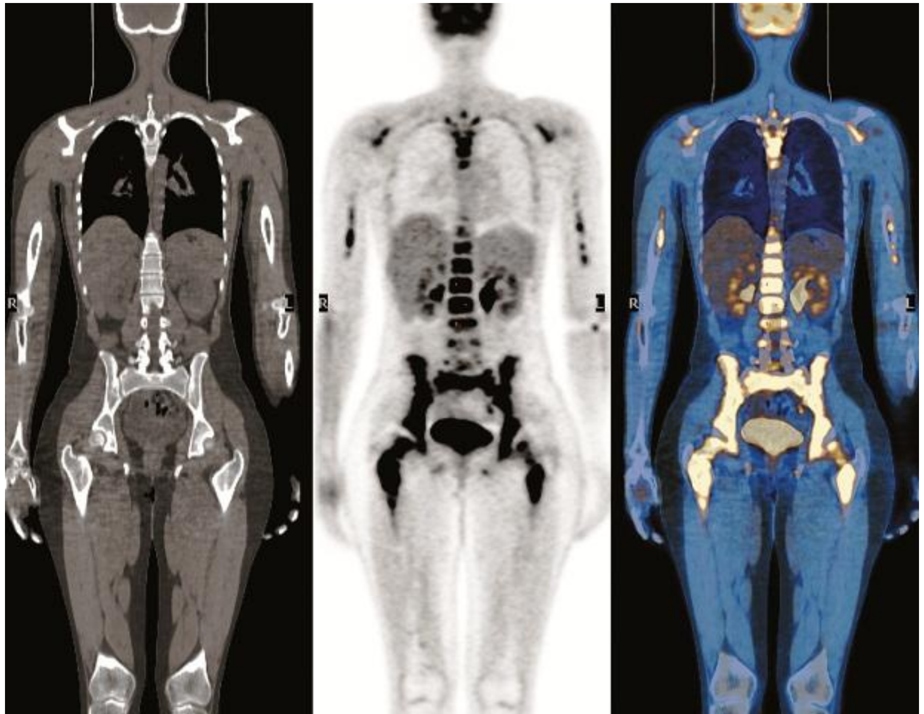
- ❑ **Inverse problem:** The inverse problem is the opposite—starting from the recorded EEG or MEG signals, infer the neural activity within the brain that generated those signals.



Asadzadeh et al., 2019

PET (Positron Emission Tomography)

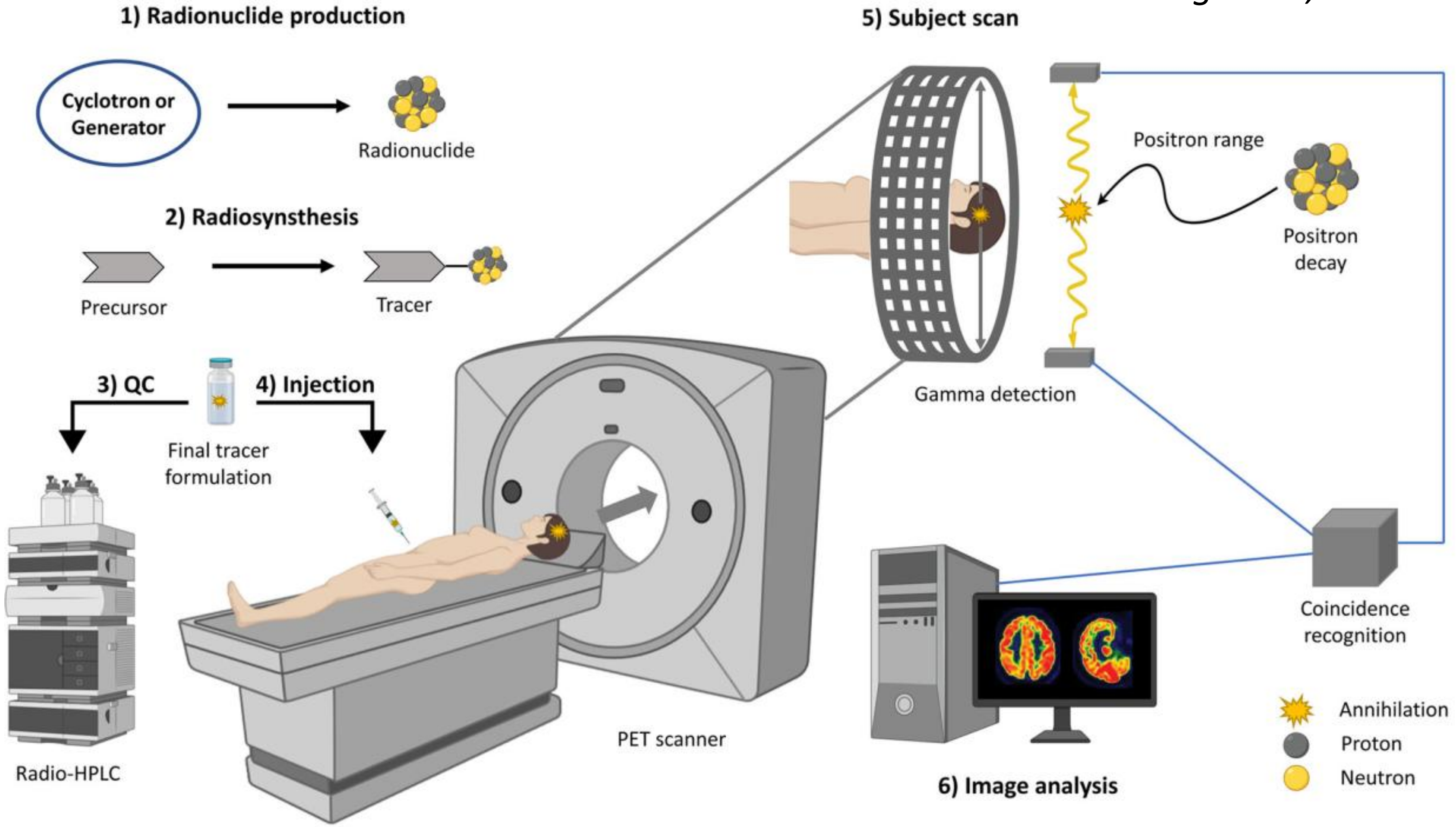
- Imaging technique for metabolic activity
- Uses radioactive tracers
- Often used in medical diagnostics



How Does PET Work?

- Radioactive tracer injected into the body → often create close in the hospital
- Tracer accumulates in active tissues
- Gamma rays emitted are detected to create images

Rong et al., 2023



How Does PET Work?

- Common elements used in PET
  - Fluorine-18 (^{18}F) s a glucose analog, meaning it behaves like glucose in the body
  - Oxygen-15-labeled water ($H_2{}^{15}O$) is used as a tracer for measuring blood flow

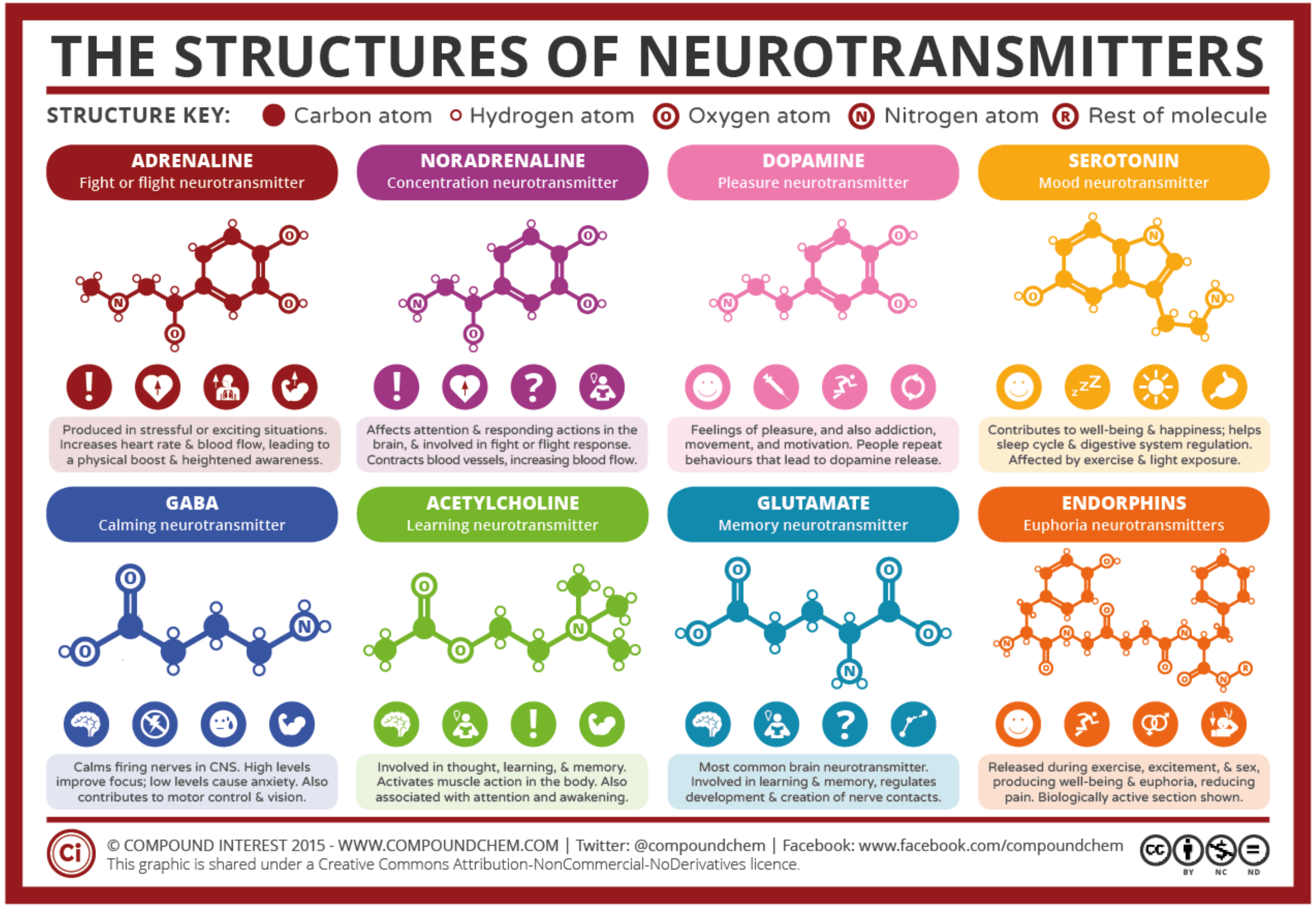
Isotope	Half-life	Production[a]	Mode of decay	Common source
^{18}F	109.8 min	$^{18}O(p,n)^{18}F$	β^+ (97%), EC (3%)	Cyclotron
^{11}C	20.4 min	$^{14}N(p,\alpha)^{11}C$	β^+ (100%)	Cyclotron
^{13}N	10 min	$^{16}O(p,\alpha)^{13}N$	β^+ (100%)	Cyclotron
^{15}O	2 min	$^{15}N(p,n)^{15}O$	β^+ (100%)	Cyclotron
^{124}I	4.2 d	$^{124}Te(p,n)^{124}I$	β^+ (23%), EC (77%)	Cyclotron
^{44}Sc	4.0 h	$^{44}Ti/^{44}Sc$	β^+ (94%), EC (6%)	Generator[b]
^{64}Cu	12.7 h	$^{64}Ni(p,n)^{64}Cu$	β^+ (17%), EC (44%), β^- (39%)	Cyclotron[c]
^{68}Ga	67.7 min	$^{68}Ge/^{68}Ga$	β^+ (89%), EC (11%)	Generator[d]
^{82}Rb	1.3 min	$^{82}Sr/^{82}Rb$	β^+ (95%), EC (5%)	Generator
^{86}Y	14.7 h	$^{86}Sr(p,n)^{86}Y$	β^+ (32%), EC (68%)	Cyclotron
^{89}Zr	78.4 h	$^{89}Y(p,n)^{89}Zr$	β^+ (23%), EC (77%)	Cyclotron

Rong et al., 2023

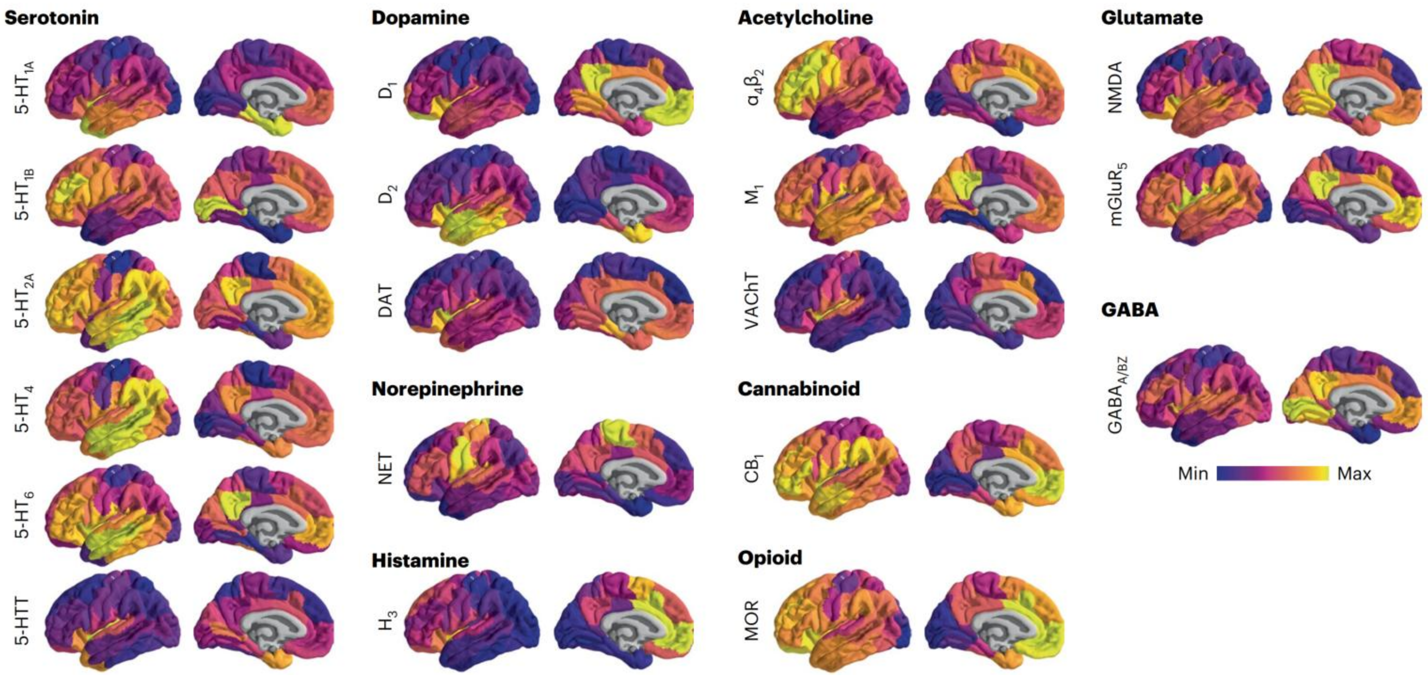
Why is PET relevant?

- Give us metabolic and structural aspect of the brain

  - For example, NEUROTRANSMITTER RECEPTOR



Why is PET relevant?



nature neuroscience

Resource

https://doi.org/10.1038/s41593-022-01186-3

Mapping neurotransmitter systems to the structural and functional organization of the human neocortex

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Published online: 27 October 2022

Check for updates

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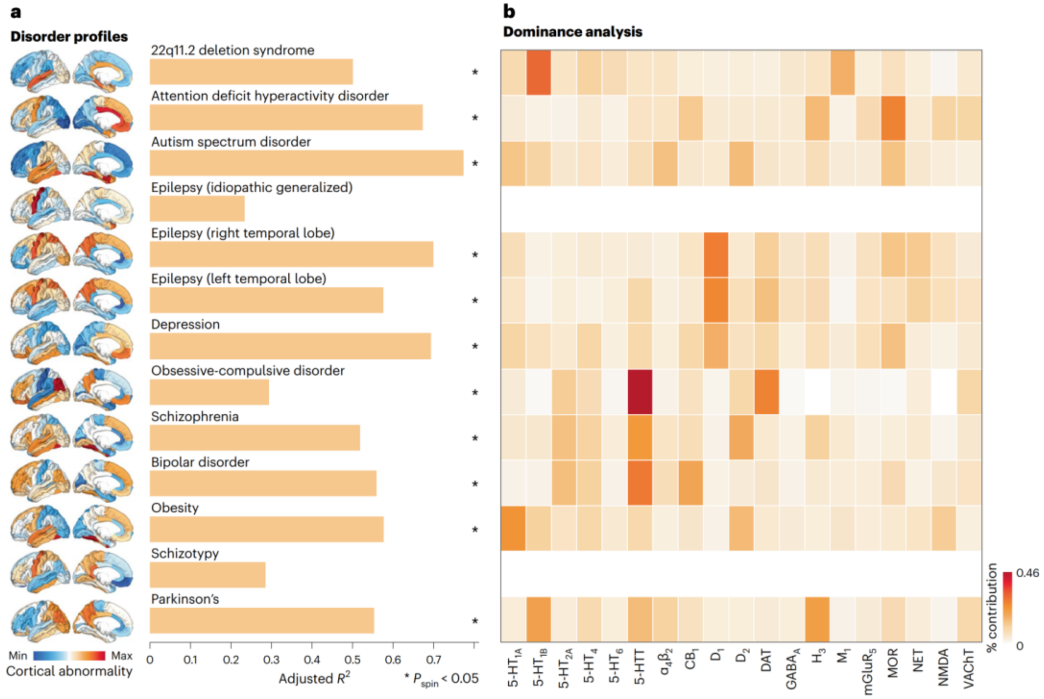
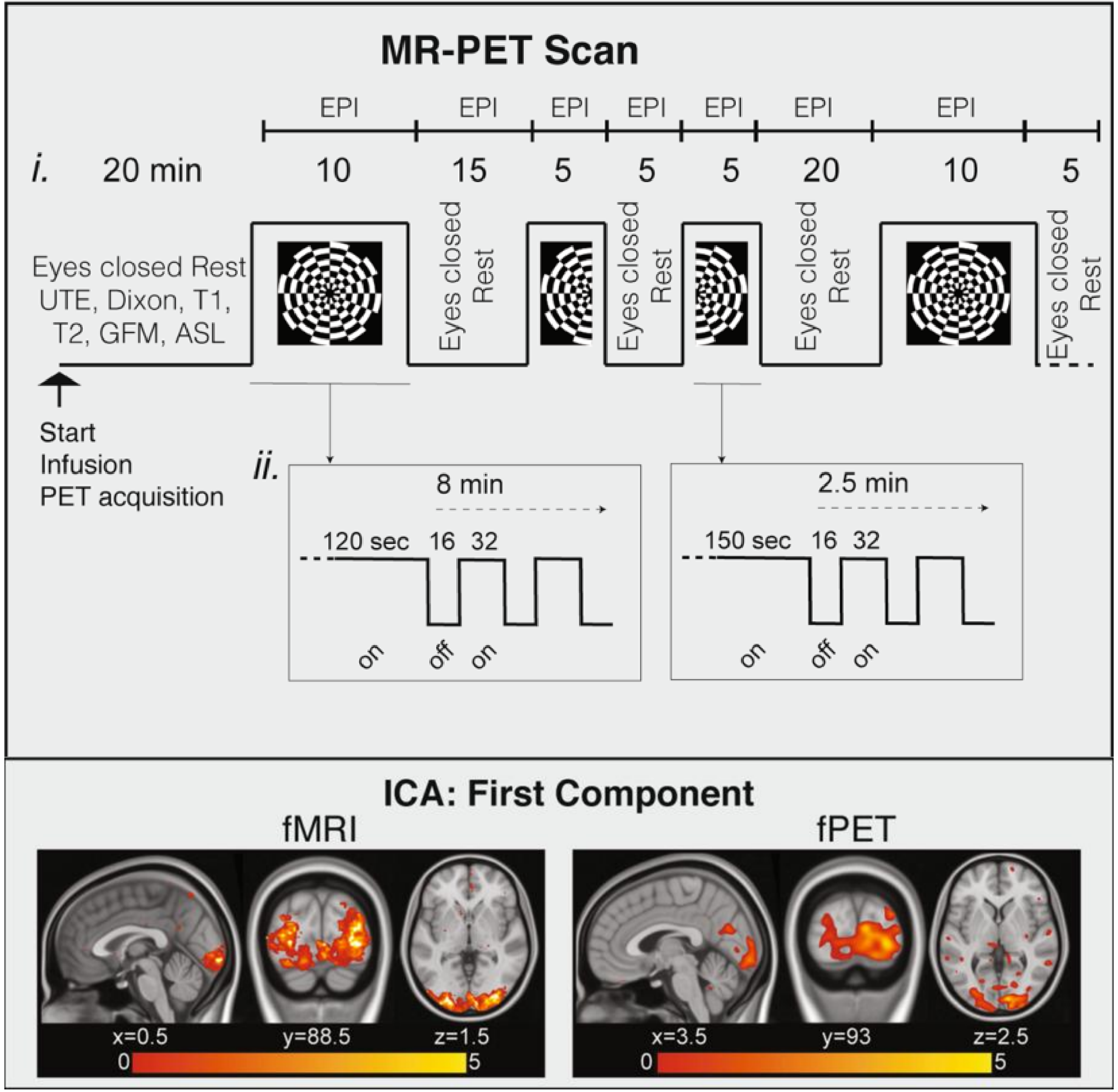


functional PET

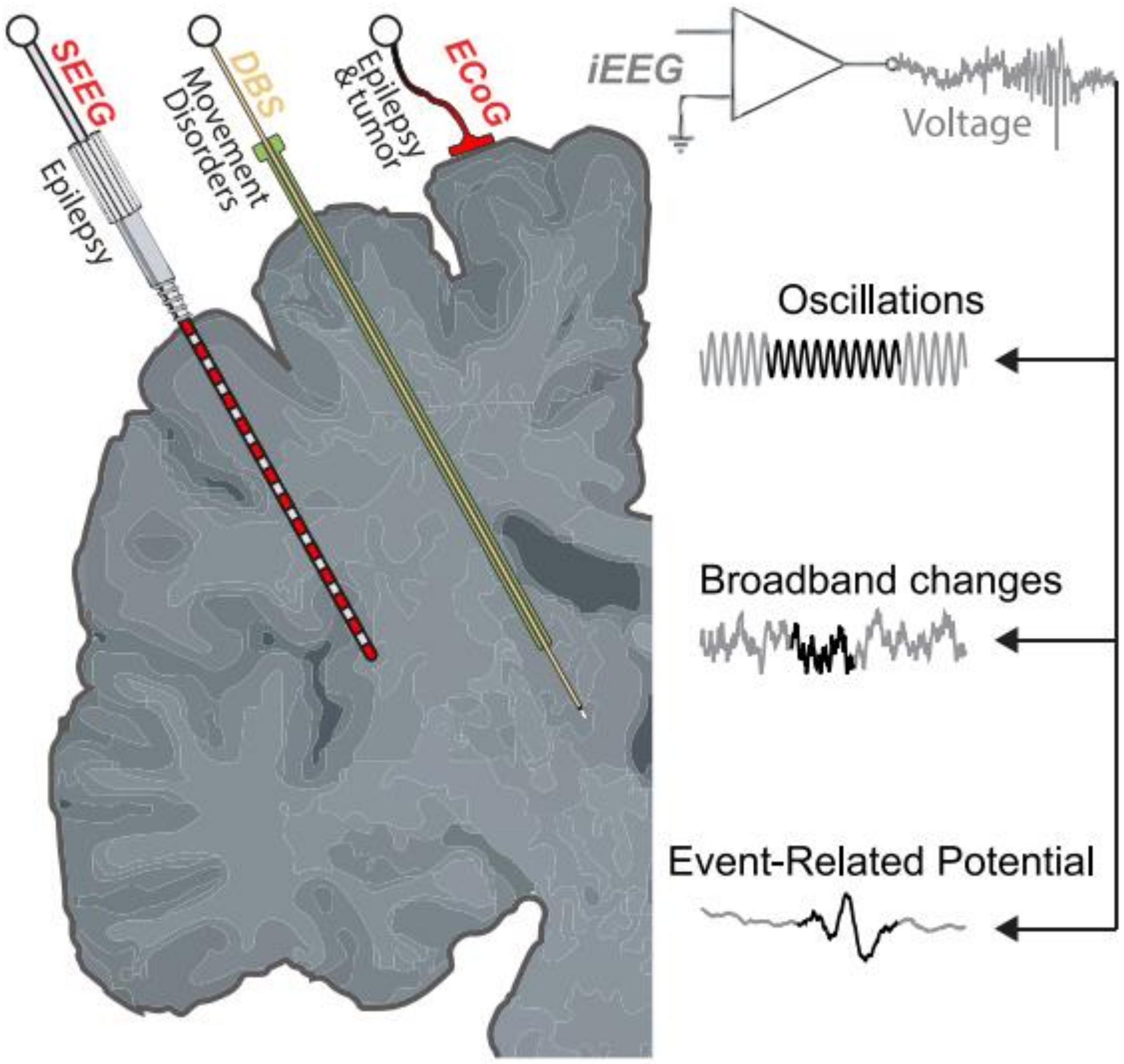
- Most of the PET image are "static" → one volume
- Low spatial resolution
- Whitin certain limitations, it is possible to study brain "continuosly"



Rong et al., 2023

Intracranial recording

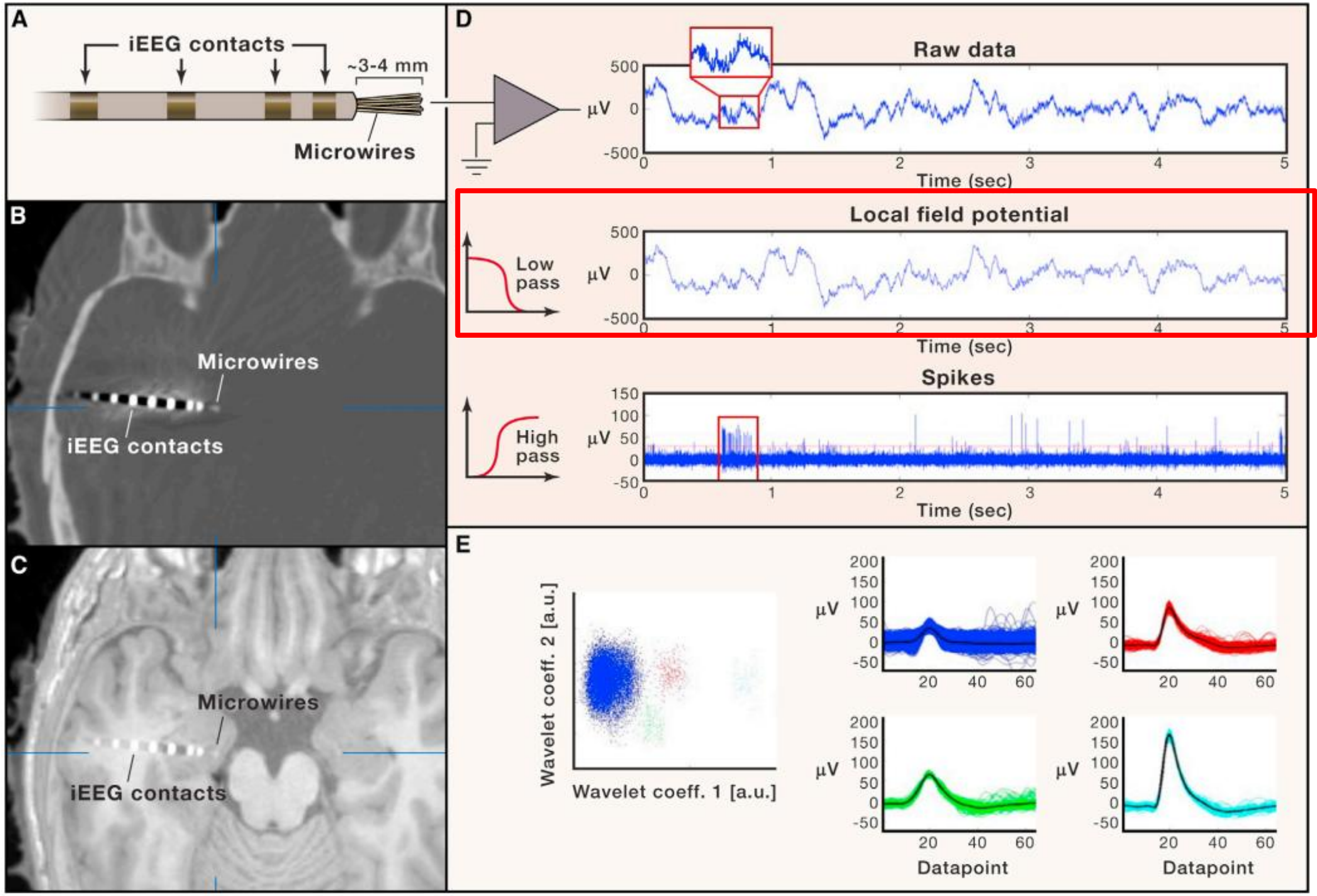
- Define microelectrodes: thin, needle-like electrodes used to record electrical activity of single neurons.
- Types: Tungsten, Platinum-iridium, Carbon fiber.
- Recording action potentials (spikes) from neurons.
- Explanation of electrical signals and signal processing.
- Equipment used: amplifier, filters, data acquisition system.



Mercier *et al.*, *2022*

Intracranial recording

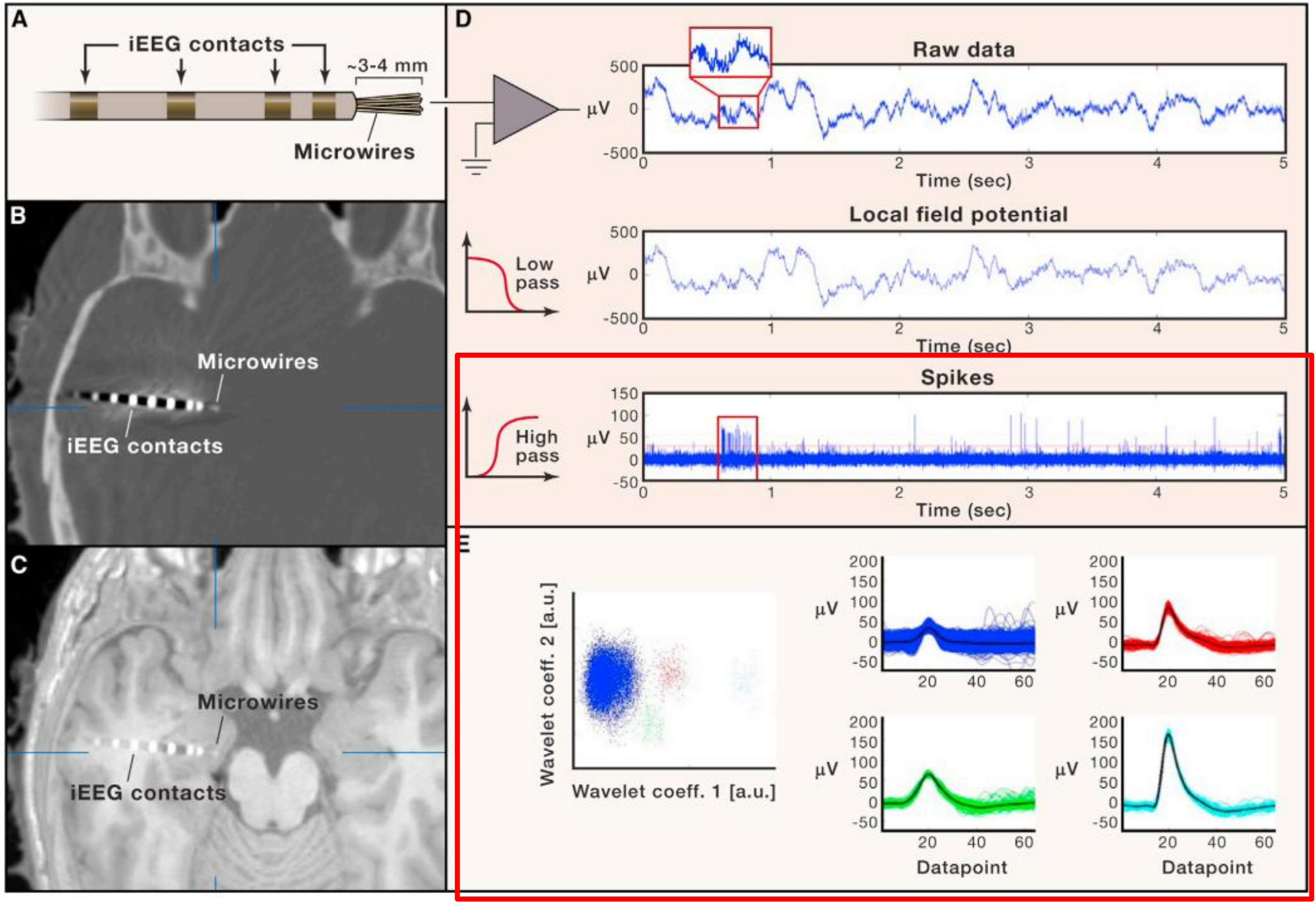
- Insertion of SEEG in temporal lobe
- Local Field Potential: refers to the electric potential in the extracellular space around neurons.
  - sources: LFPs primarily measure synaptic potentials pooled across groups of neurons near the recording electrode
  - like EEG, brainwaves



Quiroga et al., 2019

Intracranial recording

- Spikes... well Action Potential!
- Time course full of spikes > std
- **Spike sorting:** detecting and classifying action potentials (spikes) from multiple neurons recorded by the same electrode, allowing for the identification of individual neuron activity.

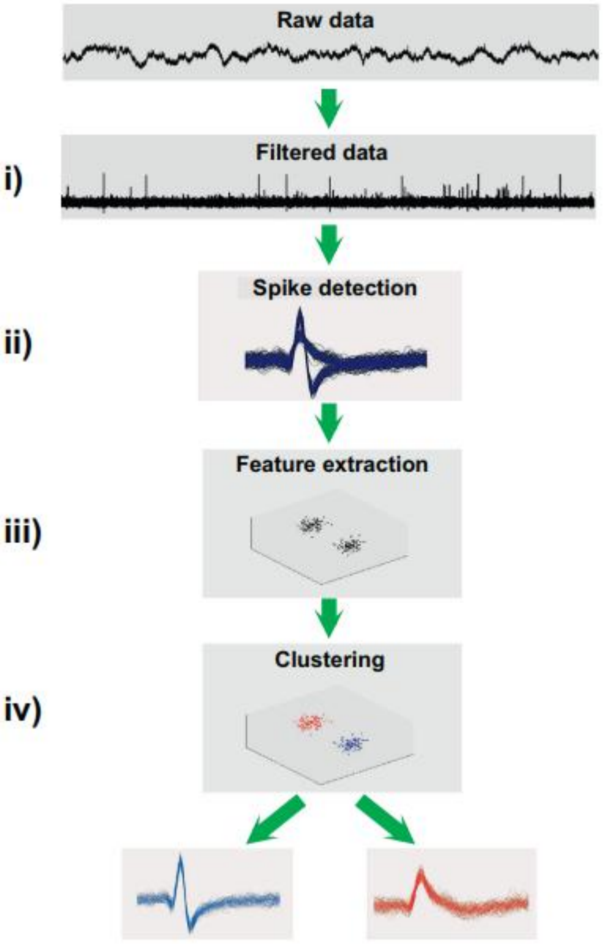


Quiroga et al., 2019

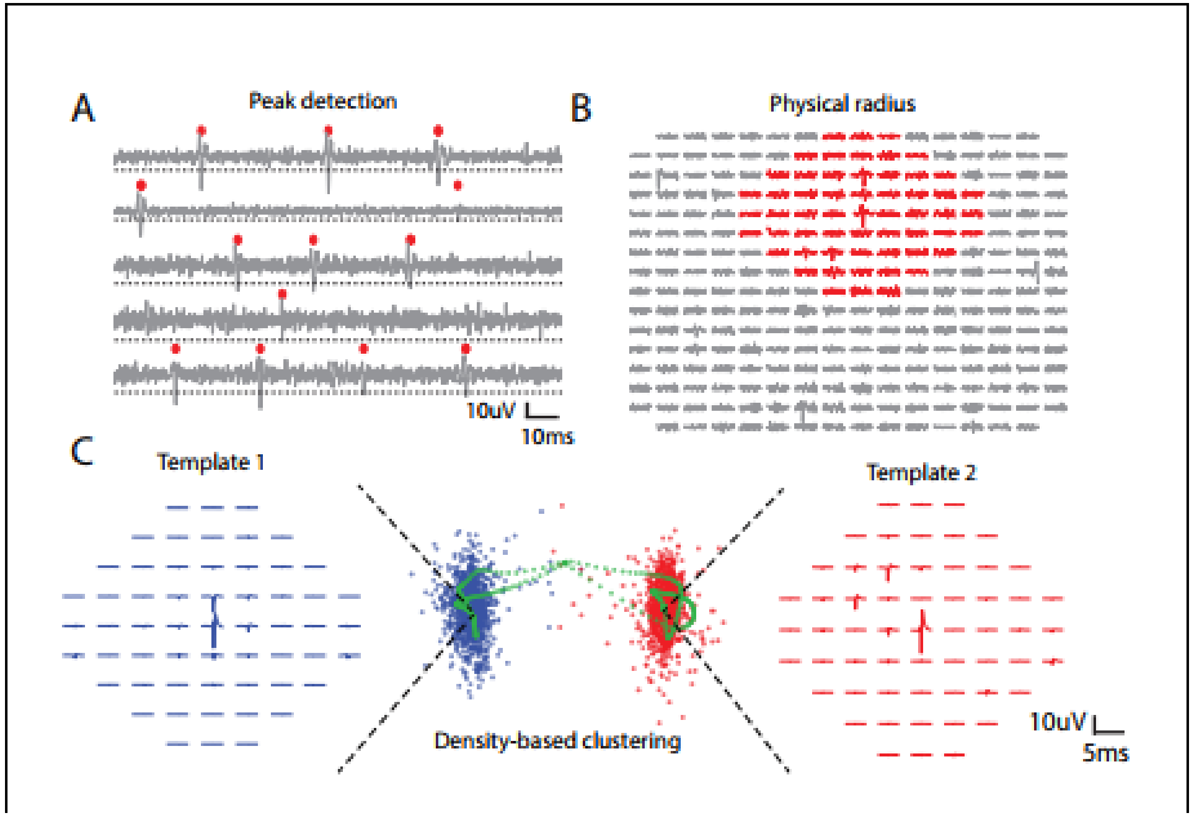
Spike sorting

- ❑ Physics kicks hard!

1) preproc data
2) detect spike
3) extract features:
   - time-to-peak
   - temporal feature
   - shape
   - ...
4) Clustering (aka sorting)



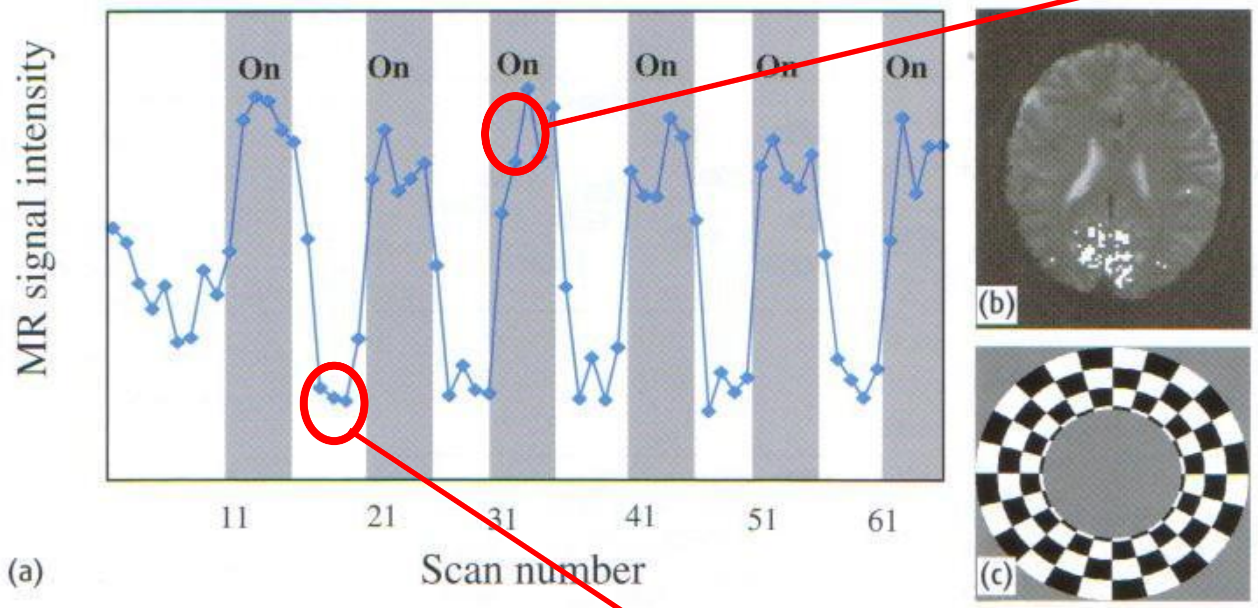
Rey et al., 2015



Yger et al., 2018

Thank you for your attention!

Example of fMRI Protocol

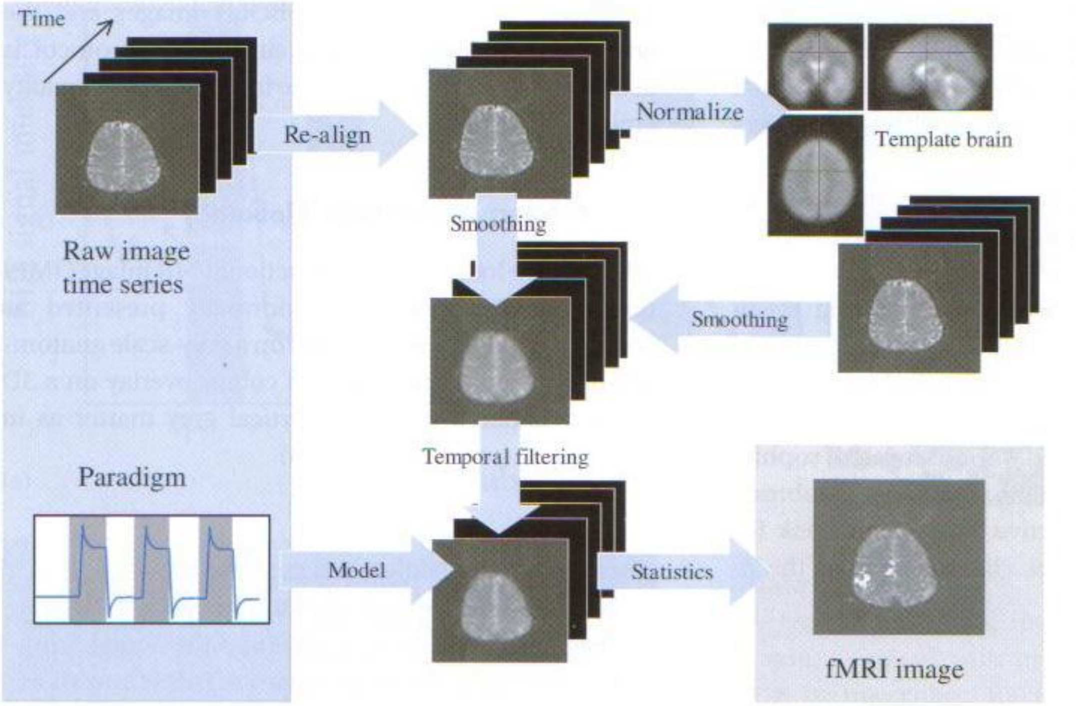


Delay between the stimulus and the vascular changes – might take up to 6 secs

Initial "Dip" – decrease in BOLD signal due to O_2 consumption

fMRI Image Processing Stages

1. Images are re-aligned
2. Spatial normalisation of images to a standard brain space
3. Smooth and normalise the data
4. Combine statistical maps with anatomical information

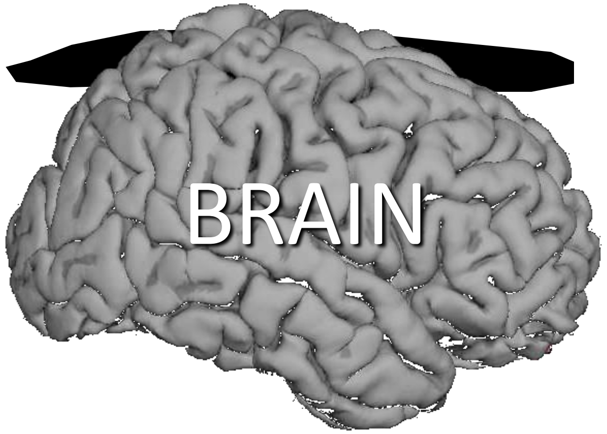


Result is a superposition of a statistical map on a raw image

fMRI is a *Measurement* Technique…

Manipulation Techniques

Lesions, TMS, Stimulation



fMRI, PET, EEG

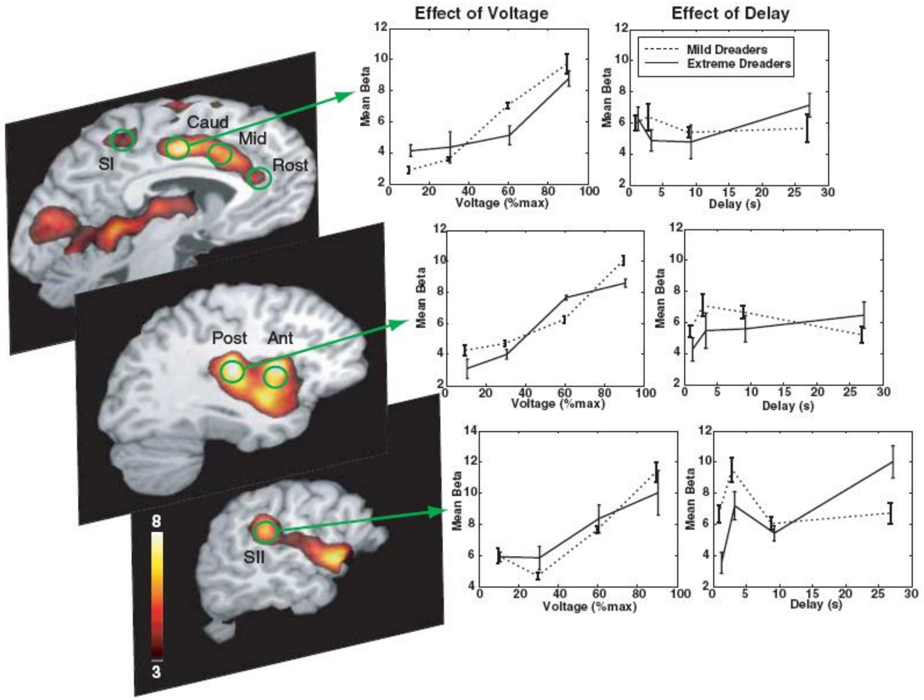
Measurement Techniques

... that provides information about a wide range of topics.

From what we see...
(ocular dominance columns)

Cheng, Waggoner, & Tanaka (2001) *Neuron*

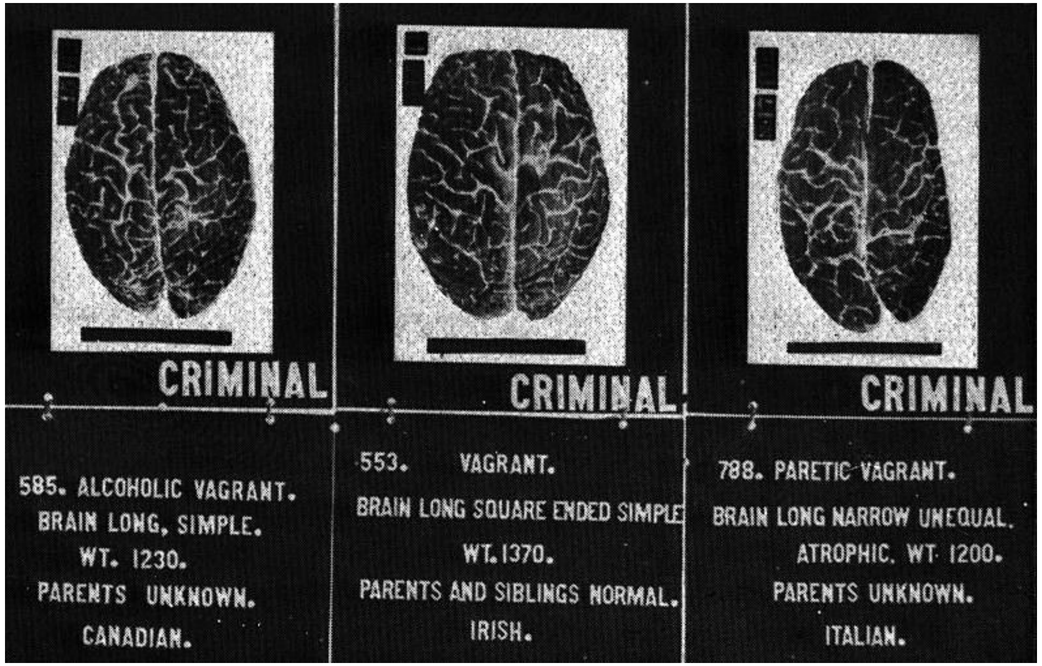
... to what we feel.
(the dread of an upcoming shock)



Berns et al. (2006) *Science*

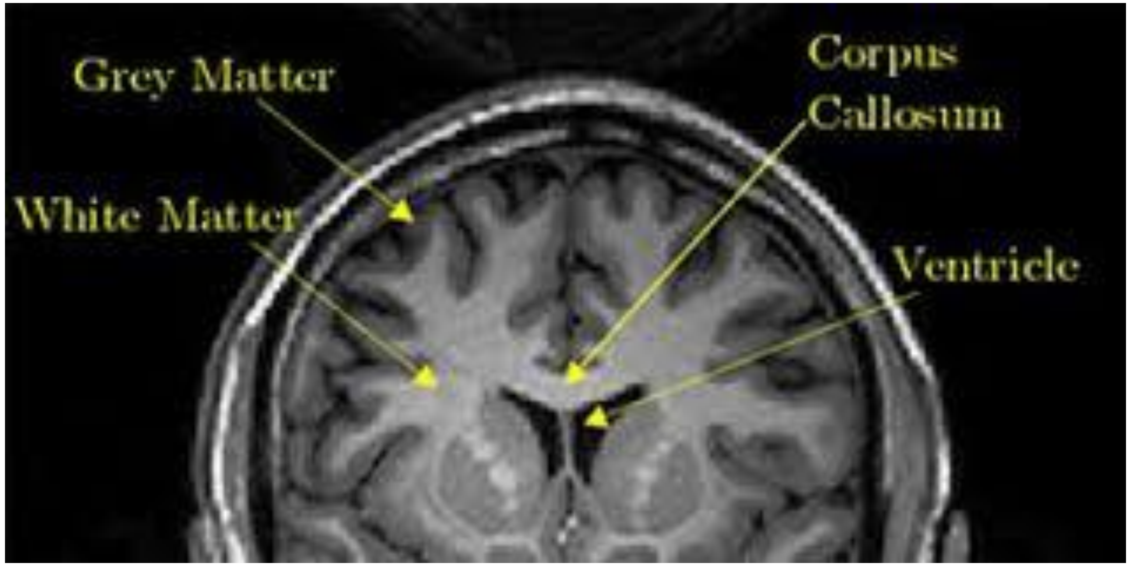
(functional) MRI and forensic application

Can we correlate brain injuries, specific cognitive function and functional impairments to "criminal" behaviors?



Voxel-Based Morphometry

Find difference within volume of the major tissue in the brain



Voxel-Based Morphometry - Case

The defendant was a 24-year-old woman, JF, who was charged with murder for smothering her newborn child to death immediately after delivery. She then wrapped the infant’s body in a towel and hid it inside a suitcase. The defendant later claimed that the newborn child was ‘born dead’ due to drug abstinence syndrome

Voxel-Based Morphometry - Case

Anamnestic information:

Heavily smoke cigarettes at the age of eleven.

Multidrug abuse

Personality profile characterized by antisocial features such as a history of illegal behavior,
sensation seeking, familial conflict, lack of sensitivity

Neuropsychological evaluation revealed impulsivity, a deficit in planning, deficits in emotional attribution and in identifying violations of social norms

Voxel-Based Morphometry - Case

Brain Diagnosis:

VBM compared with healthy women

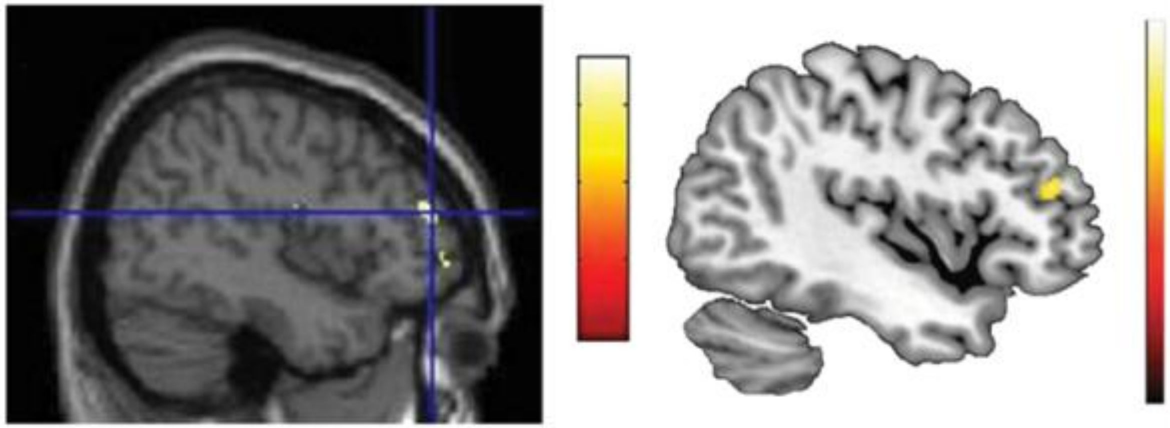


Fig. 3 VBM results. Reprinted from ref. [4]. Left panel: VBM result of the defendant's brain. Right panel: VBM results in binge drinking; image adapted from ref. [64]

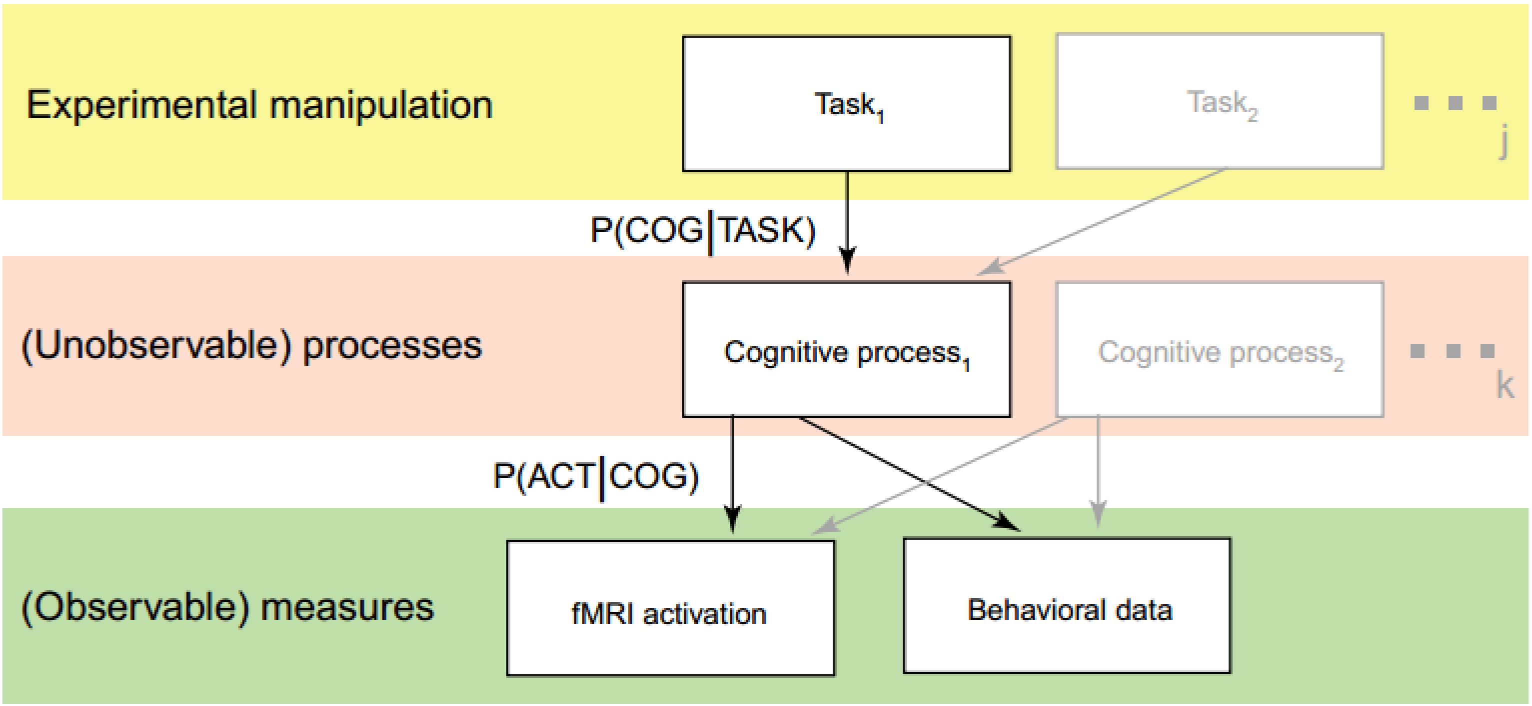
Voxel-Based Morphometry - Case

JF should be considered not responsible for her behavior because she was unable to inhibit her impulses (correlation with anatomical alteration)

Voxel-Based Morphometry - Case

However:

Problem of the **reverse-inference**



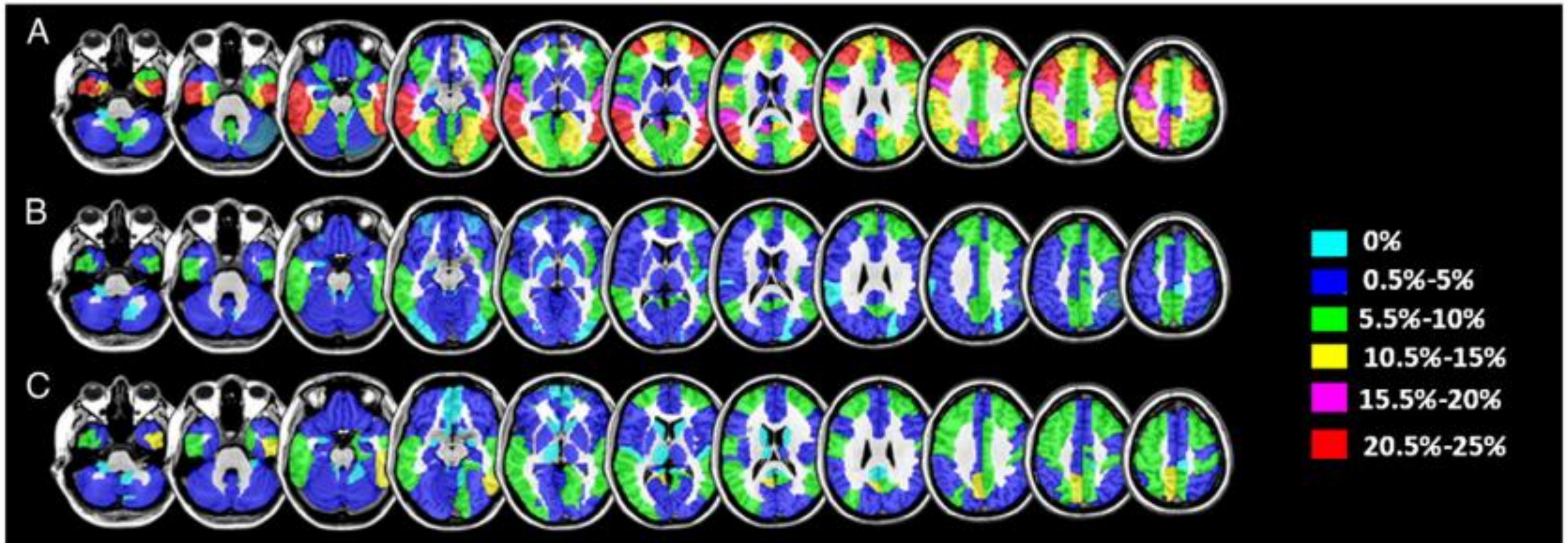
TRENDS in Cognitive Sciences

Voxel-Based Morphometry - Case

However:

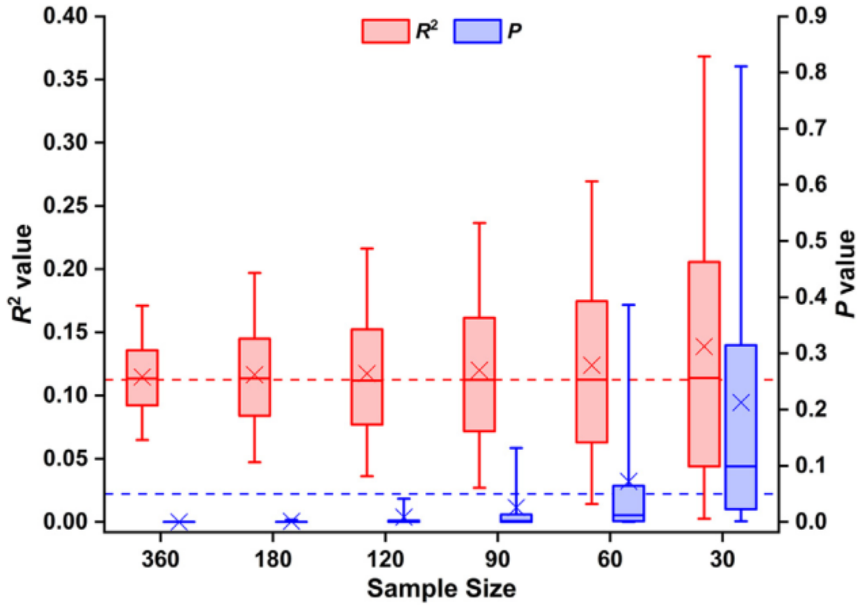
Problem of the **reverse-inference**

23% chance of being a false positive (single vs group)



Brain Lesion (and VBM) problem with sample size

- Lesion Mapping and behavioral correlation statistical significance decrease with sample size

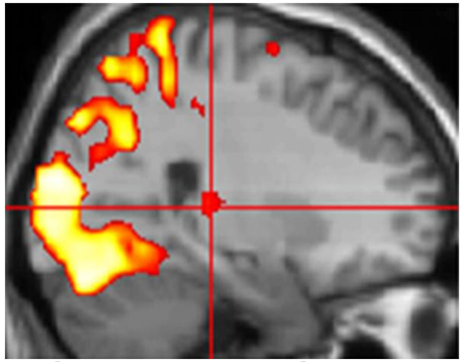


Functional Connectivity

Functional Segregation
Specialised areas exist in the cortex

Functional Integration
Networks of interactions among specialised areas

- Analyses of regionally specific effects
- Identifies regions specialized for a particular task.
- Univariate analysis



Standard SPM

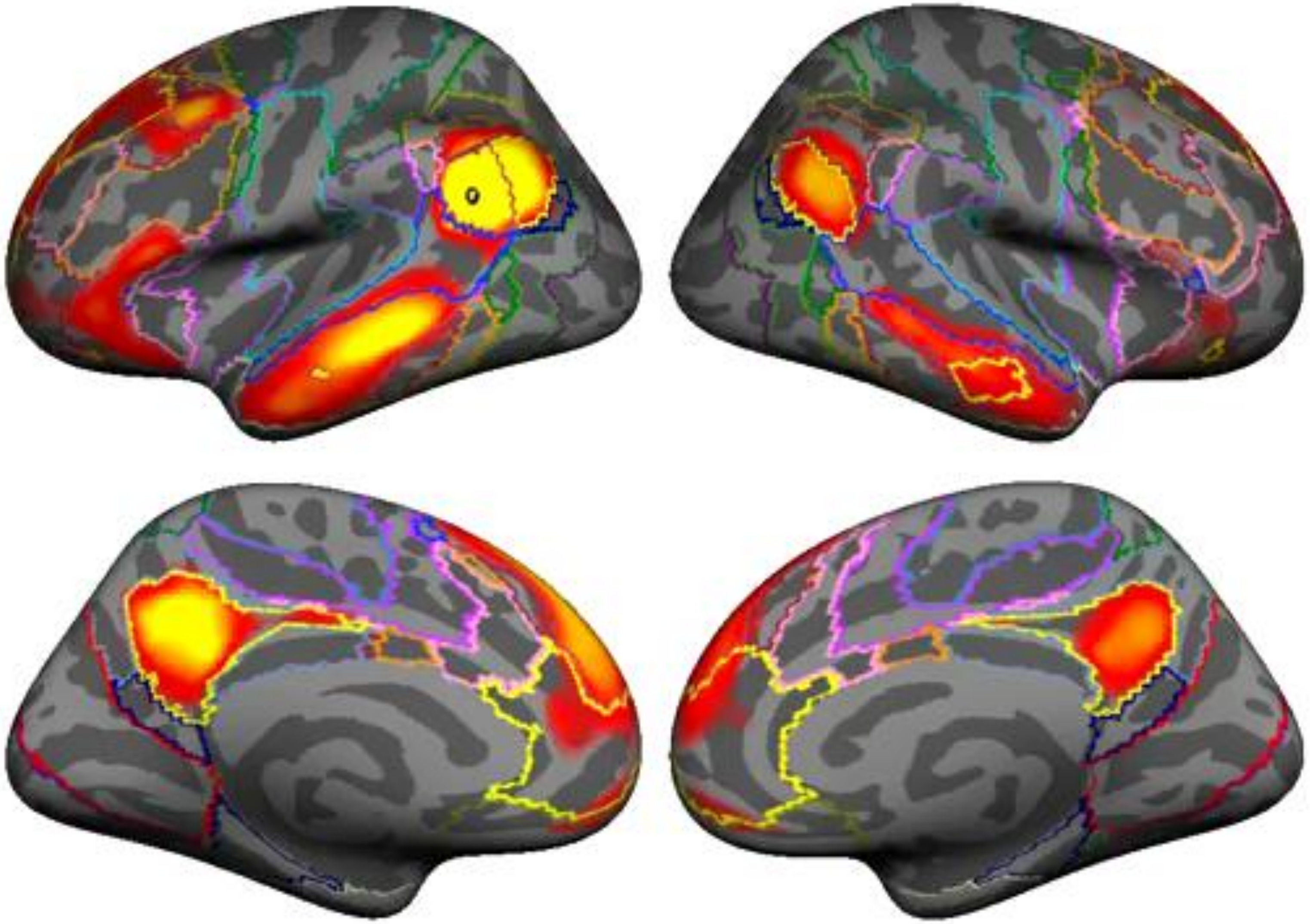
- Analysis of how different regions in a neuronal system interact (coupling).
- Determines how an experimental manipulation affects coupling between regions.
- Univariate & Multivariate analysis

Functional connectivity

Effective connectivity

Adapted from D. Gitelman, 2011

Functional Connectivity



Functional Connectivity

doi:10.1093/scan/nsr075 SCAN (2012) 7, 917–923

Breakdown in the brain network subserving moral judgment in criminal psychopathy

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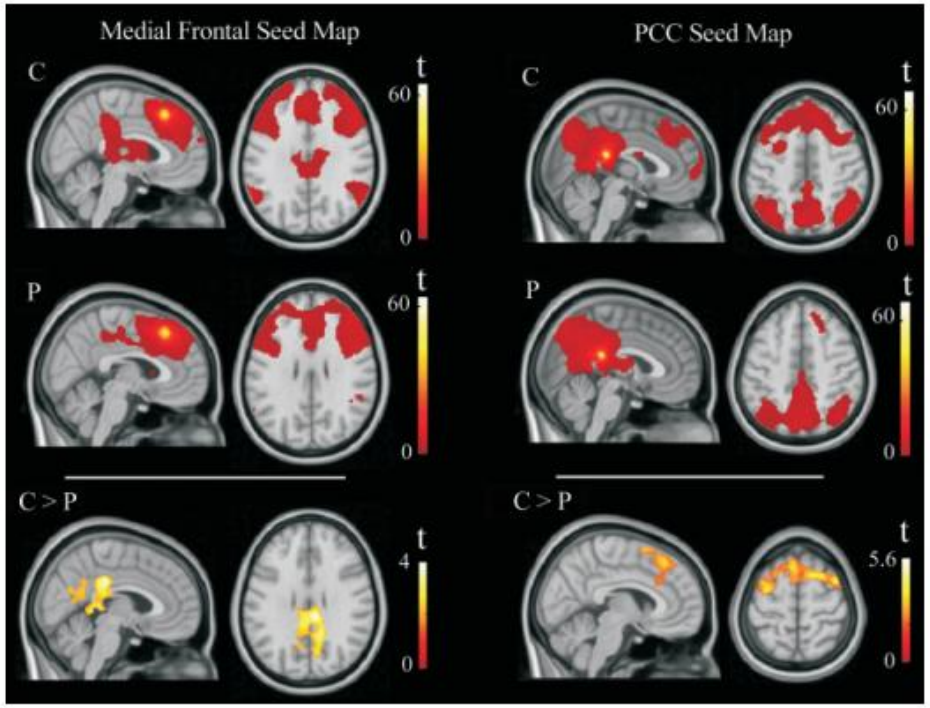

Fig. 3 Functional connectivity during brain resting-state for both medial frontal and posterior cingulate cortex (PCC) 'seed' maps. In contrast with control subjects (C), psychopathic individuals (P) showed significantly reduced functional connectivity between the medial frontal cortex and posterior brain areas and between PCC and anterior (frontal) brain areas. Right side of the Figure corresponds to the right hemisphere for axial views.

Functional Connectivity

Premotor functional connectivity predicts impulsivity in juvenile offenders

Benjamin J. Shannon[a,1], Marcus E. Raichle[a,1], Abraham Z. Snyder[a], Damien A. Fair[b], Kathryn L. Mills[b], Dongyang Zhang[a], Kevin Bache[c], Vince D. Calhoun[c,d], Joel T. Nigg[b], Bonnie J. Nagel[b], Alexander A. Stevens[b], and Kent A. Kiehl[c,d]

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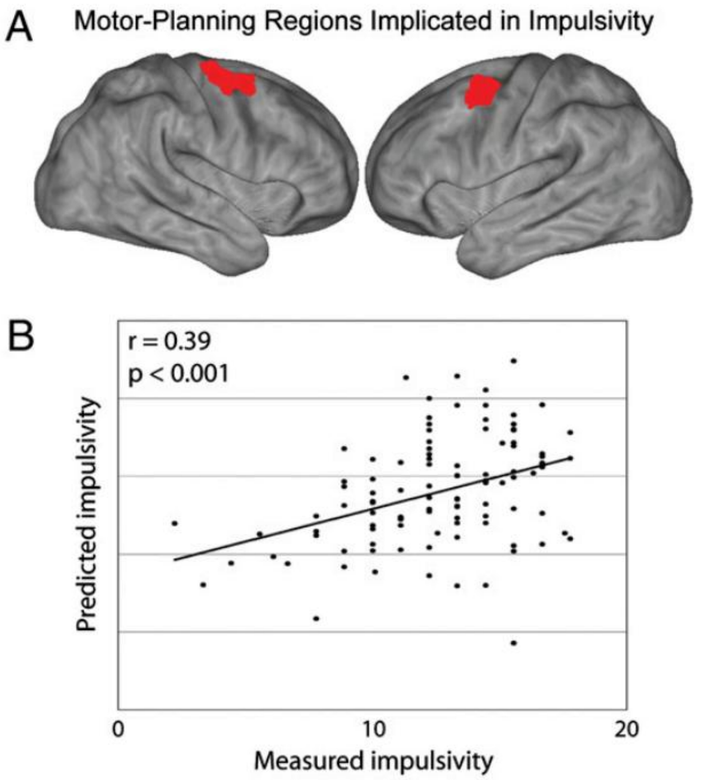
Contributed by Marcus E. Raichle, May 25, 2011 (sent for review February 19, 2011)

- Juvenile offenders: lack of self-control

- “Healty” controls ~100 subjects

- Premotor cortex



Functional Connectivity

PNAS

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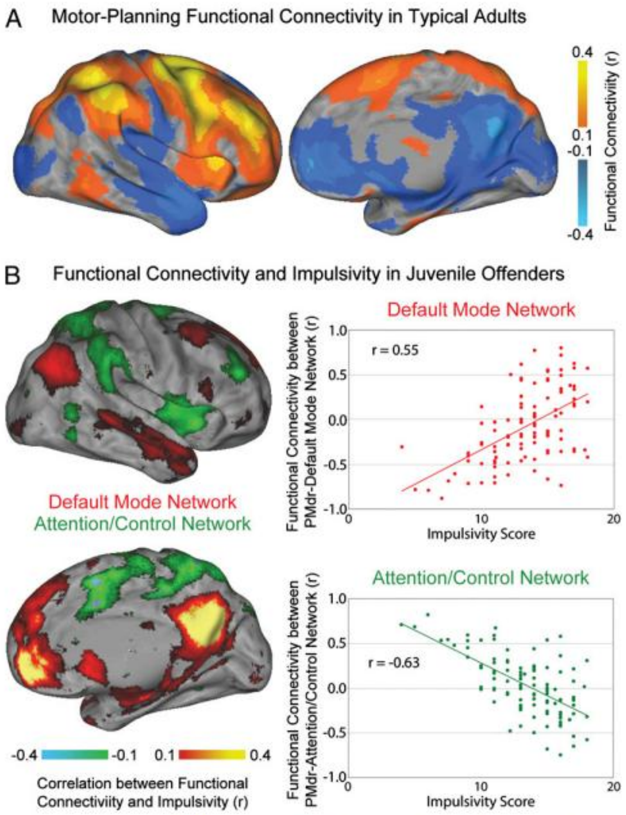


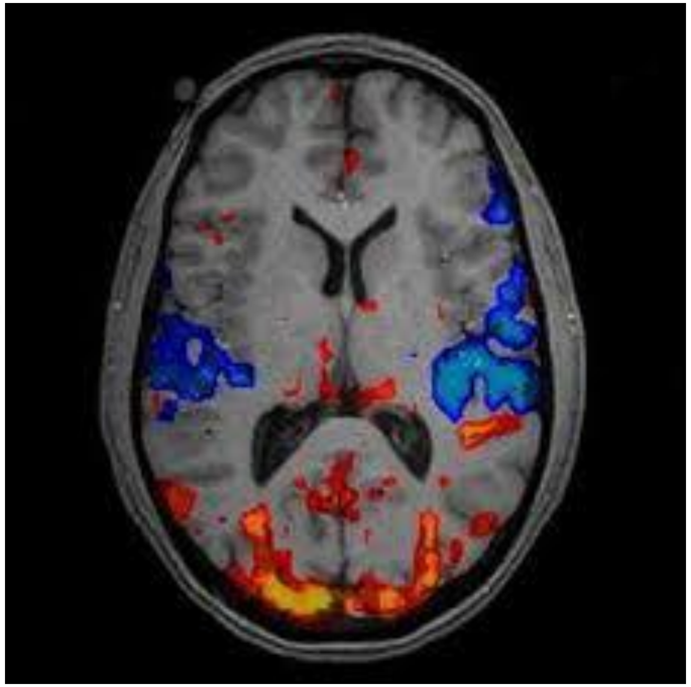
Fig. 2. (*A*) PMdr functional connectivity in typical young adults. Images are thresholded at a value of $r = 0.1$. PMdr is positively correlated with the dorsal-attention network and the executive-control network; it is anticorrelated with the default-mode network. (*B*) Correlation between PMdr functional connectivity and impulsivity in juvenile offenders. PMdr functional connectivity in less-impulsive juveniles is similar to that of adults. However, in more-impulsive individuals its relationships reverse, such that it is positively correlated with the default-mode network and anticorrelated with the dorsal-attention and executive-control networks.

-n healthy young adults and less-impulsive juveniles, PMdr was functionally connected with attention- and control-related networks.

- In more impulsive juveniles, PMdr instead was functionally connected with the default-mode network

Could we read the mind with fMRI?

I could study neural activity for each stimuli in the world



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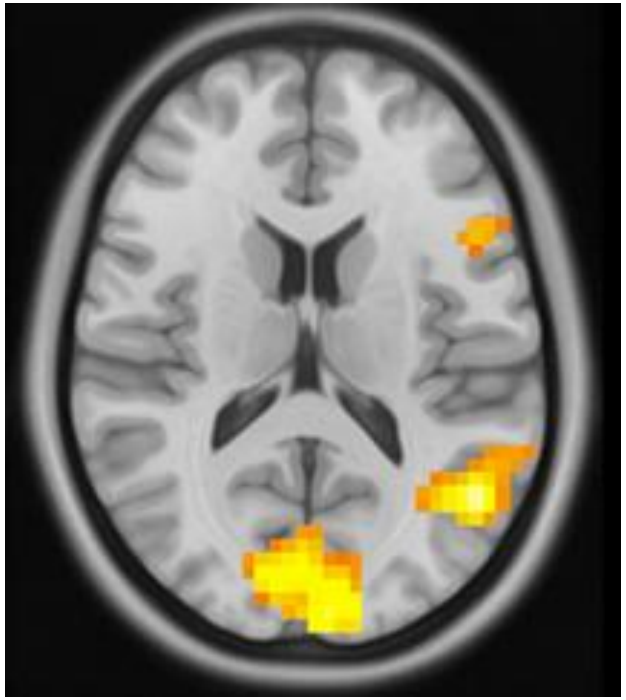
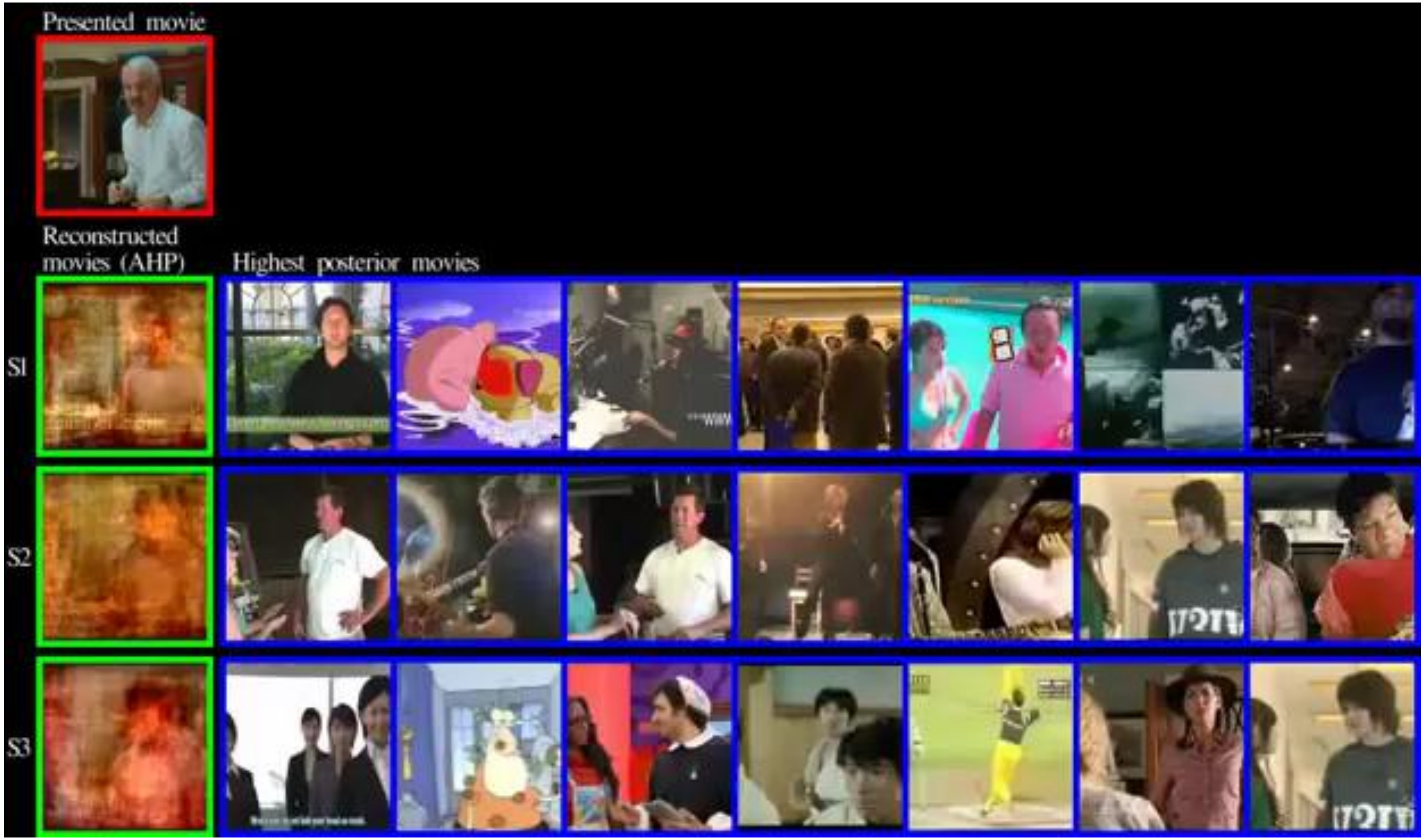


Could we read the mind with fMRI?

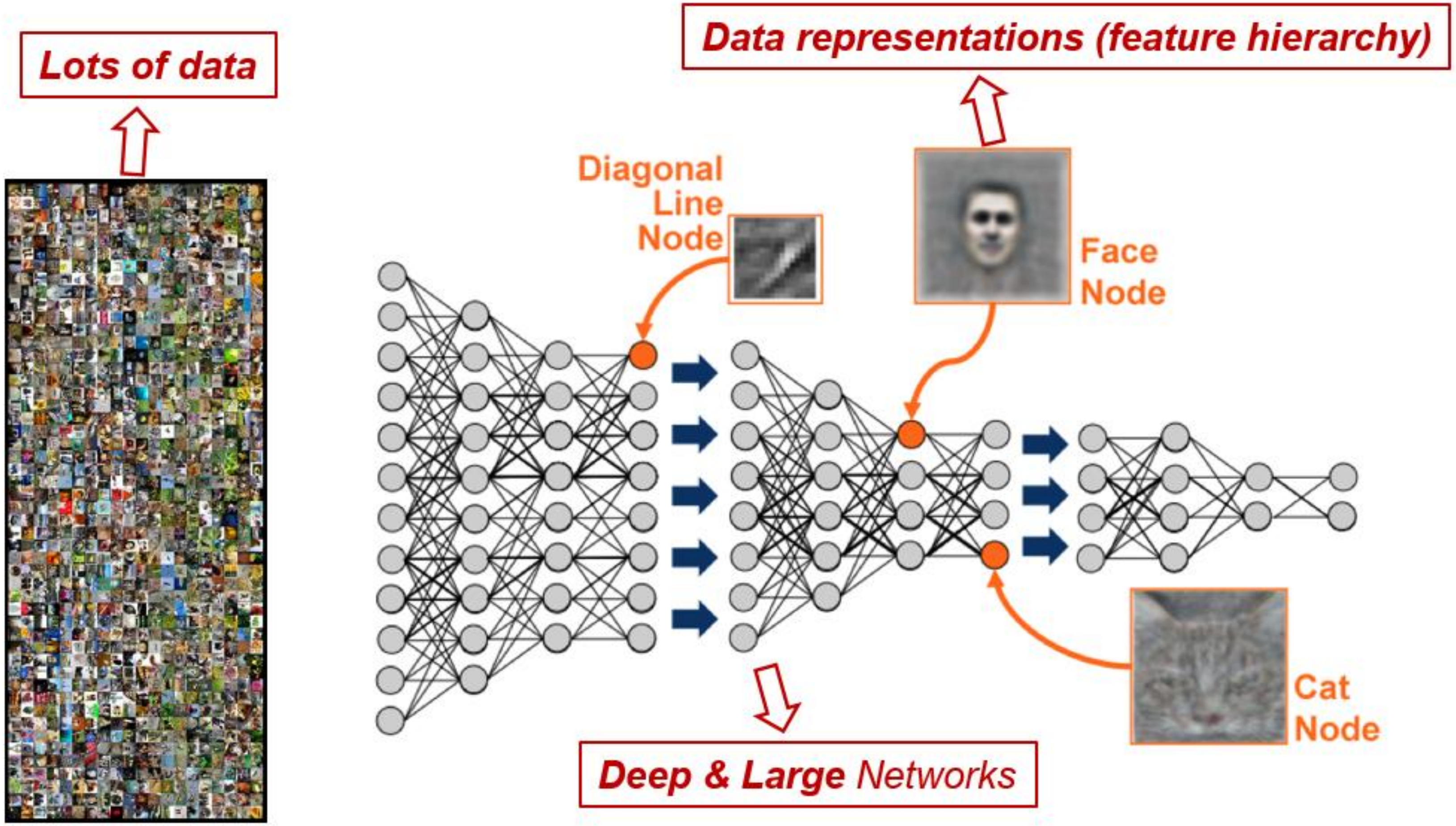
Gallant et al Nature Neuroscience 2016

Could we read the mind with fMRI?



Nishimoto et al., 2011, Current Biology

Deep Learning



Take home message

(f)MRI is quite useful to study brain functions

Every single assumption should be CAREFULY considered

Can we apply these techniques to single subject?

- Yes and No

(F)MRI helps diagnosis, <u>not</u> determines it

Prediction and criminal behaviors

Neuroprediction of future rearrest

Eyal Aharoni[a,b,1,2], Gina M. Vincent[c], Carla L. Harenski[a], Vince D. Calhoun[a,d], Walter Sinnott-Armstrong[e], Michael S. Gazzaniga[f], and Kent A. Kiehl[a,b,2]

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- 96 women after prison release
- ACC is central to an error-monitoring circuit wherein it relays error information from the basal ganglia and inferior frontal cortex to motor area → RESPONSE INIBITION
- Go/No-Go Task

Prediction and criminal behaviors

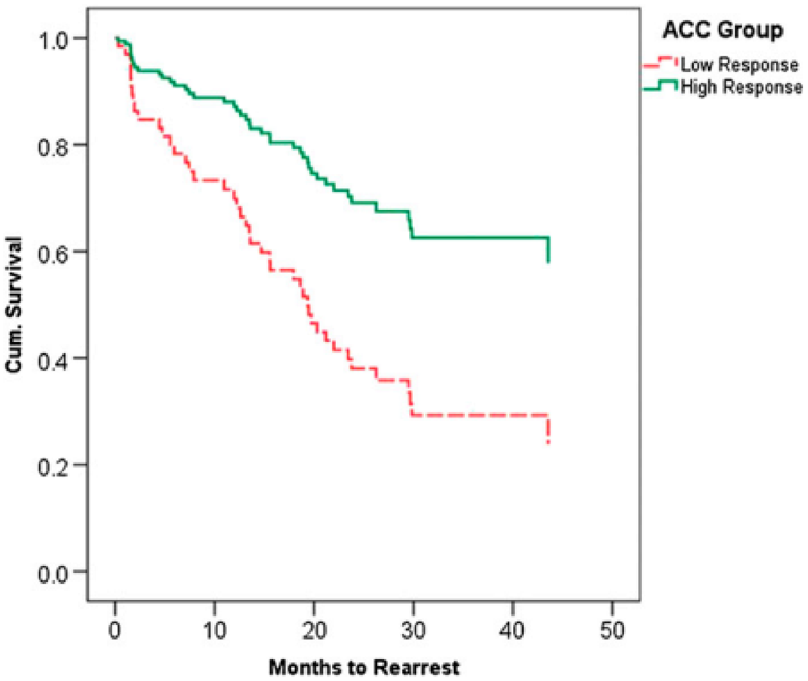

Fig. 1. Cox survival function showing proportional rearrest survival rates of high (solid green) vs. low (dashed red) ACC response groups for any crime over a 4-y period. Results of this median split analysis were equivalent to that of the parametric model: bootstrapped $B = 0.96$; SE = 0.40; $P < 0.01$; 95% CI, 0.29–1.84. The mean survival times to rearrest for the low and high ACC activity groups were 25.27 (2.80) mo and 32.42 (2.73) mo, respectively. The overall probabilities of rearrest were 60% for the low ACC group and 46% for the high ACC group.

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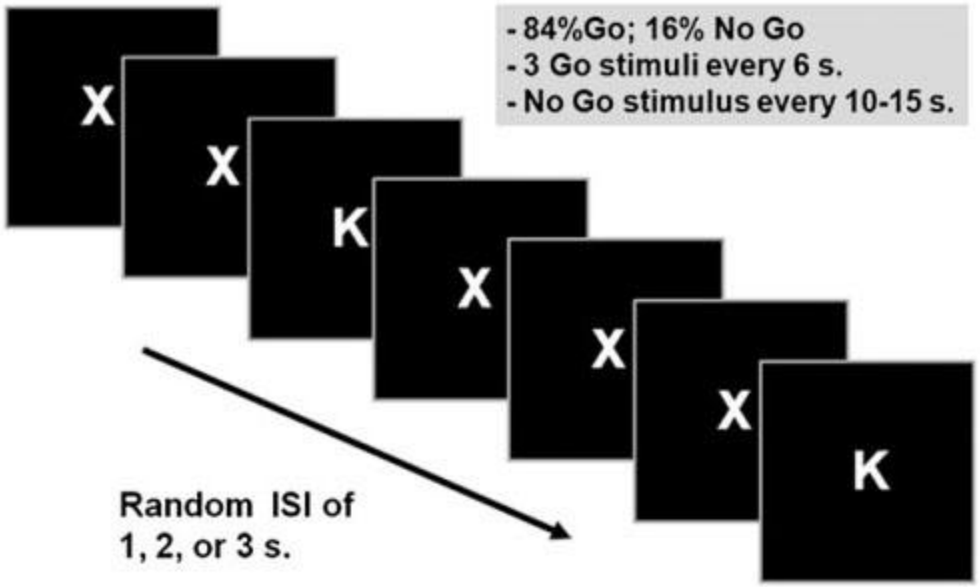
Prediction and criminal behaviors

- The GNG task is enough?

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References

http://mriphysics.github.io/

https://www.nmr.hhu.de/main/mtheorie_3_e.html

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2799958/

Psychedelics and fNIRS neuroimaging: exploring new opportunities
Felix Scholkmann, Franz X. Vollenweider

The Clinical Neurophysiology Primer

https://www.learningeeg.com/artifacts

MEG - EEG primer